Physics Lectures

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.

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Contents

Preface: What These Notes Are For

Remark

These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether's theorem and conserved currents,
- Green's functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

$$\text{classical field} \longrightarrow \text{mode expansion} \longrightarrow \text{operators} \longrightarrow \text{states} \longrightarrow \text{correlators} \longrightarrow \text{amplitudes.}$$

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

A free field is the continuum limit of a system with many coupled degrees of freedom. Its normal modes behave like independent harmonic oscillators. The first idea of quantum field theory is to quantize these oscillators and to interpret their excitations as field quanta.

1.1 The Purpose of the First Step

The transition from classical field theory to quantum field theory can look too large if one begins immediately with operator-valued distributions, Fock spaces, and relativistic commutators. This chapter takes a slower route. It explains why the harmonic oscillator algebra appears at the beginning of quantum field theory.

The guiding chain of ideas is

$$\text{classical field} \longrightarrow \text{normal modes} \longrightarrow \text{classical oscillators} \longrightarrow \text{quantum oscillators} \longrightarrow \text{field quanta.}$$

The goal is not yet to construct the full canonical field algebra. That will be done in the next chapter. Here we only identify the mechanical structure hidden inside a free field.

1.2 A Finite System of Coupled Oscillators

Consider a finite chain of N identical masses connected by springs. Let $q_j(t)$ be the displacement of the j-th mass from equilibrium. A typical quadratic Lagrangian has the form

$$L = \sum_{j=1}^{N} \frac{m}{2}\dot{q}_j^2 - \sum_{j=1}^{N} \frac{\kappa}{2}(q_{j+1} - q_j)^2,$$

with boundary conditions chosen so that the notation is meaningful. The coordinates q_j are coupled because the potential energy depends on differences between neighbouring displacements.

The normal-mode method replaces the original coordinates by new coordinates $Q_r(t)$,

$$q_j(t) = \sum_r Q_r(t) u_r(j),$$

where $u_r(j)$ are spatial mode functions. With an appropriate normalization, the Lagrangian becomes diagonal:

$$L = \sum_r \left(\frac{1}{2}\dot{Q}_r^2 - \frac{1}{2}\omega_r^2 Q_r^2 \right).$$

Thus the coupled system is equivalent to a collection of independent harmonic oscillators.

Example

The original variables q_j describe local displacements of the masses. The normal coordinates Q_r describe collective oscillation patterns of the whole chain. Quantization is easiest in the normal coordinates because the modes are independent.

1.3 The Field as a Continuum Limit

A field is the continuum analogue of a system with one degree of freedom at each spatial point. In a one-dimensional chain with lattice spacing a, the site position is $x_j = ja$, and one may think schematically of

$$q_j(t) \approx \phi(t, x_j).$$

As $a \to 0$, sums over sites become spatial integrals, and discrete normal modes become Fourier modes.

For a real scalar field in a finite box of volume V, periodic boundary conditions make the allowed momenta discrete. A schematic Fourier expansion is

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}}.$$

For a real field the complex coefficients are not all independent; one must impose the appropriate reality condition. Equivalently, one may use real sine and cosine modes. The important point is independent of this choice: after the free field is decomposed into normal modes, each independent mode behaves like a classical oscillator.

1.4 Each Free Mode Is a Harmonic Oscillator

The free real scalar field has Lagrangian

$$L = \int \mathrm{d}^3x \left(\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 \right).$$

Its equation of motion is the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

After the free field is diagonalized into normal modes, the Lagrangian takes the schematic oscillator form

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2}\dot{q}_{\mathbf{k}}^2 - \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The canonical momentum of a mode is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian is

$$H = \sum_{\mathbf{k}} H_{\mathbf{k}}, \qquad H_{\mathbf{k}} = \frac{1}{2}p_{\mathbf{k}}^2 + \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2.$$

A free scalar field is therefore a collection of independent harmonic oscillators, one for each independent normal mode.

Remark

Interactions destroy this simple diagonal form. Perturbative QFT begins by quantizing the free oscillators exactly and then treating interactions as corrections to the free theory.

1.5 Quantizing One Harmonic Oscillator

Before quantizing all field modes, recall the single harmonic oscillator. The classical variables q and p become operators $\hat{q}$ and $\hat{p}$ satisfying

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

Introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega}{2}}\,\hat{q} + \frac{i}{\sqrt{2\omega}}\,\hat{p}, \qquad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\,\hat{q} - \frac{i}{\sqrt{2\omega}}\,\hat{p}.$$

Then

$$[\hat{a}, \hat{a}^\dagger] = 1,$$

and

$$\hat{H} = \omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right).$$

The operator $\hat{a}^\dagger\hat{a}$ counts oscillator excitations.

1.6 Number States and the Oscillator Vacuum

The oscillator vacuum is defined by

$$\hat{a}|0\rangle = 0.$$

The excited states are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \qquad n = 0, 1, 2, \ldots.$$

If

$$\hat{N} = \hat{a}^\dagger\hat{a},$$

then

$$\hat{N}|n\rangle = n|n\rangle, \qquad \hat{H}|n\rangle = \omega\left(n + \frac{1}{2}\right)|n\rangle.$$

The vacuum is not a state of zero energy. It has the zero-point energy $\omega/2$. This fact becomes more delicate for fields because a field has infinitely many oscillator modes.

1.7 From One Oscillator to Many Field Modes

For a free field, every independent mode is quantized as an oscillator. Thus one introduces a pair of operators for each mode,

$$\hat{a}_{\mathbf{k}}, \qquad \hat{a}^\dagger_{\mathbf{k}}.$$

In a finite box their algebra is

$$[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = \delta_{\mathbf{k},\mathbf{p}}, \qquad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \qquad [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = 0.$$

The free-field vacuum is the state annihilated by all annihilation operators,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for all } \mathbf{k}.$$

A one-quantum state is obtained by applying one creation operator:

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

A many-quantum state is obtained by applying several creation operators.

Definition

The *Fock space* of the free field is the Hilbert space generated from the vacuum by applying creation operators for all modes. It naturally contains states with zero, one, two, and more field quanta.

1.8 Field Quanta as Mode Excitations

For the free theory, the Hamiltonian has the schematic form

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The operator $\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}$ counts how many quanta occupy the mode $\mathbf{k}$. The state $\hat{a}^\dagger_{\mathbf{k}}|0\rangle$ has one quantum of energy $\omega_{\mathbf{k}}$ above the free vacuum and momentum $\mathbf{k}$. In the free theory, this is what one means by a one-particle state.

This interpretation is precise for a free field. In interacting QFT, particles can be dressed by interactions, unstable particles appear as resonances, and the vacuum itself can be nontrivial. Nevertheless, the free Fock-space picture is the starting point for perturbation theory and scattering.

1.9 What Is Still Missing?

The oscillator construction explains why creation and annihilation operators appear, but it is not yet the full canonical quantization of a field. We have not yet imposed commutation relations directly on $\hat{\phi}(t, \mathbf{x})$ and its conjugate momentum $\hat{\pi}(t, \mathbf{x})$. We have also not fixed the continuum normalization of the mode expansion, nor have we discussed fields as operator-valued distributions.

These are the tasks of the next chapter. Chapter 2 turns the oscillator picture into the canonical quantization of the real scalar field.

Summary

A free classical field can be diagonalized into independent normal modes. Each mode is a harmonic oscillator. Quantizing those oscillators gives creation and annihilation operators. Acting with creation operators on the vacuum builds Fock space, and the resulting mode excitations are interpreted as free particles. This oscillator picture is the conceptual bridge to canonical field quantization.

1.10 Chapter Checklist

After reading this chapter, one should be able to:

- explain why a free field decomposes into normal modes;
- explain why each independent mode behaves as a harmonic oscillator;
- quantize a single harmonic oscillator using $\hat{a}$ and $\hat{a}^\dagger$;
- define the oscillator vacuum and number states;
- describe the free-field vacuum and one-particle states;
- explain why free particles are mode excitations;
- state what remains to be done in canonical field quantization.

Chapter 2

Canonical Quantization of the Real Scalar Field

Remark

Chapter 1 explained why a free field should be thought of as a collection of oscillators. This chapter makes that statement precise. We impose canonical commutation relations on the field and its conjugate momentum, derive the standard mode expansion, and recover the Fock-space interpretation from the field algebra itself.

2.1 From the Oscillator Picture to Field Operators

The previous chapter started from normal modes. It argued that a free scalar field is a collection of harmonic oscillators labelled by momentum. That is the right intuition, but it is not yet the full canonical quantization of a field.

The full canonical construction begins before the normal-mode expansion. It starts with the classical field $\phi(t, \mathbf{x})$, its conjugate momentum $\pi(t, \mathbf{x})$, and their Poisson brackets. Quantization then promotes ϕ and π to operators and replaces the Poisson brackets by commutators.

The logic of this chapter is therefore

classical field variables $\longrightarrow$ equal-time commutators $\longrightarrow$ operator mode expansion $\longrightarrow$ creation and annihilatio

This is the formal version of the oscillator picture developed in Chapter 1.

2.2 Classical Starting Point

We use units $c = \hbar = 1$ and the metric convention $(+, -, -, -)$. The free real scalar field is described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2.$$

The conjugate momentum is

$$\pi(t, \mathbf{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}).$$

The Euler–Lagrange equation is the Klein–Gordon equation,

$$(\Box + m^2)\phi(x) = 0,$$

where $x = (t, \mathbf{x})$. In components this reads

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

The classical Hamiltonian density is

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2,$$

so the Hamiltonian is

$$H = \int \mathrm{d}^3x\,\left(\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2\right).$$

This is the continuum field-theoretic analogue of the oscillator Hamiltonian $p^2/2 + \omega^2 q^2/2$.

2.3 Equal-Time Commutation Relations

The classical canonical Poisson bracket is

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\}_{\mathrm{PB}} = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Canonical quantization promotes ϕ and π to operators $\hat{\phi}$ and $\hat{\pi}$, and imposes

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

The remaining equal-time commutators vanish:

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0, \qquad [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

These relations are the field version of

$$[\hat{q}, \hat{p}] = i.$$

Instead of one coordinate and one momentum, there is one canonical pair at every point of space.

Definition

The *equal-time canonical commutation relations* are the basic operator relations between $\hat{\phi}$ and $\hat{\pi}$ at a fixed time. For bosonic fields they are commutators; for fermionic fields they will be replaced by anticommutators.

2.4 Finite Box and Continuum Normalization

Chapter 1 used a finite box to keep the momentum labels discrete. In a finite box, one writes sums over momenta and Kronecker deltas. In infinite volume, one uses integrals and Dirac deltas. The replacement rules are schematically

$$\frac{1}{V}\sum_{\mathbf{k}} \longrightarrow \int \frac{\mathrm{d}^3k}{(2\pi)^3}, \qquad V\delta_{\mathbf{k},\mathbf{p}} \longrightarrow (2\pi)^3\delta^{(3)}(\mathbf{k} - \mathbf{p}).$$

The continuum formulas below use the second convention. They look slightly more complicated than finite-box formulas, but they are the standard relativistic normalization used in scattering theory.

2.5 Mode Expansion of the Field Operator

The quantum field operator is expanded in positive- and negative-frequency solutions of the Klein–Gordon equation:

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{a}^\dagger_{\mathbf{k}} e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}}\right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

Since the field is real, $\hat{\phi}$ is Hermitian:

$$\hat{\phi}(t, \mathbf{x})^\dagger = \hat{\phi}(t, \mathbf{x}).$$

The appearance of both $\hat{a}_{\mathbf{k}}$ and $\hat{a}^\dagger_{\mathbf{k}}$ is the operator version of the reality condition on a classical real field.

The conjugate momentum operator is

$$\hat{\pi}(t, \mathbf{x}) = \partial_t \hat{\phi}(t, \mathbf{x}),$$

therefore

$$\hat{\pi}(t, \mathbf{x}) = -i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}} t + i\mathbf{k}\cdot\mathbf{x}} - \hat{a}^\dagger_{\mathbf{k}} e^{i\omega_{\mathbf{k}} t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The normalization factors are chosen so that the equal-time field commutators are satisfied.

2.6 Ladder-Operator Algebra

The canonical field commutators imply

$$[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \qquad [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = 0.$$

Conversely, these ladder-operator relations imply the equal-time commutators of $\hat{\phi}$ and $\hat{\pi}$. This is the precise connection between the canonical field algebra and the oscillator algebra of Chapter 1.

The operator $\hat{a}^\dagger_{\mathbf{k}}$ creates a quantum in the momentum mode $\mathbf{k}$, while $\hat{a}_{\mathbf{k}}$ annihilates such a quantum. For the real scalar field there is only one kind of neutral scalar quantum.

2.7 Fock Space Revisited

The free-field vacuum is defined by

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \qquad \text{for every } \mathbf{k}.$$

A one-particle momentum eigenstate is

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}} |0\rangle.$$

A two-particle state is

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}^\dagger_{\mathbf{k}_1} \hat{a}^\dagger_{\mathbf{k}_2} |0\rangle,$$

up to normalization. Since the creation operators commute, the scalar quanta are bosons.

A realistic one-particle state is usually a wave packet rather than a single momentum eigenstate:

$$|f\rangle = \int \frac{d^3k}{(2\pi)^3} f(\mathbf{k}) \hat{a}^\dagger_{\mathbf{k}} |0\rangle.$$

The function $f(\mathbf{k})$ describes the momentum profile of the state. This is the first place where the particle idea becomes more refined than the simple phrase "one quantum of one mode."

2.8 Hamiltonian and Momentum Operators

The quantum Hamiltonian is obtained from the classical Hamiltonian by replacing ϕ and π by field operators:

$$\hat{H} = \int d^3x \left(\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2 \right).$$

Substituting the mode expansion gives the formal expression

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \frac{1}{2}(2\pi)^3 \delta^{(3)}(0) \right).$$

The first term counts excitations; the second term is the continuum notation for the sum of oscillator zero-point energies.

The spatial momentum operator is

$$\hat{\mathbf{P}} = \int \mathrm{d}^3x\, \hat{\pi}\nabla\hat{\phi}.$$

With the same ordering convention, it becomes

$$\hat{\mathbf{P}} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \mathbf{k}\, \hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}.$$

Thus

$$\hat{H}\, \hat{a}^\dagger_{\mathbf{k}}|0\rangle = \omega_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{k}}|0\rangle \quad \text{above the vacuum energy,}$$

and

$$\hat{\mathbf{P}}\, \hat{a}^\dagger_{\mathbf{k}}|0\rangle = \mathbf{k}\, \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

The dispersion relation is

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + m^2.$$

This is why m is interpreted as the mass of the scalar quantum.

2.9 Normal Ordering

The vacuum-energy term in $\hat{H}$ is infinite in the continuum limit. For the free theory, it is common to define the normal-ordered Hamiltonian by moving all creation operators to the left of all annihilation operators:

$$:\hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}}: = \hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{k}}.$$

Then

$$:\hat{H}: = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \omega_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}},$$

and

$$\langle 0| :\hat{H}: |0\rangle = 0.$$

Normal ordering removes the constant free vacuum energy from this Hamiltonian. It does not remove all divergences of interacting quantum field theory. It is a useful convention at the free-field level, not a substitute for renormalization.

2.10 Zero-Point Energy

In a finite box, the vacuum energy of the free scalar field is

$$E_0 = \frac{1}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

In infinite-volume notation this becomes

$$E_0 = \frac{V}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \omega_{\mathbf{k}}.$$

The integral diverges at large momentum. The divergence is a reminder that a field has infinitely many short-distance degrees of freedom.

For many non-gravitational scattering problems, only energy differences above the free vacuum matter, and normal ordering is sufficient. In contexts where vacuum energy gravitates, or where the vacuum itself changes, the issue becomes more subtle. The present course will return to this point when discussing renormalization and vacuum structure.

2.11 Fields as Operator-Valued Distributions

The notation $\hat{\phi}(x)$ suggests an operator at a point. This notation is useful, but local quantum fields are more accurately understood as operator-valued distributions. They become well-defined only after smearing with test functions.

At fixed time, a smeared field is

$$\hat{\phi}(f) = \int \mathrm{d}^3x\, f(\mathbf{x})\hat{\phi}(t, \mathbf{x}).$$

Products such as $\hat{\phi}(x)^2$ at the same spacetime point are generally singular and require a prescription. This is one of the structural reasons why renormalization appears in interacting field theory.

This point also clarifies the role of the mode expansion. The formula for $\hat{\phi}(t, \mathbf{x})$ is a compact way to encode how the field acts after it has been smeared. It should not be treated as an ordinary function with operator coefficients.

2.12 Microcausality

Relativistic quantum field theory must respect the causal structure of spacetime. For a scalar field, the basic local causality condition is

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \qquad \text{if} \qquad (x-y)^2 < 0.$$

This condition is called *microcausality*. It says that field observables associated with spacelike separated regions commute.

For the free scalar field one can write

$$[\hat{\phi}(x), \hat{\phi}(y)] = i\Delta(x-y),$$

where Δ is the Pauli–Jordan commutator function. This function vanishes outside the light cone. The equal-time relation

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0$$

is therefore only the simplest special case of the full spacetime causality condition.

Summary

Canonical quantization begins with $\hat{\phi}$ and $\hat{\pi}$, not directly with particles. The equal-time commutation relations determine the ladder operator algebra in the mode expansion. The free-field vacuum and Fock space are then recovered from that algebra, matching the oscillator picture of Chapter 1. The new lessons are the precise continuum normalization, the Hamiltonian and momentum operators, normal ordering, the distributional nature of fields, and microcausality.

2.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the classical Lagrangian and Hamiltonian of the free real scalar field;
- derive the conjugate momentum $\pi = \dot{\phi}$;
- state the equal-time canonical commutation relations;

- write the mode expansions of $\hat{\phi}$ and $\hat{\pi}$;
- derive the ladder-operator commutators from the canonical field algebra;
- construct the vacuum, one-particle states, wave packets, and many-particle states;
- express $\hat{H}$ and $\hat{\mathbf{P}}$ in terms of ladder operators;
- explain what normal ordering removes and what it does not remove;
- explain why fields are operator-valued distributions;
- state the microcausality condition for a scalar field.

Chapter 3

The Complex Scalar Field and Antiparticles

Remark

The real scalar field has one neutral quantum for each momentum mode. The complex scalar field is the next step: the field is no longer Hermitian, and its quantization produces two independent kinds of quanta. They are naturally interpreted as particles and antiparticles.

3.1 From a Real Field to a Complex Field

Chapter 2 quantized a real scalar field. The reality condition $\hat{\phi}^\dagger = \hat{\phi}$ forced the same field operator to contain both annihilation and creation operators. A complex scalar field is different. The field ϕ and its Hermitian conjugate $\phi^\dagger$ are independent classical variables before quantization.

One may think of a complex scalar field as two real scalar fields packaged into one object:

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \qquad \phi^\dagger = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2).$$

The important new feature is that the theory has a global phase symmetry. The field may be rotated by a constant phase,

$$\phi \longrightarrow e^{-i\alpha}\phi, \qquad \phi^\dagger \longrightarrow e^{i\alpha}\phi^\dagger.$$

By Noether's theorem this symmetry gives a conserved charge. In the quantum theory, that charge will distinguish particles from antiparticles.

3.2 Classical Lagrangian and Canonical Momenta

The free complex scalar field is described by

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi.$$

In terms of time and space derivatives,

$$\mathcal{L} = \dot{\phi}^\dagger \dot{\phi} - \nabla\phi^\dagger \cdot \nabla\phi - m^2 \phi^\dagger \phi.$$

The conjugate momenta are

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}^\dagger, \qquad \pi^\dagger = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger} = \dot{\phi}.$$

Thus π is conjugate to ϕ, while $\pi^\dagger$ is conjugate to $\phi^\dagger$.

The equations of motion are

$$(\Box + m^2)\phi = 0, \qquad (\Box + m^2)\phi^\dagger = 0.$$

Both components satisfy the Klein–Gordon equation, but the quantum interpretation will be richer than in the real case.

3.3 Canonical Commutation Relations

Canonical quantization promotes $\phi, \phi^\dagger, \pi, \pi^\dagger$ to operators. The nonzero equal-time commutators are

$$[\hat{\phi}(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}),$$

and

$$[\hat{\phi}^\dagger(t,\mathbf{x}), \hat{\pi}^\dagger(t,\mathbf{y})] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}).$$

All other equal-time commutators vanish. In particular,

$$[\hat{\phi}(t,\mathbf{x}), \hat{\pi}^\dagger(t,\mathbf{y})] = 0, \qquad [\hat{\phi}^\dagger(t,\mathbf{x}), \hat{\pi}(t,\mathbf{y})] = 0.$$

This is the canonical field algebra for a charged bosonic scalar field.

3.4 Mode Expansion

The complex field is not Hermitian, so its mode expansion cannot be the same as the real scalar expansion. The standard continuum expansion is

$$\hat{\phi}(t,\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_\mathbf{k}}} \left(\hat{a}_\mathbf{k} e^{-i\omega_\mathbf{k} t + i\mathbf{k}\cdot\mathbf{x}} + \hat{b}^\dagger_\mathbf{k} e^{i\omega_\mathbf{k} t - i\mathbf{k}\cdot\mathbf{x}} \right),$$

where

$$\omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}.$$

The Hermitian conjugate field is

$$\hat{\phi}^\dagger(t,\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_\mathbf{k}}} \left(\hat{a}^\dagger_\mathbf{k} e^{i\omega_\mathbf{k} t - i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_\mathbf{k} e^{-i\omega_\mathbf{k} t + i\mathbf{k}\cdot\mathbf{x}} \right).$$

There are now two independent sets of ladder operators: the a-operators and the b-operators. This is the algebraic origin of antiparticles in the complex scalar theory.

3.5 Two Independent Oscillator Families

The nonzero ladder-operator commutators are

$$[\hat{a}_\mathbf{k}, \hat{a}^\dagger_\mathbf{p}] = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}),$$

and

$$[\hat{b}_\mathbf{k}, \hat{b}^\dagger_\mathbf{p}] = (2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p}).$$

The two families commute with each other:

$$[\hat{a}_\mathbf{k}, \hat{b}_\mathbf{p}] = 0, \qquad [\hat{a}_\mathbf{k}, \hat{b}^\dagger_\mathbf{p}] = 0,$$

with similar relations for their Hermitian conjugates.

The vacuum is defined by

$$\hat{a}_\mathbf{k}|0\rangle = 0, \qquad \hat{b}_\mathbf{k}|0\rangle = 0 \qquad \text{for every } \mathbf{k}.$$

There are two basic one-particle states:

$$|\mathbf{k}; a\rangle = \hat{a}^\dagger_\mathbf{k}|0\rangle, \qquad |\mathbf{k}; b\rangle = \hat{b}^\dagger_\mathbf{k}|0\rangle.$$

They have the same mass and the same energy-momentum relation, but opposite charges.

3.6 Hamiltonian and Momentum

After normal ordering, the Hamiltonian of the free complex scalar field is

$$:\hat{H} := \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \hat{b}^\dagger_{\mathbf{k}} \hat{b}_{\mathbf{k}} \right).$$

The momentum operator is

$$:\hat{\mathbf{P}} := \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \mathbf{k} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} + \hat{b}^\dagger_{\mathbf{k}} \hat{b}_{\mathbf{k}} \right).$$

Both kinds of quanta carry positive energy. The difference between them is not energy or mass, but charge.

3.7 The Charge Operator

The global phase symmetry gives a conserved current. With the convention $\phi \to e^{-i\alpha}\phi$, the corresponding quantum charge may be written, after normal ordering, as

$$\hat{Q} = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \left(\hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}} - \hat{b}^\dagger_{\mathbf{k}} \hat{b}_{\mathbf{k}} \right).$$

This convention gives

$$\hat{Q}\, \hat{a}^\dagger_{\mathbf{k}} |0\rangle = +\hat{a}^\dagger_{\mathbf{k}} |0\rangle, \qquad \hat{Q}\, \hat{b}^\dagger_{\mathbf{k}} |0\rangle = -\hat{b}^\dagger_{\mathbf{k}} |0\rangle.$$

Thus the a-quanta have charge $+1$, while the b-quanta have charge -1. They are particle and antiparticle excitations of the same complex field.

The charge operator also generates the phase transformation of the field:

$$[\hat{Q}, \hat{\phi}(x)] = -\hat{\phi}(x), \qquad [\hat{Q}, \hat{\phi}^\dagger(x)] = +\hat{\phi}^\dagger(x).$$

This is the operator statement that $\hat{\phi}$ carries charge -1 as an operator, even though $\hat{a}^\dagger$ creates a charge $+1$ state in the chosen convention.

3.8 Positive and Negative Frequency Revisited

In the real scalar field, the negative-frequency part was simply the Hermitian conjugate of the positive-frequency part of the same field. In the complex field, the negative-frequency part of $\hat{\phi}$ contains $\hat{b}^\dagger$, not $\hat{a}^\dagger$. This is the crucial difference.

The field $\hat{\phi}$ can annihilate an a-particle or create a b-antiparticle. The conjugate field $\hat{\phi}^\dagger$ can create an a-particle or annihilate a b-antiparticle. Symbolically,

$$\hat{\phi} \sim \hat{a} + \hat{b}^\dagger, \qquad \hat{\phi}^\dagger \sim \hat{a}^\dagger + \hat{b}.$$

This is one of the cleanest ways to see why relativistic quantum fields contain antiparticles.

3.9 Why Antiparticles Appear Naturally

The appearance of antiparticles is not an optional addition. It follows from combining three requirements:

- relativistic positive-energy modes;
- canonical quantization;

- a complex field with a conserved charge.

The field must contain both positive- and negative-frequency solutions in order to be local and to satisfy the canonical commutation relations. For a charged field, those two frequency sectors are associated with two independent oscillator families. The quanta created by the second family have the opposite charge and are interpreted as antiparticles.

3.10 Complex Scalar Propagator

The natural two-point function of the complex scalar field is

$$D_F(x-y) = \langle 0|T\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle.$$

It has the same momentum-space denominator as the real scalar propagator:

$$D_F(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

The difference is in the charge flow. The propagator connects ϕ with $\phi^\dagger$, because charge must be conserved. Correlators such as $\langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle$ vanish in the free theory when the vacuum has zero charge.

3.11 Physical Interpretation of the Fock Space

The Fock space of the complex scalar field contains particles, antiparticles, and arbitrary multiparticle states. Examples are

$$\hat{a}^\dagger_{\mathbf{k}}|0\rangle, \qquad \hat{b}^\dagger_{\mathbf{p}}|0\rangle, \qquad \hat{a}^\dagger_{\mathbf{k}}\hat{b}^\dagger_{\mathbf{p}}|0\rangle.$$

The last state contains one particle and one antiparticle. Its total charge is zero, but it is not the vacuum. Charge conservation restricts which transitions are possible once interactions are introduced.

Summary

A complex scalar field is equivalent to two real scalar degrees of freedom, but its global phase symmetry gives a conserved charge. Quantization produces two independent oscillator families. One creates particles and the other creates antiparticles, with equal mass and opposite charge. The field $\hat{\phi}$ contains an annihilation operator for particles and a creation operator for antiparticles, while $\hat{\phi}^\dagger$ does the reverse.

3.12 Chapter Checklist

After reading this chapter, one should be able to:

- explain why ϕ and $\phi^\dagger$ are independent field variables before quantization;
- write the canonical momenta of the complex scalar field;
- state the equal-time commutation relations;
- write the mode expansions of $\hat{\phi}$ and $\hat{\phi}^\dagger$;
- explain why two independent ladder-operator families appear;

- construct particle and antiparticle states;
- write the charge operator and interpret its eigenvalues;
- explain why antiparticles are required in a relativistic charged field;
- identify the charge structure of the complex scalar propagator.

Chapter 4

Quantization of the Dirac Field

Remark

Scalar fields describe spin-zero quanta and are quantized with commutators. The Dirac field describes spin-1/2 quanta. Its consistent quantization requires anticommutators, and its mode expansion naturally contains particle and antiparticle operators with spin labels.

4.1 Why Fermionic Quantization Is Different

Chapters 2 and 3 quantized bosonic scalar fields. The ladder operators obeyed commutation relations, and many-particle states were symmetric under exchange. The Dirac field is different because its quanta are fermions. Fermionic states must be antisymmetric under exchange, and no two identical fermions can occupy the same one-particle state.

This difference is implemented algebraically by replacing commutators with anticommutators. For operators $\hat{A}$ and $\hat{B}$, the anticommutator is

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

The quantized Dirac field will be built from creation and annihilation operators whose anticommutation relations enforce Fermi statistics.

There is also a second important difference. The Dirac equation is first order in time derivatives, unlike the Klein–Gordon equation. This changes the canonical structure and makes the Dirac field look less like a collection of ordinary oscillator coordinates. Nevertheless, the final Fock-space structure is parallel to the scalar case: there are annihilation operators, creation operators, a vacuum, particles, antiparticles, and number operators.

4.2 Classical Dirac Field

The Dirac field is a spinor field $\psi(x)$. Its Dirac adjoint is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

The free Dirac Lagrangian density is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

The equation of motion is the Dirac equation,

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.$$

The adjoint field satisfies

$$\bar{\psi}(x)(i\overleftarrow{\partial}_\mu \gamma^\mu + m) = 0,$$

where the arrow indicates that the derivative acts to the left.

The matrices γ^μ obey the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}.$$

This algebra is what makes the Dirac equation compatible with the relativistic mass-shell relation $p^2 = m^2$.

4.3 Canonical Structure and Anticommutation Relations

Because the Lagrangian is first order in $\dot{\psi}$, the momentum conjugate to ψ is proportional to $\psi^\dagger$:

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger.$$

The quantum theory is defined by equal-time anticommutation relations,

$$\{\hat{\psi}_\alpha(t,\mathbf{x}), \hat{\psi}^\dagger_\beta(t,\mathbf{y})\} = \delta_{\alpha\beta}\delta^{(3)}(\mathbf{x}-\mathbf{y}),$$

with

$$\{\hat{\psi}_\alpha(t,\mathbf{x}), \hat{\psi}_\beta(t,\mathbf{y})\} = 0, \qquad \{\hat{\psi}^\dagger_\alpha(t,\mathbf{x}), \hat{\psi}^\dagger_\beta(t,\mathbf{y})\} = 0.$$

The spinor indices α, β label the components of the Dirac spinor.

Remark

The replacement of commutators by anticommutators is not a cosmetic change. It is what produces fermionic Fock space, positive-energy particle interpretation, and the Pauli exclusion principle.

4.4 Mode Expansion of the Dirac Field

The free Dirac equation has positive-energy spinor solutions $u_s(\mathbf{p})$ and negative-frequency spinor solutions $v_s(\mathbf{p})$, where s labels spin. A standard expansion is

$$\hat{\psi}(x) = \sum_s \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \left(\hat{b}_s(\mathbf{p}) u_s(\mathbf{p}) e^{-iE_\mathbf{p}t + i\mathbf{p}\cdot\mathbf{x}} + \hat{d}^\dagger_s(\mathbf{p}) v_s(\mathbf{p}) e^{iE_\mathbf{p}t - i\mathbf{p}\cdot\mathbf{x}} \right),$$

where

$$E_\mathbf{p} = \sqrt{\mathbf{p}^2 + m^2}.$$

The adjoint field is

$$\hat{\bar{\psi}}(x) = \sum_s \int \frac{\mathrm{d}^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \left(\hat{b}^\dagger_s(\mathbf{p}) \bar{u}_s(\mathbf{p}) e^{iE_\mathbf{p}t - i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s(\mathbf{p}) \bar{v}_s(\mathbf{p}) e^{-iE_\mathbf{p}t + i\mathbf{p}\cdot\mathbf{x}} \right).$$

The structure is analogous to the complex scalar field:

$$\hat{\psi} \sim \hat{b} + \hat{d}^\dagger, \qquad \hat{\bar{\psi}} \sim \hat{b}^\dagger + \hat{d}.$$

The operators $\hat{b}^\dagger$ create particles, while $\hat{d}^\dagger$ create antiparticles.

4.5 Spinor Creation and Annihilation Operators

The nonzero anticommutators are

$$\{\hat{b}_s(\mathbf{p}), \hat{b}^\dagger_r(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p}-\mathbf{q}),$$

and

$$\{\hat{d}_s(\mathbf{p}), \hat{d}^\dagger_r(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p}-\mathbf{q}).$$

All other anticommutators vanish. In particular,

$$\{\hat{b}_s(\mathbf{p}), \hat{d}_r(\mathbf{q})\} = 0, \qquad \{\hat{b}_s(\mathbf{p}), \hat{d}^\dagger_r(\mathbf{q})\} = 0.$$

The label s is the spin label. For a massive Dirac particle there are two spin states for the particle and two spin states for the antiparticle.

Because the creation operators anticommute, applying the same creation operator twice gives zero:

$$\left(\hat{b}_s^\dagger(\mathbf{p})\right)^2 = 0, \qquad \left(\hat{d}_s^\dagger(\mathbf{p})\right)^2 = 0.$$

This is the algebraic form of the Pauli exclusion principle.

4.6 Particle and Antiparticle States

The vacuum is defined by

$$\hat{b}_s(\mathbf{p})|0\rangle = 0, \qquad \hat{d}_s(\mathbf{p})|0\rangle = 0 \qquad \text{for all } \mathbf{p}, s.$$

A one-particle state is

$$|\mathbf{p}, s; b\rangle = \hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

A one-antiparticle state is

$$|\mathbf{p}, s; d\rangle = \hat{d}_s^\dagger(\mathbf{p})|0\rangle.$$

For the electron field, the $b^\dagger$-states are electron states and the $d^\dagger$-states are positron states. More generally, a Dirac field creates and annihilates a charged fermion and its antiparticle.

A two-fermion state changes sign when the order of the creation operators is exchanged:

$$\hat{b}_s^\dagger(\mathbf{p})\hat{b}_r^\dagger(\mathbf{q})|0\rangle = -\hat{b}_r^\dagger(\mathbf{q})\hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

This antisymmetry is the defining feature of fermionic Fock space.

4.7 Hamiltonian, Momentum, and Charge

After normal ordering, the free Dirac Hamiltonian is

$$:\hat{H} := \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, E_{\mathbf{p}} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

The momentum operator is

$$:\hat{\mathbf{P}} := \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3}\, \mathbf{p} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

Both particles and antiparticles carry positive energy.

For a Dirac field with unit charge convention, the normal-ordered charge has the form

$$\hat{Q} = \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) - \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p})\right).$$

For the electron field one often multiplies this operator by $-e$, so that particles created by $\hat{b}^\dagger$ have charge $-e$, and antiparticles created by $\hat{d}^\dagger$ have charge $+e$.

4.8 Why Anticommutators Give the Correct Energy Spectrum

The negative-frequency part of the Dirac field is not interpreted as a tower of negative-energy particles. After quantization, it is reinterpreted as the creation of antiparticles with positive energy. The anticommutation relations are essential for this reinterpretation to give a Hamiltonian bounded below.

Schematically, the Dirac Hamiltonian before normal ordering contains terms of the form

$$\hat{b}^\dagger \hat{b} - \hat{d}\hat{d}^\dagger.$$

Using anticommutators,

$$\hat{d}\hat{d}^\dagger = 1 - \hat{d}^\dagger \hat{d},$$

so, up to an infinite constant that is removed by normal ordering, the antiparticle contribution becomes positive:

$$\hat{d}^\dagger \hat{d}.$$

This is the algebraic reason the free Dirac Hamiltonian counts both particles and antiparticles with positive energy.

4.9 Spinor Sums and Completeness

The spinor wave functions are normalized so that they satisfy completeness relations. A common convention is

$$\sum_s u_s(\mathbf{p})\bar{u}_s(\mathbf{p}) = \gamma^\mu p_\mu + m,$$

and

$$\sum_s v_s(\mathbf{p})\bar{v}_s(\mathbf{p}) = \gamma^\mu p_\mu - m,$$

where $p^0 = E_\mathbf{p}$. These identities are used constantly when computing fermion propagators and scattering amplitudes.

The spinor sums play for the Dirac field a role analogous to polarization sums for vector fields. They allow spin labels on internal lines to be summed in a Lorentz-covariant way.

4.10 The Dirac Propagator

The Feynman propagator of the Dirac field is

$$S_F(x-y) = \langle 0|T\hat{\psi}(x)\hat{\bar{\psi}}(y)|0\rangle.$$

In momentum space it is

$$S_F(x-y) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)}.$$

Equivalently, the propagator is the Green function of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m)S_F(x-y) = i\delta^{(4)}(x-y).$$

Compared with the scalar propagator, the denominator is the same mass-shell factor, but the numerator carries spinor structure.

4.11 Chirality, Helicity, and the Massless Limit

The Dirac field contains spin degrees of freedom. For massive fermions, the spin label may be chosen in several equivalent ways. For massless fermions, helicity and chirality become especially important.

The chirality matrix γ^5 defines projection operators

$$P_L = \frac{1}{2}(1-\gamma^5), \qquad P_R = \frac{1}{2}(1+\gamma^5).$$

They split a Dirac field into left- and right-chiral components:

$$\psi_L = P_L\psi, \qquad \psi_R = P_R\psi.$$

For a massless Dirac field, the left- and right-chiral parts decouple. This will be essential later in the electroweak theory, where left- and right-handed fermions transform differently under gauge symmetries.

4.12 Majorana Fields: Conceptual Overview

A Dirac field describes a charged fermion and a distinct antiparticle. A Majorana field is different: it is a fermionic field that is equal to its charge conjugate in an appropriate sense. Informally, a Majorana fermion is its own antiparticle.

The Majorana idea is possible for neutral fermions, not for fields carrying an ordinary conserved electric charge. The full construction requires charge conjugation matrices and reality conditions on spinors. For now, the important contrast is simple:

$$\text{Dirac field: particle and antiparticle are distinct,}$$

while

$$\text{Majorana field: particle and antiparticle are identified.}$$

Majorana fields will become relevant when discussing neutrino masses and extensions of the Standard Model.

Summary

The Dirac field is the first fermionic quantum field in the course. Its quantization uses anticommutators rather than commutators. The mode expansion contains particle annihilation operators and antiparticle creation operators, now with spin labels. Anticommutation gives fermionic Fock space, the Pauli exclusion principle, and a Hamiltonian with positive particle and antiparticle energies after normal ordering.

4.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the free Dirac Lagrangian and Dirac equation;
- explain why fermionic fields are quantized with anticommutators;
- write the mode expansion of $\hat{\psi}$ and $\hat{\bar{\psi}}$;
- state the anticommutation relations of $\hat{b}, \hat{b}^\dagger$ and $\hat{d}, \hat{d}^\dagger$;
- construct fermion and antifermion states;
- explain how the Hamiltonian counts positive-energy particles and antiparticles;
- write the charge operator and interpret the sign convention;
- state the standard spinor completeness relations;
- write the Dirac propagator;
- distinguish Dirac and Majorana fermions at the conceptual level.

Chapter 5

Quantization of the Electromagnetic Field

Remark

The scalar and Dirac fields taught us how particles and antiparticles arise from mode expansions. The electromagnetic field adds a new feature: gauge redundancy. A photon is not an excitation of four independent components of A_μ, but of the physical transverse degrees of freedom of a gauge field.

5.1 Why Gauge Fields Are Harder to Quantize

The real scalar, complex scalar, and Dirac fields all had independent field components that could be quantized after imposing the appropriate commutation or anticommutation relations. The electromagnetic field is different. It is described by a vector potential A_μ, but not all components of A_μ represent physical degrees of freedom.

The field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

The free electromagnetic Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.$$

It is invariant under the gauge transformation

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda.$$

The potentials A_μ and $A_\mu + \partial_\mu \Lambda$ describe the same physical electromagnetic field. Therefore quantizing every component of A_μ as if it were an independent scalar field would overcount states.

5.2 Physical and Redundant Degrees of Freedom

A massless vector field in four spacetime dimensions has two physical polarizations. This is already suggested classically by electromagnetic waves: the electric and magnetic fields of a plane wave are transverse to the direction of propagation.

The four components of A_μ contain redundant information. Gauge freedom removes one part, and the equations of motion impose a constraint. The result is that only two transverse photon polarizations are physical.

This distinction is central:

$$\text{four components of } A_\mu \neq \text{four physical photon polarizations.}$$

The task of quantization is to keep Lorentz covariance and locality under control while ensuring that only the physical states contribute to observable processes.

5.3 Canonical Obstruction

The canonical momentum conjugate to A_μ is

$$\Pi^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}.$$

For the time component one finds

$$\Pi^0 = 0.$$

This means that A_0 has no independent time derivative in the Lagrangian. It is not an ordinary dynamical coordinate. Instead, it acts as a Lagrange multiplier imposing Gauss's law.

This is the first signal that gauge fields require constraints or gauge fixing. One cannot simply impose four independent canonical commutation relations for all components of A_μ.

5.4 Coulomb Gauge Quantization

A physically transparent gauge choice is Coulomb gauge,

$$\nabla \cdot \mathbf{A} = 0.$$

For the free electromagnetic field, one may also set the nondynamical scalar potential to zero after solving the constraint. The remaining field is the transverse vector potential $\mathbf{A}_T$.

The transverse field may be expanded as

$$\hat{\mathbf{A}}(t, \mathbf{x}) = \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \frac{1}{\sqrt{2|\mathbf{k}|}} \left(\epsilon_\lambda(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}) e^{-i|\mathbf{k}|t + i\mathbf{k}\cdot\mathbf{x}} + \epsilon_\lambda^*(\mathbf{k}) \hat{a}_\lambda^\dagger(\mathbf{k}) e^{i|\mathbf{k}|t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The polarization vectors are transverse:

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0.$$

The index $\lambda = 1, 2$ labels the two physical polarizations.

5.5 Photon Creation and Annihilation Operators

The photon ladder operators satisfy

$$[\hat{a}_\lambda(\mathbf{k}), \hat{a}_\sigma^\dagger(\mathbf{p})] = (2\pi)^3 \delta_{\lambda\sigma} \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with all other commutators vanishing. The vacuum is defined by

$$\hat{a}_\lambda(\mathbf{k})|0\rangle = 0 \qquad \text{for all } \mathbf{k}, \lambda.$$

A one-photon state is

$$|\mathbf{k}, \lambda\rangle = \hat{a}_\lambda^\dagger(\mathbf{k})|0\rangle.$$

Its energy is

$$\omega_\mathbf{k} = |\mathbf{k}|,$$

which is the massless dispersion relation.

The normal-ordered Hamiltonian is

$$: \hat{H} := \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} |\mathbf{k}| \, \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

The momentum operator is

$$: \hat{\mathbf{P}} := \sum_{\lambda=1}^{2} \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \mathbf{k} \, \hat{a}_\lambda^\dagger(\mathbf{k}) \hat{a}_\lambda(\mathbf{k}).$$

5.6 Polarization Vectors

For each nonzero momentum $\mathbf{k}$, one chooses two transverse polarization vectors. They obey

$$\mathbf{k} \cdot \epsilon_\lambda(\mathbf{k}) = 0, \qquad \epsilon^*_\lambda(\mathbf{k}) \cdot \epsilon_\sigma(\mathbf{k}) = \delta_{\lambda\sigma}.$$

Linear polarizations are useful for many classical comparisons. Circular polarizations are useful for describing helicity. They are built from two linear polarizations by

$$\epsilon_\pm = \frac{1}{\sqrt{2}} \left(\epsilon_1 \pm i\epsilon_2\right).$$

The labels $+$ and $-$ correspond to the two helicity states of the photon.

5.7 Photon States and Helicity

A massive spin-one particle would have three spin polarizations. A photon is massless and gauge invariant, so only two physical helicities remain. A one-photon state may be labelled as

$$|\mathbf{k}, +\rangle, \qquad |\mathbf{k}, -\rangle.$$

These are helicity eigenstates. Helicity is the projection of angular momentum onto the direction of motion. For a massless particle, helicity is Lorentz invariant under proper orthochronous Lorentz transformations.

The absence of a longitudinal physical photon is not an accident. The longitudinal component of $\mathbf{A}$ is gauge redundancy, not a separate particle.

5.8 Covariant Quantization

Coulomb gauge makes the physical degrees of freedom visible, but it hides manifest Lorentz covariance. For perturbative QFT it is often more convenient to use a covariant gauge-fixing term,

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The gauge-fixed Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The parameter ξ labels a family of gauges. The choice $\xi = 1$ is called Feynman gauge.

Covariant quantization introduces unphysical components into intermediate calculations. These components must not appear in physical observables. In QED, current conservation ensures that unphysical polarizations cancel from scattering amplitudes. In non-Abelian gauge theory, the corresponding statement requires ghosts and BRST symmetry, which will be developed later.

5.9 Gauge Fixing and the Photon Propagator

In covariant gauges, the momentum-space photon propagator is

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 + i\epsilon}\right).$$

In Feynman gauge, this simplifies to

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The pole at $k^2 = 0$ reflects the masslessness of the photon. The numerator carries the vector structure of the field.

The gauge-dependent part proportional to $k_\mu k_\nu$ does not change physical QED amplitudes coupled to conserved currents. This is the practical reason covariant gauges are useful: they simplify calculations without changing observable predictions.

5.10 Unphysical Polarizations

Covariant quantization treats A_μ as a four-component field during intermediate steps. This introduces time-like and longitudinal polarizations that are not physical photon states. The physical Hilbert space is obtained only after imposing the appropriate condition that removes the unphysical sector.

In elementary QED calculations, this removal is often hidden inside Ward identities and current conservation. If an external photon polarization vector is shifted by a multiple of its momentum,

$$\epsilon_\mu(k) \longrightarrow \epsilon_\mu(k) + c\, k_\mu,$$

physical amplitudes are unchanged. This is the amplitude-level expression of gauge redundancy.

5.11 Gauge-Invariant Observables

The potential A_μ is not gauge invariant. The field strength $F_{\mu\nu}$ is gauge invariant, and so are the electric and magnetic fields. Observables must ultimately be built from gauge-invariant quantities or from gauge-invariant combinations of charged fields and gauge fields.

Examples include

$$F_{\mu\nu}(x), \qquad F_{\mu\nu}(x)F^{\mu\nu}(x),$$

and, in more advanced settings, Wilson line and Wilson loop operators. The important lesson at this stage is that the photon is described by a gauge potential, but physical information cannot depend on the redundant gauge choice.

5.12 What Is a Photon?

In free QED, a photon is a one-quantum excitation of a transverse mode of the electromagnetic field:

$$|\mathbf{k}, \lambda\rangle = \hat{a}^\dagger_\lambda(\mathbf{k})|0\rangle, \qquad \lambda = 1, 2.$$

Equivalently, one may label the two states by helicity $\lambda = \pm$. The photon is massless, has energy $|\mathbf{k}|$, and carries no electric charge.

This definition is sharp in the free theory and in scattering theory with asymptotic photon states. In interacting QED, soft photons and infrared effects make the exact particle interpretation more subtle. Those subtleties will be revisited when discussing scattering and loop corrections.

Summary

The electromagnetic field is the first gauge field in the course. Gauge redundancy means that not all components of A_μ are physical. In Coulomb gauge the two transverse photon polarizations are explicit. In covariant gauges one keeps Lorentz covariance at the cost of introducing unphysical components, which cancel from physical observables. A photon is a transverse, massless quantum of the electromagnetic field.

5.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain why gauge fields are harder to quantize than scalar fields;
- state the gauge redundancy of A_μ;
- explain why the photon has two physical polarizations;
- write the Coulomb-gauge mode expansion of the transverse vector potential;
- construct one-photon states;
- distinguish linear polarization from helicity;
- write the covariant gauge-fixing term;
- write the photon propagator in a covariant gauge;
- explain why unphysical polarizations do not appear as external photon states;
- identify gauge-invariant electromagnetic observables.

Chapter 6

Propagators as Quantum Correlation Functions

Remark

The previous chapters constructed free quantum fields and their particle states. This chapter introduces the next central object: the propagator. A propagator is not merely a line in a Feynman diagram. It is a vacuum correlation function of quantum fields, and it encodes both the equation of motion and the particle content of the theory.

6.1 From Classical Green Functions to Quantum Correlators

In classical field theory, a Green function solves an inhomogeneous differential equation. For the Klein–Gordon operator, one writes

$$(\Box_x + m^2)G(x-y) = \delta^{(4)}(x-y).$$

Different boundary conditions give different Green functions: retarded, advanced, or other choices.

In quantum field theory, a closely related object appears, but with a different interpretation. The basic two-point function is a vacuum expectation value of a product of field operators. For a real scalar field, the simplest one is

$$\langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

This is not the value of a classical field at two points. It is a quantum correlation between two field operators in the vacuum.

6.2 Wightman Functions

The positive-frequency Wightman function of the real scalar field is

$$\Delta^+(x-y) = \langle 0|\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

Using the mode expansion, one finds

$$\Delta^+(x-y) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{k}}}\,e^{-i\omega_{\mathbf{k}}(x^0-y^0)+i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}.$$

The reversed ordering gives

$$\Delta^-(x-y) = \langle 0|\hat{\phi}(y)\hat{\phi}(x)|0\rangle.$$

These functions measure vacuum correlations. Even in the vacuum, field operators at different spacetime points are correlated.

The commutator function is the difference

$$[\hat{\phi}(x),\hat{\phi}(y)] = \Delta^+(x-y) - \Delta^-(x-y) = i\Delta(x-y).$$

Microcausality says that this commutator vanishes for spacelike separation.

6.3 Time-Ordered Products

Scattering theory is built from time-ordered products. For two bosonic field operators, the time-ordered product is defined by

$$T\hat{\phi}(x)\hat{\phi}(y) = \begin{cases} \hat{\phi}(x)\hat{\phi}(y), & x^0 > y^0, \\ \hat{\phi}(y)\hat{\phi}(x), & y^0 > x^0. \end{cases}$$

Equivalently,

$$T\hat{\phi}(x)\hat{\phi}(y) = \theta(x^0 - y^0)\hat{\phi}(x)\hat{\phi}(y) + \theta(y^0 - x^0)\hat{\phi}(y)\hat{\phi}(x).$$

For fermionic fields, time ordering includes a minus sign when two fermion operators are exchanged. This sign is one of the first places where fermionic statistics enters perturbation theory.

6.4 The Feynman Propagator

The scalar Feynman propagator is the vacuum expectation value of the time-ordered two-point function:

$$D_F(x - y) = \langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle.$$

Using the Wightman functions, this may be written as

$$D_F(x - y) = \theta(x^0 - y^0)\Delta^+(x - y) + \theta(y^0 - x^0)\Delta^+(y - x).$$

It has the four-dimensional momentum-space representation

$$D_F(x - y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

The small $i\epsilon$ prescription specifies how the poles are passed in the complex k^0-plane. This prescription is what makes the propagator the vacuum time-ordered correlation function rather than some other Green function.

6.5 The Propagator as a Green Function

The Feynman propagator satisfies

$$(\Box_x + m^2)D_F(x - y) = -i\delta^{(4)}(x - y),$$

with the sign convention following from the definition above. Thus the Feynman propagator is also a Green function of the Klein–Gordon operator, but with a specific quantum boundary condition.

This dual nature is important:

$$\text{propagator} = \text{Green function} = \text{vacuum two-point correlation function.}$$

The first viewpoint emphasizes equations of motion. The second emphasizes quantum states and operator expectation values.

6.6 Retarded and Advanced Propagators

The retarded propagator is built from the commutator:

$$D_R(x - y) = i\theta(x^0 - y^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 < y^0$, so it describes causal response to a disturbance in the past. The advanced propagator is

$$D_A(x-y) = -i\theta(y^0 - x^0)[\hat{\phi}(x), \hat{\phi}(y)].$$

It vanishes when $x^0 > y^0$.

Retarded and advanced propagators are especially natural in classical response problems and nonequilibrium physics. The Feynman propagator is the object that appears in ordinary perturbative scattering theory.

6.7 The $i\epsilon$ Prescription

The expression

$$\frac{i}{k^2 - m^2 + i\epsilon}$$

contains more information than the algebraic denominator. The poles are located near

$$k^0 = \pm\omega_{\mathbf{k}}.$$

The $i\epsilon$ prescription tells us how to deform the integration contour around these poles. It is the momentum-space encoding of time ordering.

For positive time separation, the contour picks up the positive-energy pole. For negative time separation, it picks up the negative-energy pole in the way appropriate to a time-ordered vacuum expectation value. This is how the compact four-dimensional expression reproduces the two time orderings in position space.

6.8 Momentum-Space Propagators for Free Fields

For the real scalar field,

$$D_F(k) = \frac{i}{k^2 - m^2 + i\epsilon}.$$

For the complex scalar field,

$$\langle 0|T\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle$$

has the same momentum-space denominator. The difference is that the complex scalar propagator carries charge flow from $\phi^\dagger$ to ϕ.

For the Dirac field,

$$S_F(k) = \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}.$$

The denominator is the same mass-shell structure, while the numerator carries spinor information.

For the photon in Feynman gauge,

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

In a general covariant gauge, the numerator changes by gauge-dependent terms. Physical amplitudes do not depend on that gauge choice.

6.9 Pole Structure and Particle Interpretation

The location of poles in a propagator reveals the particle content of the free theory. The scalar propagator has poles at

$$k^2 = m^2,$$

which is the relativistic mass shell. This is the analytic trace of a particle of mass m.

For a massless photon, the pole is at

$$k^2 = 0.$$

For a free Dirac field, the denominator again has poles at $k^2 = m^2$, while the numerator encodes spinor structure.

In an interacting theory, the pole structure becomes even more informative. A shift in the pole position corresponds to a mass correction. A pole with an imaginary part corresponds to an unstable resonance. Branch cuts correspond to multiparticle thresholds. These ideas will be developed later, but the free propagator already contains the basic principle: particles appear as singular structures of correlation functions.

6.10 Propagation Versus Correlation

The word "propagator" can be misleading. It suggests that a particle literally travels from y to x. In QFT the primary object is a correlation function of fields. It measures the amplitude associated with inserting one field at y and another at x, with the vacuum in between.

For spacelike separation, the Feynman propagator need not vanish. This does not violate causality, because the commutator vanishes outside the light cone. The causal statement is not that all vacuum correlations vanish at spacelike separation, but that spacelike separated observables commute.

Thus one should distinguish

$$\text{vacuum correlation} \neq \text{signal propagation}.$$

The propagator is a correlation function used in the computation of amplitudes; it is not by itself a directly observable signal.

6.11 What a Propagator Means in Perturbation Theory

In perturbation theory, propagators connect field insertions. Wick's theorem will show that time-ordered products of free fields can be reduced to sums of contractions, and each contraction is a Feynman propagator.

This is why propagators become internal lines in Feynman diagrams. A diagram is a compact bookkeeping device for a term in the perturbative expansion. The line stands for a two-point function of the corresponding free field.

For example, a scalar internal line carries

$$\frac{i}{k^2 - m^2 + i\epsilon},$$

a fermion internal line carries

$$\frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon},$$

and a photon internal line in Feynman gauge carries

$$\frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The diagrammatic rules are therefore not arbitrary. They are built from the free-field correlation functions developed in this chapter.

6.12 Summary of the Main Propagators

The most important free propagators introduced so far are

$$
\begin{aligned}
\text{real scalar} \quad &: \quad \frac{i}{k^2 - m^2 + i\epsilon}, \\
\text{complex scalar} \quad &: \quad \frac{i}{k^2 - m^2 + i\epsilon} \text{ with charge flow}, \\
\text{Dirac field} \quad &: \quad \frac{i(\gamma^\mu k_\mu + m)}{k^2 - m^2 + i\epsilon}, \\
\text{photon, Feynman gauge} \quad &: \quad \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.
\end{aligned}
$$

The denominators encode the mass shell. The numerators encode spin, vector, or gauge structure.

Summary

A propagator is a time-ordered vacuum two-point function. It is also a Green function with a specific quantum boundary condition. Its poles encode particle content, while its numerator encodes spin and gauge structure. Propagators are the building blocks of perturbative QFT because Wick's theorem turns contractions of free fields into propagators.

6.13 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish classical Green functions from quantum two-point functions;
- define Wightman functions;
- define the time-ordered product;
- write the scalar Feynman propagator in position and momentum space;
- explain the role of the $i\epsilon$ prescription;
- distinguish Feynman, retarded, and advanced propagators;
- write the free scalar, Dirac, and photon propagators;
- explain how pole structure encodes particle content;
- explain why vacuum correlation is not the same as signal propagation;
- explain why propagators become internal lines in Feynman diagrams.

Chapter 7

The Interaction Picture

Remark

The previous chapter introduced propagators of free fields. Perturbation theory uses those free fields as the starting point and treats interactions as a controlled deformation. The interaction picture is the framework that makes this separation precise.

7.1 Why We Need a New Picture

In quantum mechanics one may describe time evolution in several equivalent ways. The Schroedinger picture puts time dependence in the states. The Heisenberg picture puts time dependence in the operators. The interaction picture splits the time dependence between them.

This split is useful when the Hamiltonian has the form

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}.$$

Here $\hat{H}_0$ is the exactly solvable free Hamiltonian, and $\hat{H}_{\text{int}}$ contains interactions. In QFT, $\hat{H}_0$ is built from the free scalar, spinor, or photon fields of the previous chapters. The interaction part describes processes such as scattering, decay, emission, and absorption.

The guiding idea is simple:

solve the free theory exactly, then expand in the interaction.

The interaction picture is the operator language in which this idea becomes a systematic perturbation theory.

7.2 Schroedinger, Heisenberg, and Interaction Pictures

Let $|\Psi_S(t)\rangle$ be a Schroedinger-picture state. It satisfies

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\Psi_S(t)\rangle = \hat{H}|\Psi_S(t)\rangle.$$

Operators in this picture are usually time independent unless they have explicit time dependence.

In the Heisenberg picture, the state is fixed and operators evolve according to

$$\hat{O}_H(t) = e^{i\hat{H}t}\hat{O}_S e^{-i\hat{H}t}.$$

This picture is elegant but difficult for interacting QFT because exact interacting operators are rarely known.

In the interaction picture, free time evolution is assigned to the operators:

$$\hat{O}_I(t) = e^{i\hat{H}_0 t}\hat{O}_S e^{-i\hat{H}_0 t}.$$

The states evolve only with the interaction. Thus interaction-picture fields are free fields, even though the states know about the interaction.

7.3 Free Fields as the Starting Point

The scalar interaction-picture field satisfies the free Klein–Gordon equation,

$$(\Box + m^2)\hat{\phi}_I(x) = 0.$$

Its mode expansion is the same free expansion used in Chapter 2:

$$\hat{\phi}_I(x) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-ik\cdot x} + \hat{a}^\dagger_{\mathbf{k}} e^{ik\cdot x} \right),$$

where $k^0 = \omega_{\mathbf{k}}$. Similarly, interaction-picture Dirac and photon fields are free Dirac and free photon fields.

This is why the propagators of Chapter 6 are the basic building blocks of perturbation theory. The fields appearing inside the perturbative expansion are free fields, so their time-ordered products can be evaluated using free-field correlation functions.

7.4 The Interaction Hamiltonian

The interaction Hamiltonian in the interaction picture is

$$\hat{H}_I(t) = e^{i\hat{H}_0 t} \hat{H}_{\text{int}} e^{-i\hat{H}_0 t}.$$

It is written in terms of interaction-picture fields. For example, a scalar ϕ^4 interaction is described by

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

For this case, the interaction Hamiltonian density is

$$\mathcal{H}_{\text{int}} = \frac{\lambda}{4!}\phi_I^4,$$

and

$$\hat{H}_I(t) = \int \mathrm{d}^3x \, \frac{\lambda}{4!} \hat{\phi}_I(t, \mathbf{x})^4.$$

The subscript I reminds us that the fields evolve freely. The coupling λ measures the strength of the interaction.

For QED, the interaction Lagrangian is

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu \psi A_\mu,$$

so the interaction Hamiltonian is commonly written as

$$\hat{H}_I(t) = e \int \mathrm{d}^3x \, \hat{\bar{\psi}}_I \gamma^\mu \hat{\psi}_I \hat{A}_{I\mu}.$$

This term is responsible for the elementary QED vertex.

7.5 Time Evolution Operator

Interaction-picture states satisfy

$$i\frac{\mathrm{d}}{\mathrm{d}t}|\Psi_I(t)\rangle = \hat{H}_I(t)|\Psi_I(t)\rangle.$$

Define the time evolution operator $\hat{U}_I(t, t_0)$ by

$$|\Psi_I(t)\rangle = \hat{U}_I(t, t_0)|\Psi_I(t_0)\rangle.$$

Then $\hat{U}_I(t, t_0)$ obeys

$$i\frac{\partial}{\partial t}\hat{U}_I(t, t_0) = \hat{H}_I(t)\hat{U}_I(t, t_0), \qquad \hat{U}_I(t_0, t_0) = 1.$$

Formally, this is solved by a time-ordered exponential:

$$\hat{U}_I(t, t_0) = T \exp\left(-i\int_{t_0}^{t} \mathrm{d}t'\, \hat{H}_I(t')\right).$$

The time-ordering symbol is essential because $\hat{H}_I(t_1)$ and $\hat{H}_I(t_2)$ need not commute at different times.

7.6 The Dyson Series

Expanding the time-ordered exponential gives the Dyson series:

$$\hat{U}_I(t, t_0) = 1 - i\int_{t_0}^{t} \mathrm{d}t_1\, \hat{H}_I(t_1) + \frac{(-i)^2}{2!}\int_{t_0}^{t} \mathrm{d}t_1\, \mathrm{d}t_2\, T\{\hat{H}_I(t_1)\hat{H}_I(t_2)\} + \cdots.$$

Equivalently,

$$\hat{U}_I(t, t_0) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!}\int_{t_0}^{t} \mathrm{d}t_1 \cdots \mathrm{d}t_n\, T\{\hat{H}_I(t_1)\cdots\hat{H}_I(t_n)\}.$$

Each power of $\hat{H}_I$ brings one power of the coupling. In ϕ^4 theory, the n-th order term contains λ^n. In QED, the n-th order term contains e^n.

This is the algebraic origin of perturbation theory: physical quantities are computed as series in coupling constants.

7.7 Perturbation Theory and Time-Ordered Correlators

Suppose we want the interacting time-ordered two-point function of a scalar field. In the interaction picture it is expressed as

$$\langle\Omega|T\hat{\phi}_H(x)\hat{\phi}_H(y)|\Omega\rangle = \frac{\langle 0|T\left\{\hat{\phi}_I(x)\hat{\phi}_I(y)\exp\left[-i\int \mathrm{d}^4z\,\mathcal{H}_{\text{int}}(z)\right]\right\}|0\rangle}{\langle 0|T\exp\left[-i\int \mathrm{d}^4z\,\mathcal{H}_{\text{int}}(z)\right]|0\rangle}.$$

Here $|0\rangle$ is the free vacuum and $|\Omega\rangle$ is the interacting vacuum. The denominator removes disconnected vacuum factors. This formula is a central bridge between operator quantization and Feynman diagrams.

The fields inside the expectation value are free fields. Therefore Wick's theorem can reduce the time-ordered products to sums of propagators. This will be the subject of the next chapter.

7.8 Vacuum Persistence Amplitude

The vacuum persistence amplitude is

$$\langle 0|\hat{U}_I(\infty, -\infty)|0\rangle.$$

It is the amplitude for the free vacuum in the far past to become the free vacuum in the far future in the presence of interactions. Its perturbative expansion contains vacuum bubble diagrams: diagrams with no external legs.

These vacuum bubbles multiply correlation functions by an overall factor. In normalized expectation values, they cancel against the denominator. This is why connected diagrams, rather than all diagrams, carry the main physical content of scattering amplitudes.

7.9 Adiabatic Switching and In/Out States

Scattering theory compares states in the far past with states in the far future. The idealization is that interactions are effectively switched off at early and late times, so that asymptotic states can be described by free particles. This is expressed schematically by replacing the interaction with

$$\hat{H}_I(t) \longrightarrow e^{-\epsilon|t|}\hat{H}_I(t),$$

and taking $\epsilon \to 0^+$ at the end.

This adiabatic switching is a mathematical device. It helps define the connection between free Fock states and interacting scattering amplitudes. The states in the far past are called in-states, and those in the far future are called out-states.

The scattering operator is

$$\hat{S} = \hat{U}_I(\infty, -\infty) = T\exp\left(-i\int_{-\infty}^{\infty} \mathrm{d}t\, \hat{H}_I(t)\right).$$

Matrix elements of $\hat{S}$ between in- and out-states are the basic objects of perturbative scattering theory.

7.10 Why Exact Interacting QFT Is Difficult

The free theory is exactly solvable because it is a collection of independent oscillators. Interactions couple those oscillators. Once interactions are present, particle number may change, the vacuum may differ from the free vacuum, and exact operator solutions are usually unavailable.

Perturbation theory avoids solving the interacting theory exactly. It expands around the free theory and organizes corrections in powers of the coupling. This strategy is extremely powerful when the coupling is small and the relevant states are close to free asymptotic particle states.

However, perturbation theory has limits. Strong coupling, bound states, confinement, and nonperturbative vacuum effects require additional methods. The interaction picture is therefore not the whole of QFT, but it is the standard entry point into calculational quantum field theory.

Summary

The interaction picture separates the exactly solvable free Hamiltonian from the interaction Hamiltonian. Interaction-picture fields are free fields, while states evolve with $\hat{H}_I(t)$. The time evolution operator is a time-ordered exponential, and its expansion gives the Dyson series. This is the operator foundation of perturbation theory and the starting point for Wick's theorem and Feynman diagrams.

7.11 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish the Schroedinger, Heisenberg, and interaction pictures;
- explain why interaction-picture fields are free fields;
- write the interaction Hamiltonian for ϕ^4 theory and QED;
- define the interaction-picture time evolution operator;
- expand the time-ordered exponential as the Dyson series;
- explain perturbation theory as an expansion in couplings;

- describe the role of the vacuum persistence amplitude;
- explain why vacuum bubbles cancel in normalized correlators;
- define in-states, out-states, and the scattering operator;
- explain why exact interacting QFT is difficult.

Chapter 8

Wick Theorem and Diagrammatics

Remark

The Dyson series expresses interacting quantities as time-ordered products of free fields. Wick's theorem is the algebraic tool that evaluates those products. It reduces them to normal-ordered products plus contractions, and each contraction is a propagator.

8.1 Why Wick's Theorem Appears

In Chapter 7, perturbation theory led to expressions of the form

$$\langle 0|T\{\hat{\phi}_I(x_1)\cdots\hat{\phi}_I(x_n)\}|0\rangle.$$

The fields are free interaction-picture fields, but there may be many of them. To compute perturbative corrections, one needs a systematic way to evaluate such vacuum expectation values.

Wick's theorem provides exactly this. It says that a time-ordered product of free fields can be rewritten as a sum of normal-ordered products multiplied by contractions. In vacuum expectation values, the normal-ordered parts vanish unless all fields are contracted. Thus vacuum expectation values reduce to sums of products of propagators.

This is the algebraic origin of Feynman diagrams.

8.2 Normal Ordering Revisited

Normal ordering places all creation operators to the left of all annihilation operators. For a scalar field, we write

$$:\hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}}:=\hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{k}}.$$

For fields, normal ordering means applying this rule after expanding the field into creation and annihilation parts. Schematically,

$$\hat{\phi}(x)=\hat{\phi}^{(+)}(x)+\hat{\phi}^{(-)}(x),$$

where $\hat{\phi}^{(+)}$ contains annihilation operators and $\hat{\phi}^{(-)}$ contains creation operators.

A normal-ordered product has zero vacuum expectation value unless it is just a number:

$$\langle 0|:\hat{\phi}(x_1)\cdots\hat{\phi}(x_n):|0\rangle=0$$

for $n>0$. This is why normal ordering is useful in perturbation theory.

8.3 Contractions

For two scalar fields, the contraction is defined as the difference between the time-ordered product and the normal-ordered product:

$$\mathrm{contr}(\hat{\phi}(x),\hat{\phi}(y))\equiv T\{\hat{\phi}(x)\hat{\phi}(y)\}-:\hat{\phi}(x)\hat{\phi}(y):.$$

For a free real scalar field, this contraction is the Feynman propagator:

$$\operatorname{contr}(\hat{\phi}(x), \hat{\phi}(y)) = D_F(x-y).$$

Equivalently,

$$\langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle = D_F(x-y).$$

A contraction is therefore not a new physical object. It is the propagator written as an algebraic operation inside a time-ordered product.

8.4 Wick Theorem for Scalar Fields

For free scalar fields, Wick's theorem states that

$$T\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} =: \hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n : + \text{all single contractions} + \text{all double contractions} + \cdots,$$

where $\hat{\phi}_i = \hat{\phi}(x_i)$. The expansion continues until all possible contractions have been included.

For two fields,

$$T\{\hat{\phi}_1\hat{\phi}_2\} =: \hat{\phi}_1\hat{\phi}_2 : + D_F(x_1 - x_2).$$

For four fields,

$$\begin{aligned}\langle 0|T\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\}|0\rangle &= D_F(x_1-x_2)D_F(x_3-x_4)\\ &\quad + D_F(x_1-x_3)D_F(x_2-x_4)\\ &\quad + D_F(x_1-x_4)D_F(x_2-x_3).\end{aligned}$$

Only fully contracted terms survive between vacuum states.

Example

The vacuum expectation value of an odd number of free real scalar fields vanishes. There is no way to pair all fields into contractions.

8.5 Application to the Dyson Series

Consider ϕ^4 theory with

$$\mathcal{H}_{\text{int}}(x) = \frac{\lambda}{4!}\hat{\phi}_I(x)^4.$$

The first-order correction to a two-point function contains

$$\frac{-i\lambda}{4!}\int \mathrm{d}^4 z\, \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\hat{\phi}(z)^4\}|0\rangle.$$

Wick's theorem reduces this six-field expectation value to a sum of products of three propagators. Some contractions connect x and y to the interaction point z; others form vacuum bubbles or tadpole factors.

This example illustrates the general rule: interaction terms add field insertions, and Wick's theorem pairs those insertions into propagators.

8.6 Wick Theorem for Fermions

For fermionic fields, Wick's theorem has the same structure but includes signs. The basic fermion contraction is

$$\operatorname{contr}(\hat{\psi}(x), \hat{\bar{\psi}}(y)) = S_F(x-y),$$

where S_F is the Dirac propagator. Time ordering of fermions includes a minus sign when two fermionic operators are exchanged.

For example,

$$T\{\hat{\psi}(x)\hat{\bar{\psi}}(y)\} = \theta(x^0 - y^0)\hat{\psi}(x)\hat{\bar{\psi}}(y) - \theta(y^0 - x^0)\hat{\bar{\psi}}(y)\hat{\psi}(x).$$

The minus sign is not optional. It is required by the anticommutation relations of fermionic fields.

In diagrammatic language, every closed fermion loop contributes an additional minus sign. This sign has its origin in the permutations required to contract fermionic operators.

8.7 Fermionic Sign Factors

When applying Wick's theorem to fermions, one must count how many exchanges of fermionic operators are needed to bring the contracted fields next to each other. Each exchange contributes a minus sign.

A useful rule is:

$$\text{fermionic sign} = (-1)^{N_{\text{exchanges}}}.$$

For practical Feynman rules, these signs are packaged into standard rules: closed fermion loops give a minus sign, and external fermion lines must be ordered consistently with the chosen amplitude convention.

The sign bookkeeping is one of the main reasons diagrammatic notation is useful. It turns long operator manipulations into a structured set of graphical and algebraic rules.

8.8 From Algebra to Diagrams

Each term produced by Wick's theorem can be represented graphically. The rules are:

- each field insertion is represented by a point or an external leg;
- each contraction is represented by a line;
- each interaction Hamiltonian insertion is represented by a vertex;
- each vertex is integrated over spacetime;
- each line contributes the propagator of the corresponding field.

For ϕ^4 theory, each interaction vertex has four scalar lines attached to it. For QED, the interaction

$$e\bar{\psi}\gamma^\mu\psi A_\mu$$

has one photon line and two fermion lines attached to the vertex.

A Feynman diagram is therefore a picture of a term in the Wick expansion of the Dyson series. It is not a literal photograph of particles moving in spacetime.

8.9 External Lines, Internal Lines, and Vertices

External lines represent field insertions associated with initial or final states, or with operators whose correlation function is being computed. Internal lines represent contractions between fields inside the perturbative expansion. They carry propagators.

Vertices represent interaction terms. In ϕ^4 theory, the vertex factor is related to $-i\lambda$. In QED, the vertex factor is related to $-ie\gamma^\mu$, with conventions depending on the sign chosen for the interaction Lagrangian.

In momentum space, each internal line carries a momentum variable. Momentum is conserved at each vertex. Integrations over undetermined internal momenta become loop integrals.

8.10 Connected and Disconnected Diagrams

A diagram is connected if every part of it is linked to every other part by lines and vertices. A disconnected diagram factorizes into independent pieces.

In correlation functions, disconnected pieces often include vacuum bubbles, which are diagrams with no external lines. The normalized generating formulas of Chapter 7 divide by the vacuum persistence amplitude, causing pure vacuum bubbles to cancel.

Connected diagrams are the ones that carry the essential correlation between external insertions. Later, when we construct scattering amplitudes, connected amputated diagrams will play the central role.

8.11 Vacuum Bubbles

Vacuum bubbles arise from contractions among fields belonging only to interaction vertices, with no connection to external fields. For example, in ϕ^4 theory, a first-order vacuum bubble comes from contracting the four fields at a single interaction point among themselves.

Vacuum bubbles contribute to

$$\langle 0|T\exp\left[-i\int \mathrm{d}^4x\,\mathcal{H}_{\mathrm{int}}(x)\right]|0\rangle.$$

They affect the vacuum persistence amplitude but cancel from normalized correlation functions. This cancellation is why one often focuses on connected diagrams attached to the external points.

8.12 Symmetry Factors

The same diagram can arise from many equivalent Wick contractions. The symmetry factor accounts for this overcounting. In a perturbative expansion, the factors $1/n!$ from the Dyson series and factors such as $1/4!$ in the interaction Lagrangian combine with the number of contractions to produce the final coefficient of a diagram.

A symmetry factor is the inverse of the number of graph automorphisms that leave the diagram unchanged while preserving its external structure. In practice, one often learns symmetry factors by examples.

For instance, in ϕ^4 theory, the factor $1/4!$ is chosen so that the basic four-point vertex has a simple Feynman rule. Without this factorial, every vertex would carry extra combinatorial factors from permuting identical scalar fields.

8.13 Interpretation Limits of Feynman Diagrams

Feynman diagrams are extraordinarily useful, but they should be interpreted carefully. A diagram represents a term in a perturbative expansion, not a unique classical history. Internal lines do not represent directly observed particles. They represent propagators, or equivalently contractions of free fields.

This distinction matters especially for virtual particles. A virtual particle is not a particle detected in an experiment. It is a feature of a diagrammatic representation of a correlation function or amplitude. Different gauges or field variables can reorganize the same physical amplitude into different sets of intermediate diagrammatic terms.

The physical content lies in gauge-invariant amplitudes, cross sections, decay rates, and correlation functions, not in any single diagram by itself.

Summary

Wick's theorem converts time-ordered products of free fields into sums of normal-ordered products and contractions. Vacuum expectation values keep only fully contracted terms, and each contraction is a propagator. Applying Wick's theorem to the Dyson series produces Feynman diagrams: vertices come from interactions, internal lines from propagators, external lines from field insertions, and symmetry factors from combinatorics.

8.14 Chapter Checklist

After reading this chapter, one should be able to:

- explain why Wick's theorem is needed in perturbation theory;
- define normal ordering and contractions;
- state Wick's theorem for free scalar fields;
- evaluate a four-point free scalar vacuum expectation value;
- explain how contractions become propagators;
- state the fermionic version of Wick's theorem qualitatively;
- explain the origin of fermionic sign factors;
- translate Wick contractions into Feynman diagrams;
- distinguish external lines, internal lines, and vertices;
- explain connected diagrams, vacuum bubbles, and symmetry factors;
- state why Feynman diagrams should not be interpreted as literal particle trajectories.

Chapter 9

Scattering Theory and the S-Matrix

Remark

The previous chapters built the machinery of perturbative QFT: free fields, propagators, the interaction picture, Wick's theorem, and Feynman diagrams. Scattering theory explains how this machinery becomes physics. It connects correlation functions and diagrams to transition amplitudes, cross sections, and decay rates.

9.1 In-States and Out-States

Scattering theory compares what comes in from the far past with what goes out in the far future. The basic idealization is that particles are well separated at very early and very late times, so that they may be described by free Fock states.

An initial state is called an in-state and is denoted

$$|\alpha, \mathrm{in}\rangle.$$

A final state is called an out-state and is denoted

$$|\beta, \mathrm{out}\rangle.$$

For example, a two-particle initial state may be labelled by momenta and other quantum numbers,

$$|p_1, p_2, \mathrm{in}\rangle,$$

while a final state may contain different momenta or even a different number of particles.

The central physical question is: what is the amplitude for the initial state α to become the final state β?

9.2 The S-Matrix

The scattering matrix is defined by the overlap

$$S_{\beta\alpha} = \langle \beta, \mathrm{out}|\alpha, \mathrm{in}\rangle.$$

Equivalently, one introduces an operator $\hat{S}$ acting on free asymptotic states:

$$S_{\beta\alpha} = \langle \beta|\hat{S}|\alpha\rangle.$$

In the interaction picture,

$$\hat{S} = T \exp\left(-i\int_{-\infty}^{\infty} \mathrm{d}t\, \hat{H}_I(t)\right).$$

This is the same operator introduced at the end of Chapter 7. The novelty here is that its matrix elements are interpreted as scattering amplitudes.

It is useful to separate the identity contribution from the interacting part:

$$\hat{S} = 1 + i\hat{T}.$$

The identity term describes no scattering. The operator $\hat{T}$ contains the nontrivial transition information.

9.3 Transition Amplitudes

For relativistic scattering, momentum conservation is factored out. One writes

$$\langle\beta|i\hat{T}|\alpha\rangle = i(2\pi)^4\delta^{(4)}(P_\beta - P_\alpha)\mathcal{M}_{\beta\alpha}.$$

The quantity $\mathcal{M}_{\beta\alpha}$ is the invariant matrix element, or scattering amplitude. It is what Feynman rules are designed to compute.

The delta function expresses conservation of total four-momentum. The amplitude $\mathcal{M}$ contains the dynamical information: coupling constants, propagators, spinor factors, polarization vectors, and momentum dependence.

For example, in a tree-level scalar process, $\mathcal{M}$ may be a simple polynomial in the coupling. In QED it includes spinor bilinears and photon propagators. In loop calculations it includes integrals over internal momenta.

9.4 Matrix Elements and Momentum Conservation

Translation invariance implies four-momentum conservation. If the theory is invariant under spacetime translations, then the total four-momentum operator commutes with the scattering operator:

$$[\hat{P}^\mu, \hat{S}] = 0.$$

Therefore only transitions satisfying

$$P^\mu_\alpha = P^\mu_\beta$$

can have nonzero amplitude.

This is why every Feynman diagram in momentum space carries a momentum-conserving delta function at each vertex. After all internal vertex integrations are done, one overall delta function remains. It expresses total energy-momentum conservation for the full scattering process.

9.5 Lorentz-Invariant Phase Space

Probabilities are obtained by squaring amplitudes and summing or integrating over final states. The Lorentz-invariant one-particle phase-space measure is

$$\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{2E_{\mathbf{p}}}.$$

For an n-particle final state with total incoming momentum P, the Lorentz-invariant phase space is

$$\mathrm{d}\Pi_n(P; p_1, \ldots, p_n) = (2\pi)^4\delta^{(4)}\left(P - \sum_{i=1}^n p_i\right)\prod_{i=1}^n \frac{\mathrm{d}^3p_i}{(2\pi)^3 2E_i}.$$

This measure combines the on-shell conditions for final particles with total four-momentum conservation.

The same measure appears in cross sections and decay rates. It is one of the standard bridges between amplitudes and measurable predictions.

9.6 Cross Sections

For a two-particle scattering process

$$a + b \longrightarrow 1 + 2 + \cdots + n,$$

the differential cross section has the schematic form

$$\mathrm{d}\sigma = \frac{1}{\text{flux}}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n.$$

The flux factor depends on the normalization of the incoming states. In a Lorentz-invariant form for two incoming particles with momenta p_a, p_b, it is built from

$$4\sqrt{(p_a\cdot p_b)^2 - m_a^2 m_b^2}.$$

Thus

$$\mathrm{d}\sigma = \frac{|\mathcal{M}|^2}{4\sqrt{(p_a\cdot p_b)^2 - m_a^2 m_b^2}}\,\mathrm{d}\Pi_n.$$

For massless or center-of-mass examples this expression simplifies, but the structure is always the same: amplitude squared times phase space divided by incoming flux.

9.7 Decay Rates

For a decay process

$$A \longrightarrow 1 + 2 + \cdots + n,$$

there is only one initial particle. The differential decay rate is

$$\mathrm{d}\Gamma = \frac{1}{2M_A}|\mathcal{M}|^2\,\mathrm{d}\Pi_n,$$

where M_A is the mass of the decaying particle. The total decay rate is obtained by integrating over final-state phase space:

$$\Gamma = \int \mathrm{d}\Gamma.$$

The lifetime is

$$\tau = \frac{1}{\Gamma}.$$

The same invariant amplitude technology therefore describes both scattering experiments and unstable-particle decays.

9.8 The LSZ Reduction Formula

Perturbation theory naturally computes time-ordered correlation functions. The LSZ reduction formula explains how to extract scattering amplitudes from those correlators.

For a scalar field, the rough idea is this: external particles correspond to poles of the Fourier-transformed correlation function. To obtain an amplitude, one multiplies by inverse propagator factors and takes the external momenta on shell. Schematically,

$$\mathcal{M} \sim \prod_i \left(p_i^2 - m^2\right)\widetilde{G}(p_1,\ldots,p_n)\Big|_{p_i^2=m^2},$$

with normalization factors suppressed.

The exact LSZ formula includes wave-function renormalization factors and signs that depend on conventions. Its conceptual message is more important at this stage: scattering amplitudes are obtained from the pole structure of correlation functions.

9.9 From Correlation Functions to Scattering Amplitudes

The chain of ideas is now complete:

$$\text{Lagrangian} \longrightarrow \text{interaction Hamiltonian} \longrightarrow \text{Dyson series} \longrightarrow \text{Wick contractions} \longrightarrow \text{Feynman diagrams} \longrightarrow \mathcal{M}$$

The first part of the chain is formal and operator-theoretic. The middle part is diagrammatic. The last part converts amplitudes into cross sections and decay rates.

This is why propagators and vertices are not merely mathematical decorations. They are the intermediate objects that connect a Lagrangian to measurable numbers.

9.10 Tree-Level $2 \to 2$ Scattering

A simple example is tree-level $2 \to 2$ scattering in ϕ^4 theory. The interaction is

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

At lowest order, there is one four-point vertex and no internal propagator. The amplitude is

$$i\mathcal{M} = -i\lambda, \qquad \mathcal{M} = -\lambda.$$

This example is deliberately simple. It shows that a contact interaction gives a constant tree-level amplitude. More complicated theories produce amplitudes with propagators, spinor factors, and polarization vectors.

In momentum space, the full matrix element is

$$\langle p_3, p_4 | i\hat{T} | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4)\mathcal{M}.$$

The delta function enforces momentum conservation; $\mathcal{M}$ is the part used in the cross-section formula.

9.11 Decay of an Unstable Particle

Suppose a particle A can decay into two particles 1 and 2. If the amplitude is $\mathcal{M}(A \to 1 + 2)$, then

$$\mathrm{d}\Gamma = \frac{1}{2M_A}|\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}(p_A - p_1 - p_2)\frac{\mathrm{d}^3 p_1}{(2\pi)^3 2E_1}\frac{\mathrm{d}^3 p_2}{(2\pi)^3 2E_2}.$$

This formula is the decay analogue of the scattering cross section. It says that rates are determined by amplitude squared times available phase space.

The phase space can strongly affect the physical rate. A large amplitude may produce a small rate if very little phase space is available, while a modest amplitude can produce a large rate if many final states are accessible.

9.12 From Amplitudes to Measurable Predictions

Experiments do not measure Feynman diagrams directly. They measure cross sections, angular distributions, branching ratios, decay widths, and event rates. QFT predicts these quantities by computing amplitudes and integrating them over the appropriate phase space.

A realistic prediction usually requires additional steps: spin sums, averages over initial polarizations, sums over final polarizations, symmetry factors for identical particles, detector cuts, and sometimes radiative corrections. The basic structure, however, is already visible:

$$\text{theory} \longrightarrow \mathcal{M} \longrightarrow |\mathcal{M}|^2 \longrightarrow \text{phase-space integral} \longrightarrow \text{observable}.$$

Summary

Scattering theory turns the operator and diagrammatic machinery of perturbative QFT into measurable predictions. The S-matrix connects in-states to out-states. Momentum conservation is factored out to define the invariant amplitude $\mathcal{M}$. Cross sections and decay rates are obtained from $|\mathcal{M}|^2$ integrated over Lorentz-invariant phase space. LSZ explains why scattering amplitudes are encoded in the pole structure of time-ordered correlation functions.

9.13 Chapter Checklist

After reading this chapter, one should be able to:

- define in-states, out-states, and the S-matrix;
- distinguish S, T, and the invariant amplitude $\mathcal{M}$;
- explain the origin of the momentum-conserving delta function;
- write the Lorentz-invariant phase-space measure;
- state the schematic cross-section formula;
- state the decay-rate formula;
- explain the conceptual role of LSZ reduction;
- describe how correlation functions lead to amplitudes;
- compute the tree-level ϕ^4 amplitude at lowest order;
- explain how amplitudes become measurable predictions.

Chapter 10

Path Integral for Quantum Mechanics

Remark

The operator formalism led from fields to propagators, diagrams, and scattering amplitudes. The path integral gives a different but equivalent viewpoint: a transition amplitude is computed as a sum over histories, weighted by e^{iS}. We first develop this idea in ordinary quantum mechanics before extending it to fields.

10.1 Transition Amplitudes as Sums Over Histories

In the operator formulation of quantum mechanics, the transition amplitude from position q_i at time t_i to position q_f at time t_f is

$$\langle q_f, t_f | q_i, t_i \rangle = \langle q_f | e^{-i\hat{H}(t_f - t_i)} | q_i \rangle.$$

The path-integral idea is that the same amplitude can be written as a sum over all paths connecting the endpoints:

$$\langle q_f, t_f | q_i, t_i \rangle = \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}q(t)\, e^{iS[q]}.$$

Here

$$S[q] = \int_{t_i}^{t_f} \mathrm{d}t\, L(q, \dot{q})$$

is the classical action evaluated on the path $q(t)$. The notation $\mathcal{D}q(t)$ means integration over all intermediate paths.

This formula should not be interpreted as saying that the particle secretly chooses one path. Rather, the transition amplitude is built by coherently summing contributions from all paths.

10.2 Time Slicing

The path integral can be derived by dividing the time interval into many small steps. Let

$$t_f - t_i = N\epsilon.$$

Insert resolutions of the identity in the position basis:

$$1 = \int \mathrm{d}q_j\, |q_j\rangle\langle q_j|.$$

Then

$$\langle q_f | e^{-i\hat{H}(t_f - t_i)} | q_i \rangle = \int \prod_{j=1}^{N-1} \mathrm{d}q_j \prod_{j=0}^{N-1} \langle q_{j+1} | e^{-i\epsilon\hat{H}} | q_j \rangle.$$

For a Hamiltonian

$$H = \frac{p^2}{2m} + V(q),$$

one also inserts momentum states and evaluates the short-time matrix element. In the limit $N \to \infty$, this produces

$$\int \mathcal{D}q \, e^{i \int \mathrm{d}t \, [\frac{1}{2} m \dot{q}^2 - V(q)]}.$$

Thus the path integral is not a new postulate here; it is a reorganization of operator time evolution.

10.3 Classical Limit and Stationary Phase

The factor $e^{iS[q]}$ oscillates rapidly when the action is large compared with $\hbar$. Restoring $\hbar$, the weight is

$$e^{iS[q]/\hbar}.$$

In the limit $\hbar \to 0$, most paths interfere destructively. The dominant contribution comes from paths for which the action is stationary:

$$\delta S = 0.$$

This condition gives the classical equations of motion.

Thus the classical path is not the only path included in the quantum amplitude. It is the path around which neighbouring contributions add coherently in the classical limit. Quantum mechanics is the sum over all histories; classical mechanics is the stationary-phase approximation.

10.4 Gaussian Integrals

Gaussian integrals are the technical backbone of free path integrals. The ordinary one-dimensional formula is

$$\int_{-\infty}^{\infty} \mathrm{d}x \, e^{-\frac{1}{2} a x^2 + Jx} = \sqrt{\frac{2\pi}{a}} \, e^{\frac{J^2}{2a}},$$

for $a > 0$. In oscillatory form one uses an $i\epsilon$ prescription to make the integral well defined.

The multidimensional version is

$$\int \mathrm{d}^n x \, \exp\left[-\frac{1}{2} x^T A x + J^T x\right] = \left(\frac{(2\pi)^n}{\det A}\right)^{1/2} \exp\left(\frac{1}{2} J^T A^{-1} J\right).$$

Free quantum theories lead to infinite-dimensional versions of this formula. The inverse operator A^{-1} becomes the propagator.

10.5 The Free Particle

For a free particle,

$$L = \frac{1}{2} m \dot{q}^2.$$

The path integral is Gaussian. The result is

$$K(q_f, t_f; q_i, t_i) = \left(\frac{m}{2\pi i T}\right)^{1/2} \exp\left[\frac{i m (q_f - q_i)^2}{2T}\right],$$

where $T = t_f - t_i$. The exponent is the classical action evaluated on the straight-line path connecting the endpoints.

This example illustrates a common pattern. For a quadratic action, the path integral can be evaluated exactly. The answer is a normalization determinant multiplying $e^{iS_{\mathrm{cl}}}$.

10.6 The Harmonic Oscillator

For the harmonic oscillator,

$$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2.$$

The action is still quadratic, so the path integral remains Gaussian. The transition amplitude again has the form

$$K(q_f, t_f; q_i, t_i) = \mathcal{N}(T)e^{iS_{\mathrm{cl}}(q_f,t_f;q_i,t_i)}.$$

The exact normalization $\mathcal{N}(T)$ comes from the determinant of the quadratic fluctuation operator.

The oscillator is important because a free field is a collection of oscillators. The free-field path integral will therefore be an infinite product of Gaussian oscillator path integrals.

10.7 Generating Functionals in Quantum Mechanics

To compute correlation functions, introduce a source $J(t)$ coupled to $q(t)$:

$$Z[J] = \int \mathcal{D}q \, \exp\left(iS[q] + i\int \mathrm{d}t \, J(t)q(t)\right).$$

The source is an auxiliary function. It is set to zero after differentiating. Correlation functions are obtained by functional derivatives:

$$\langle Tq(t_1)q(t_2)\rangle = \frac{1}{Z[0]}\,\frac{1}{i^2}\frac{\delta^2 Z[J]}{\delta J(t_1)\delta J(t_2)}\bigg|_{J=0}.$$

This is the prototype of the generating functional method used in QFT.

10.8 Sources and Correlation Functions

The source term keeps track of insertions. Differentiating once gives

$$\frac{1}{i}\frac{\delta Z[J]}{\delta J(t)} = \int \mathcal{D}q \, q(t) \exp\left(iS[q] + i\int \mathrm{d}t' \, J(t')q(t')\right).$$

Differentiating twice inserts two factors of q, and so on. Therefore

$$\frac{1}{i^n}\frac{\delta^n Z[J]}{\delta J(t_1)\cdots\delta J(t_n)}\bigg|_{J=0}$$

generates the n-point time-ordered correlation function.

In operator language these are vacuum expectation values of time-ordered products. In the path-integral language they are moments of a functional integral with an oscillatory weight.

10.9 Why the Path Integral Is Useful for QFT

The path integral is especially useful in field theory for several reasons. First, it makes spacetime symmetries more visible because the action appears directly. Second, it gives a compact way to derive Feynman rules. Third, it handles correlation functions and sources naturally. Fourth, it generalizes to gauge fixing, ghosts, effective actions, and nonperturbative approximations.

For a field, the single coordinate $q(t)$ is replaced by a field configuration $\phi(x)$. The ordinary source $J(t)$ is replaced by a spacetime source $J(x)$. The path integral becomes

$$\int \mathcal{D}\phi \, e^{iS[\phi]}.$$

This is the subject of the next chapter.

10.10 Relation to Canonical Quantization

The path integral and canonical quantization are not separate quantum theories. They are two formulations of the same theory. Canonical quantization emphasizes Hilbert spaces, operators, commutators, and states. The path integral emphasizes action functionals, histories, sources, and correlation functions.

The connection can be summarized as

$$\langle q_f|e^{-i\hat{H}T}|q_i\rangle = \int_{q_i}^{q_f} \mathcal{D}q\, e^{iS[q]}.$$

The left-hand side is canonical. The right-hand side is the path integral. In field theory, this equivalence underlies the use of functional integrals to compute the same time-ordered correlators that appear in the operator formalism.

Summary

The path integral expresses a transition amplitude as a sum over histories weighted by e^{iS}. Time slicing connects this formula to operator evolution. The classical equations arise by stationary phase. Quadratic actions lead to Gaussian integrals, whose inverses become propagators. Adding sources produces a generating functional whose derivatives give time-ordered correlation functions.

10.11 Chapter Checklist

After reading this chapter, one should be able to:

- write a transition amplitude as a path integral over histories;
- explain the time-slicing derivation;
- explain how the classical limit follows from stationary phase;
- evaluate the role of Gaussian integrals in free theories;
- state the path integral for the free particle and harmonic oscillator;
- define a generating functional with a source;
- obtain correlation functions by differentiating with respect to the source;
- explain why the path integral is useful for QFT;
- relate the path integral to canonical quantization.

Chapter 11

Path Integral for Scalar Fields

Remark

The path integral for a particle sums over histories $q(t)$. The path integral for a field sums over field configurations $\phi(x)$. This chapter develops the functional integral for scalar fields and shows how propagators, correlation functions, perturbation theory, and diagrams arise from a single generating functional.

11.1 From Paths to Field Configurations

In ordinary quantum mechanics, the path integral is

$$\int \mathcal{D}q(t)\, e^{iS[q]}.$$

For a scalar field, the dynamical variable is not a single function of time but a function of spacetime. The corresponding path integral is

$$\int \mathcal{D}\phi\, e^{iS[\phi]}.$$

The measure $\mathcal{D}\phi$ symbolically means integration over all field configurations.

For a real scalar field, the action is

$$S[\phi] = \int \mathrm{d}^4x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2\right)$$

for the free theory. Interactions are added by including non-quadratic terms, for example

$$S[\phi] = \int \mathrm{d}^4x \left(\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right).$$

11.2 The Generating Functional

Introduce a source $J(x)$ coupled linearly to the field:

$$Z[J] = \int \mathcal{D}\phi\, \exp\left(iS[\phi] + i\int \mathrm{d}^4x\, J(x)\phi(x)\right).$$

The source is an auxiliary function. It is used to generate correlation functions and then set to zero.

The vacuum-to-vacuum functional is

$$Z[0] = \int \mathcal{D}\phi\, e^{iS[\phi]}.$$

Normalized correlation functions are obtained by dividing by $Z[0]$. For example,

$$\langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle = \frac{1}{Z[0]}\,\frac{1}{i^2}\frac{\delta^2 Z[J]}{\delta J(x)\delta J(y)}\bigg|_{J=0}.$$

Thus the path integral generates the same time-ordered correlators that appeared in the operator formulation.

11.3 The Free Scalar Generating Functional

The free scalar action can be written, after integrating by parts and ignoring boundary terms, as

$$S_0[\phi] = -\frac{1}{2}\int \mathrm{d}^4x\, \phi(x)(\Box + m^2)\phi(x).$$

With a source, the exponent is quadratic in ϕ:

$$iS_0[\phi] + i\int \mathrm{d}^4x\, J\phi.$$

This is an infinite-dimensional Gaussian integral. The result has the form

$$Z_0[J] = Z_0[0]\exp\left(\frac{i}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right),$$

where D_F is the Feynman propagator satisfying

$$(\Box_x + m^2)D_F(x-y) = -i\delta^{(4)}(x-y).$$

The inverse of the quadratic operator in the action is therefore the propagator. This is the path-integral version of the result obtained from canonical quantization.

11.4 Sources and Functional Derivatives

Functional derivatives insert fields. The basic rule is

$$\frac{1}{i}\frac{\delta}{\delta J(x)}Z[J] = \int \mathcal{D}\phi\,\phi(x)\exp\left(iS[\phi] + i\int \mathrm{d}^4z\, J(z)\phi(z)\right).$$

Therefore

$$\langle T\phi(x_1)\cdots\phi(x_n)\rangle = \frac{1}{Z[0]}\,\frac{1}{i^n}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}.$$

For the free theory, differentiating $Z_0[J]$ reproduces Wick's theorem. The exponential quadratic in J generates all pairings of the external points, and each pair contributes one propagator.

11.5 Gaussian Functional Integrals

The free scalar generating functional is the infinite-dimensional analogue of

$$\int \mathrm{d}^nx\, \exp\left[-\frac{1}{2}x^TAx + J^Tx\right] \propto \exp\left(\frac{1}{2}J^TA^{-1}J\right).$$

In field theory, the matrix A becomes a differential operator, and A^{-1} becomes a Green function.

Schematically,

$$A = \Box + m^2, \qquad A^{-1} \sim D_F.$$

The $i\epsilon$ prescription specifies which inverse is meant. Without it, the operator has several possible Green functions. With it, the inverse is the Feynman propagator appropriate for time-ordered vacuum correlators.

The determinant of the quadratic operator contributes to $Z_0[0]$. In many normalized correlation functions this factor cancels, but determinants become important in loop effects, effective actions, and vacuum energies.

11.6 Interacting Scalar Theory

For ϕ^4 theory,

$$S[\phi] = S_0[\phi] - \int \mathrm{d}^4x \, \frac{\lambda}{4!}\phi(x)^4.$$

The generating functional is

$$Z[J] = \int \mathcal{D}\phi \, \exp\left(iS_0[\phi] - i\int \mathrm{d}^4x \, \frac{\lambda}{4!}\phi^4 + i\int \mathrm{d}^4x \, J\phi\right).$$

The interaction can be represented as functional derivatives acting on the free generating functional:

$$Z[J] = \exp\left[-i\int \mathrm{d}^4x \, \frac{\lambda}{4!}\left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)^4\right] Z_0[J].$$

Expanding the exponential in powers of λ gives perturbation theory. This is the path-integral version of the Dyson expansion.

11.7 Diagrammatic Expansion

The diagrammatic rules follow from expanding the interacting generating functional and differentiating with respect to sources. Each factor of $D_F(x-y)$ becomes a line. Each factor of $-i\lambda \int \mathrm{d}^4x$ becomes an interaction vertex integrated over spacetime.

For ϕ^4 theory:

- a propagator contributes $D_F(x-y)$ in position space or $i/(k^2 - m^2 + i\epsilon)$ in momentum space;
- a vertex contributes a factor $-i\lambda$;
- four scalar lines meet at each vertex;
- one integrates over internal vertex positions or internal momenta;
- symmetry factors account for equivalent contractions.

Thus the Feynman rules derived earlier from Wick's theorem also emerge directly from the functional integral.

11.8 Connected Generating Functional

The generating functional $Z[J]$ includes disconnected diagrams. It is useful to define the connected generating functional $W[J]$ by

$$Z[J] = e^{iW[J]}.$$

Equivalently,

$$W[J] = -i \ln Z[J].$$

Functional derivatives of $W[J]$ generate connected correlation functions. For example,

$$\frac{1}{i}\frac{\delta W[J]}{\delta J(x)}$$

gives the connected one-point function, and higher derivatives give connected higher-point functions.

The logarithm removes the disconnected multiplication structure of $Z[J]$. This is the functional-integral version of the statement that disconnected vacuum bubbles exponentiate and cancel in normalized correlators.

11.9 The Classical Field

The source-dependent expectation value of the field is defined by

$$\phi_c(x) = \frac{\delta W[J]}{\delta J(x)}.$$

It is called the classical field, although it is really the expectation value of the quantum field in the presence of a source. When the source is removed, ϕ_c may vanish or may describe a nontrivial vacuum expectation value.

This quantity is important because it is the natural variable of the effective action. Instead of describing the theory as a functional of the external source J, one can describe it as a functional of the expectation value ϕ_c.

11.10 The Effective Action

The effective action $\Gamma[\phi_c]$ is defined by a Legendre transform:

$$\Gamma[\phi_c] = W[J] - \int \mathrm{d}^4x\, J(x)\phi_c(x),$$

where J is understood as a functional of ϕ_c. It satisfies

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x).$$

When the source is set to zero, the quantum equation of motion is

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = 0.$$

At tree level, Γ is just the classical action. Loop corrections modify it. Therefore the effective action is the quantum-corrected action governing expectation values.

11.11 One-Particle-Irreducible Functions

Functional derivatives of $\Gamma[\phi_c]$ generate one-particle-irreducible correlation functions, abbreviated 1PI functions. A diagram is one-particle irreducible if it cannot be disconnected by cutting a single internal line.

The two-point 1PI function is related to the inverse full propagator. The four-point 1PI function is related to the quantum-corrected interaction vertex. This language becomes essential in renormalization, where masses, fields, and couplings are corrected by loop effects.

The hierarchy is therefore

$$Z[J] \longrightarrow \text{all diagrams}, \qquad W[J] \longrightarrow \text{connected diagrams}, \qquad \Gamma[\phi_c] \longrightarrow \text{1PI diagrams}.$$

11.12 Classical Limit and Saddle Points

The path integral also gives a useful view of the classical limit. When the action is large compared with $\hbar$, the integral is dominated by stationary points of the action:

$$\frac{\delta S}{\delta\phi(x)} = 0.$$

This is the classical field equation. Quantum corrections come from fluctuations around the saddle point.

For a free theory, fluctuations are Gaussian. For an interacting theory, fluctuations generate vertices and loop diagrams. Thus perturbation theory can be viewed as a systematic expansion around a saddle point of the action.

Summary

The scalar-field path integral sums over field configurations rather than particle paths. The source functional $Z[J]$ generates time-ordered correlation functions. For a free scalar field, the functional integral is Gaussian and the inverse quadratic operator is the Feynman propagator. Interactions are generated by functional derivatives and produce the same diagrammatic expansion as Wick's theorem. The functionals $W[J]$ and $\Gamma[\phi_c]$ organize connected and one-particle-irreducible diagrams.

11.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain the meaning of the functional measure $\mathcal{D}\phi$;
- write the scalar-field generating functional with a source;
- derive the free generating functional $Z_0[J]$;
- obtain correlation functions by differentiating with respect to J;
- explain why the free path integral is Gaussian;
- express interactions as functional derivative operators;
- derive the diagrammatic expansion from $Z[J]$;
- define $W[J]$ and explain connected diagrams;
- define the classical field ϕ_c;
- define the effective action $\Gamma[\phi_c]$;
- explain the meaning of one-particle-irreducible functions.

Chapter 12

Fermionic Path Integrals

Remark

Scalar path integrals integrate over ordinary commuting fields. Fermionic path integrals require anticommuting variables. The price is algebraic unfamiliarity; the reward is a compact functional-integral description of Dirac fields, fermion propagators, fermion loops, and determinants.

12.1 Why Fermions Need a Different Integral

The scalar path integral sums over ordinary field configurations:

$$Z[J] = \int \mathcal{D}\phi \, e^{iS[\phi]+i\int J\phi}.$$

This works because scalar fields are bosonic. Their quantum operators obey commutation relations, and the corresponding functional variables commute.

Dirac fields are different. Their operators obey anticommutation relations, and fermionic many-particle states are antisymmetric. A path integral for fermions must reproduce this algebra. The appropriate variables are therefore not ordinary complex numbers, but Grassmann variables: objects that anticommute.

The transition is

$$\text{bosons} \longrightarrow \text{commuting variables}, \qquad \text{fermions} \longrightarrow \text{anticommuting variables}.$$

This chapter develops the basic calculus needed for fermionic functional integrals.

12.2 Grassmann Variables

A Grassmann variable θ anticommutes with another Grassmann variable χ:

$$\theta\chi = -\chi\theta.$$

In particular,

$$\theta^2 = 0.$$

This nilpotency makes every function of a single Grassmann variable finite:

$$f(\theta) = a + b\theta.$$

For several Grassmann variables, polynomials terminate because each variable can appear at most once.

Grassmann variables also anticommute with fermionic sources and fermionic fields, but commute with ordinary numbers and bosonic variables. They are bookkeeping objects that encode fermionic antisymmetry inside a functional integral.

12.3 Berezin Integration

Integration over Grassmann variables is called Berezin integration. It is defined by the rules

$$\int \mathrm{d}\theta\, 1 = 0, \qquad \int \mathrm{d}\theta\, \theta = 1.$$

Thus Grassmann integration behaves like differentiation. For a function $f(\theta) = a + b\theta$,

$$\int \mathrm{d}\theta\, f(\theta) = b.$$

For two variables, the order matters:

$$\int \mathrm{d}\theta_2\, \mathrm{d}\theta_1\, \theta_1\theta_2 = 1.$$

Changing the order of variables or integration measures can introduce minus signs. This is the algebraic origin of many sign factors in fermionic path integrals.

12.4 Gaussian Integrals for Grassmann Variables

For one pair of independent Grassmann variables $\bar{\theta}, \theta$,

$$\int \mathrm{d}\bar{\theta}\, \mathrm{d}\theta\, e^{-\bar{\theta}a\theta} = a.$$

For many variables, the general formula is

$$\int \prod_i \mathrm{d}\bar{\theta}_i\, \mathrm{d}\theta_i\, \exp\left(-\bar{\theta}_i A_{ij} \theta_j\right) = \det A.$$

This differs from the bosonic Gaussian integral, which produces $(\det A)^{-1/2}$ for real variables or $(\det A)^{-1}$ for complex commuting variables. Fermions produce determinants in the numerator.

With sources, one has

$$\int \mathrm{d}\bar{\theta}\, \mathrm{d}\theta\, \exp\left(-\bar{\theta}A\theta + \bar{\eta}\theta + \bar{\theta}\eta\right) = \det A\, \exp\left(\bar{\eta}A^{-1}\eta\right),$$

up to conventions for signs and ordering. The inverse matrix A^{-1} will become the fermion propagator.

12.5 The Dirac Generating Functional

The free Dirac action is

$$S_0[\bar{\psi}, \psi] = \int \mathrm{d}^4x\, \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

In the path integral, ψ and $\bar{\psi}$ are independent Grassmann-valued fields. Introduce Grassmann-valued sources η and $\bar{\eta}$:

$$Z_0[\bar{\eta}, \eta] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\, \exp\left(iS_0[\bar{\psi}, \psi] + i\int \mathrm{d}^4x\, [\bar{\eta}\psi + \bar{\psi}\eta]\right).$$

This is a Gaussian Grassmann functional integral. It gives

$$Z_0[\bar{\eta}, \eta] = Z_0[0, 0] \exp\left(-i\int \mathrm{d}^4x\, \mathrm{d}^4y\, \bar{\eta}(x) S_F(x-y) \eta(y)\right),$$

where S_F is the Dirac Feynman propagator.

The sign in the exponent depends on source-ordering conventions. The essential point is invariant: differentiating with respect to fermionic sources produces Dirac propagators and fermionic correlation functions.

12.6 Dirac Propagator from the Path Integral

The Dirac propagator is the inverse of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m)S_F(x-y) = i\delta^{(4)}(x-y).$$

In momentum space,

$$S_F(p) = \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon}.$$

From the generating functional,

$$\langle 0|T\hat{\psi}_\alpha(x)\hat{\bar{\psi}}_\beta(y)|0\rangle = S_{F,\alpha\beta}(x-y).$$

Thus the same propagator obtained by canonical quantization appears as the inverse operator in the fermionic Gaussian integral.

The path-integral perspective makes the parallel with scalar fields clear:

$$\text{quadratic operator in the action}^{-1} = \text{free propagator}.$$

For scalars the operator is Klein–Gordon. For fermions it is Dirac.

12.7 Generating Fermionic Correlation Functions

Fermionic correlation functions are generated by differentiating with respect to Grassmann sources. Schematically,

$$\langle T\psi(x_1)\bar{\psi}(y_1)\cdots\psi(x_n)\bar{\psi}(y_n)\rangle$$

is obtained by applying derivatives with respect to $\bar{\eta}(x_i)$ and $\eta(y_i)$ to $Z[\bar{\eta},\eta]$, then setting the sources to zero.

The order of the derivatives matters. Since the sources anticommute, exchanging two source derivatives can produce a minus sign. These signs are the path-integral version of the fermionic signs in Wick's theorem.

A useful practical rule is that the path integral automatically keeps track of fermionic antisymmetry if the source ordering is used consistently.

12.8 Interactions with Fermions

For QED, the interaction term is

$$S_{\text{int}} = -e\int \mathrm{d}^4x\, \bar{\psi}\gamma^\mu\psi A_\mu.$$

The full generating functional is

$$Z[\bar{\eta},\eta,J] = \int \mathcal{D}A\mathcal{D}\bar{\psi}\mathcal{D}\psi\, \exp\left(iS[A,\bar{\psi},\psi] + i\int \mathrm{d}^4x\, [J_\mu A^\mu + \bar{\eta}\psi + \bar{\psi}\eta]\right).$$

Perturbation theory is obtained by expanding the interaction exponential or by rewriting interactions as functional derivatives with respect to sources.

The QED vertex contains one photon field, one ψ, and one $\bar{\psi}$. Therefore each vertex joins one photon line and two fermion lines. This is the path-integral origin of the usual QED diagrammatic vertex.

12.9 Fermion Loops and Minus Signs

Closed fermion loops carry an extra minus sign. In the operator formalism this comes from permuting fermionic operators in Wick contractions. In the path integral it comes from the algebra of Grassmann variables and the determinant produced by integrating out fermions.

A closed fermion loop is associated with a trace over spinor indices and an integral over loop momentum. The additional sign is part of the Feynman rule:

$$\text{each closed fermion loop contributes a factor } -1.$$

This rule is not an arbitrary convention. It is a direct consequence of fermionic anticommutation.

12.10 Determinants from Integrating Out Fermions

Because the fermionic path integral is Gaussian when the action is quadratic in $\psi, \bar{\psi}$, one can formally integrate out fermions. For a Dirac operator D,

$$\int \mathcal{D}\bar{\psi}\mathcal{D}\psi \, e^{i\int \bar{\psi}D\psi} \propto \det D.$$

If the Dirac operator depends on a background gauge field, the determinant also depends on that background:

$$\det(i\gamma^\mu D_\mu - m).$$

This determinant encodes the effect of virtual fermion loops on the gauge-field dynamics.

For bosonic Gaussian integrals, one typically gets inverse determinants. This difference between bosonic and fermionic determinants is one of the key structural distinctions between commuting and anticommuting fields.

12.11 Majorana Fermions in the Path Integral

A Majorana fermion is constrained to be related to its charge conjugate. In the path integral this means that the independent fermionic variables are reduced relative to a Dirac field. The Gaussian integral no longer gives an ordinary determinant in the same way; it gives a Pfaffian.

Schematically, for an antisymmetric operator A,

$$\int \mathcal{D}\chi \, e^{-\frac{1}{2}\chi A\chi} \propto \mathrm{Pf}(A),$$

and

$$\mathrm{Pf}(A)^2 = \det A.$$

The details require a careful treatment of charge conjugation and spinor conventions. At this stage, the main lesson is that Majorana path integrals are fermionic Gaussian integrals with fewer independent variables than Dirac path integrals.

12.12 Physical Interpretation and Common Pitfalls

Grassmann variables are not classical values of a fermion field. They are anticommuting integration variables designed to reproduce fermionic quantum statistics. One should not imagine a fermionic path integral as a sum over ordinary classical spinor configurations.

Another common pitfall is to treat ψ and $\bar{\psi}$ as complex conjugates inside the functional integral. In the fermionic path integral they are independent Grassmann variables. Their relation as adjoint operators belongs to the canonical operator formulation, while the functional integral uses an independent-variable representation.

Finally, fermionic determinants and loop signs are not optional decorations. They are the mechanism by which the Pauli principle and virtual fermion effects enter the path-integral formalism.

Summary

Fermionic path integrals use Grassmann variables. Berezin integration turns fermionic Gaussian integrals into determinants, and source derivatives generate fermionic time-ordered correlation functions. The inverse Dirac operator is the Dirac propagator. Fermion loops acquire minus signs, and integrating out fermions produces determinants that encode virtual fermion effects. Majorana fermions require a related Pfaffian structure.

12.13 Chapter Checklist

After reading this chapter, one should be able to:

- define Grassmann variables and state their anticommutation property;
- state the basic rules of Berezin integration;
- evaluate the structure of a Gaussian Grassmann integral;
- write the free Dirac generating functional with Grassmann sources;
- identify the Dirac propagator as the inverse Dirac operator;
- explain how fermionic correlation functions are generated by source derivatives;
- explain why closed fermion loops carry a minus sign;
- describe how integrating out fermions produces determinants;
- explain the basic path-integral difference between Dirac and Majorana fermions;
- avoid interpreting Grassmann fields as ordinary classical spinor configurations.

Chapter 13

Gauge Fields in the Path Integral

Remark

Gauge fields require special care in the path integral because the functional integral over A_μ overcounts physically equivalent configurations. Gauge fixing removes this overcounting. In non-Abelian theories, the gauge-fixing procedure introduces ghost fields and leads naturally to BRST symmetry.

13.1 Overcounting Gauge-Equivalent Configurations

For electromagnetism, the gauge potential transforms as

$$A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda.$$

The two potentials describe the same physical electromagnetic field. Therefore a naive path integral

$$\int \mathcal{D}A_\mu \, e^{iS[A]}$$

integrates over infinitely many copies of the same physical configuration.

This is not merely an aesthetic problem. The redundancy makes the quadratic operator non-invertible, so the propagator of A_μ is not defined until one chooses a gauge. In operator language this was the statement that not all components of A_μ are physical. In the path-integral language it becomes a problem of dividing by the volume of the gauge group.

13.2 Gauge Fixing in the Functional Integral

A gauge condition is a function $G[A]$ imposed on gauge fields. A common covariant choice is

$$G[A] = \partial_\mu A^\mu.$$

Instead of integrating over all representatives of each gauge orbit, one wants to integrate once over each physical configuration. Formally,

$$\int \mathcal{D}A_\mu \quad \longrightarrow \quad \int \mathcal{D}A_\mu \, \delta(G[A]) \, \Delta[A],$$

where $\Delta[A]$ is a determinant that compensates for the change of variables from gauge orbits to gauge-fixed representatives.

In practice, one often replaces the sharp delta-function condition by a Gaussian average over gauges. This leads to the covariant gauge-fixing term

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

The parameter ξ labels the gauge. The choice $\xi = 1$ is Feynman gauge.

13.3 The Faddeev–Popov Method

The Faddeev–Popov method inserts a carefully chosen form of unity into the path integral:

$$1 = \Delta_{\mathrm{FP}}[A] \int \mathcal{D}\alpha \, \delta(G[A^\alpha]).$$

Here A^α denotes the gauge transform of A, and $\Delta_{\mathrm{FP}}[A]$ is the Faddeev–Popov determinant.

The determinant is given formally by

$$\Delta_{\mathrm{FP}}[A] = \det \left(\frac{\delta G[A^\alpha]}{\delta \alpha} \right).$$

This factor tells us how the gauge condition changes as one moves along a gauge orbit. It is the Jacobian associated with gauge fixing.

For Abelian gauge theory in a linear covariant gauge, this determinant is field-independent. It contributes only an overall constant and is usually ignored. For non-Abelian gauge theory, the determinant depends on the gauge field and has dynamical consequences.

13.4 Ghost Fields

A determinant can be represented as a Grassmann integral. For an operator M,

$$\det M = \int \mathcal{D}\bar{c}\mathcal{D}c \, \exp \left(i \int \mathrm{d}^4x \, \bar{c} M c \right),$$

with appropriate conventions. The Grassmann fields c and $\bar{c}$ are called ghost fields.

Ghosts are anticommuting scalar fields. This phrase sounds paradoxical: they have scalar Lorentz transformation properties but fermionic statistics. They are not introduced because the theory has new physical particles. They are introduced because the gauge-fixing determinant must be represented inside the functional integral.

For non-Abelian gauge theory, ghosts interact with gauge fields. Their diagrams are essential for unitarity and gauge invariance in covariant gauges.

13.5 Why Ghosts Are Not Physical Particles

Ghost fields appear as internal lines in Feynman diagrams, but they do not appear as physical external states. They are artifacts of gauge fixing, not quanta in the physical Hilbert space.

This is analogous to the unphysical photon polarizations discussed in Chapter 5. Covariant quantization introduces redundant components for calculational convenience. The formalism must then ensure that unphysical states do not appear in physical observables.

Ghosts are part of that mechanism. Their contributions cancel gauge-dependent and unphysical pieces of gauge-field loops. Removing them would spoil the consistency of non-Abelian perturbation theory.

13.6 Gauge-Fixed Photon Path Integral

For QED, the gauge-fixed action is

$$S[A] = \int \mathrm{d}^4x \left[-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right].$$

The corresponding generating functional is

$$Z[J] = \int \mathcal{D}A \, \exp \left(iS[A] + i \int \mathrm{d}^4x \, J_\mu A^\mu \right).$$

Because the Abelian Faddeev–Popov determinant is field-independent in this gauge, ghosts decouple from ordinary QED in covariant gauges.

The photon propagator in momentum space is

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon} \left(\eta_{\mu\nu} - (1 - \xi) \frac{k_\mu k_\nu}{k^2 + i\epsilon} \right).$$

This is the same propagator introduced in Chapter 5, now derived as the inverse of the gauge-fixed quadratic operator.

13.7 Gauge-Fixed Yang–Mills Path Integral

For a non-Abelian gauge field A_μ^a, the field strength is

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c.$$

The Yang–Mills action is

$$S_{\text{YM}} = -\frac{1}{4} \int \mathrm{d}^4x \, F_{\mu\nu}^a F^{a\mu\nu}.$$

A covariant gauge-fixing term is

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial_\mu A^{a\mu})^2.$$

The Faddeev–Popov ghost Lagrangian is

$$\mathcal{L}_{\text{gh}} = \bar{c}^a \, \partial^\mu D_\mu^{ab} c^b,$$

where

$$D_\mu^{ab} = \delta^{ab} \partial_\mu + g f^{acb} A_\mu^c$$

is the covariant derivative in the adjoint representation.

The full gauge-fixed path integral is schematically

$$Z[J] = \int \mathcal{D}A \mathcal{D}\bar{c} \mathcal{D}c \, e^{i \int \mathrm{d}^4x (\mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} + J \cdot A)}.$$

Unlike QED, the ghosts do not decouple. They interact with the gauge field.

13.8 First Introduction to BRST Symmetry

Gauge fixing seems to break gauge invariance because a particular gauge condition has been chosen. However, the gauge-fixed theory has a remnant symmetry called BRST symmetry. It mixes gauge fields, ghosts, and auxiliary fields.

At this stage, the details are less important than the role of the symmetry. BRST symmetry keeps track of the original gauge redundancy after gauge fixing. It ensures that physical observables are gauge independent and that unphysical states decouple from the physical spectrum.

The physical state space can be characterized in terms of BRST cohomology: physical states are BRST-closed, and states differing by BRST-exact terms are identified. This language will become more important in the later chapter on BRST symmetry.

13.9 Gauge Independence of Physical Observables

Intermediate quantities can depend on the gauge parameter ξ. Propagators, individual diagrams, and off-shell Green functions may change when ξ changes. Physical observables should not.

Examples of physical observables include S-matrix elements between physical states, cross sections, decay rates, and expectation values of gauge-invariant operators. Gauge independence of these quantities is a consistency requirement of the theory.

In perturbation theory, gauge independence may involve cancellations among many diagrams, including ghost diagrams. Therefore it is generally wrong to assign physical meaning to a single gauge-dependent diagram in isolation.

13.10 Meaning of Gauge Fixing in Quantum Theory

Gauge fixing is not the selection of a physically preferred gauge. It is a method for coordinatizing the space of physically distinct configurations. Just as one may choose coordinates on an ordinary manifold without changing the geometry, one may choose a gauge without changing physical predictions.

The path integral makes this especially clear. The original integral over A_μ includes redundant gauge copies. Gauge fixing removes the redundancy and supplies the correct determinant. Different gauges are different ways of performing the same physical integral.

This point is crucial for non-Abelian theories. Without gauge fixing and ghosts, the perturbative expansion would not correctly represent the quantum gauge system.

Summary

Gauge fields cannot be integrated naively because gauge-equivalent configurations are overcounted. Gauge fixing inserts a condition selecting one representative per gauge orbit, together with the Faddeev–Popov determinant. In Abelian covariant gauges the determinant decouples, while in non-Abelian gauge theory it becomes a ghost-field action. Ghosts are not physical particles; they are the functional-integral mechanism that preserves consistency of covariant quantization. BRST symmetry records the original gauge redundancy after gauge fixing.

13.11 Chapter Checklist

After reading this chapter, one should be able to:

- explain why the naive gauge-field path integral overcounts configurations;
- describe the role of a gauge-fixing condition;
- state the basic Faddeev–Popov identity;
- explain why determinants can be represented by ghost fields;
- distinguish ghosts from physical particles;
- write the gauge-fixed photon path integral;
- write the schematic gauge-fixed Yang–Mills path integral;
- explain why ghosts decouple in Abelian theory but not in non-Abelian theory;
- state the conceptual role of BRST symmetry;
- explain why physical observables must be gauge independent.

Chapter 14

ϕ^4 Theory as the First Interacting QFT

Remark

The previous chapters built the general machinery: canonical quantization, propagators, perturbation theory, scattering, and path integrals. The theory with interaction $\lambda\phi^4/4!$ is the simplest laboratory where all of these ideas become concrete.

14.1 Why Start with ϕ^4 Theory?

The free scalar field is exactly solvable because it is a collection of independent oscillators. To study genuine quantum field theory, we need an interaction. The simplest Lorentz-invariant self-interaction of a real scalar field in four dimensions is ϕ^4.

The theory is simple enough to compute with, but rich enough to display the main features of perturbative QFT:

- interaction vertices;
- scattering amplitudes;
- loop corrections;
- ultraviolet divergences;
- counterterms;
- renormalized masses and couplings.

For this reason, ϕ^4 theory is the standard first interacting quantum field theory.

14.2 The Lagrangian

The classical Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

The free part is

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2,$$

and the interaction part is

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4.$$

The factor 4! is conventional. It cancels the number of permutations of four identical scalar fields at a vertex, giving a simple Feynman rule.

The classical equation of motion is

$$(\Box + m^2)\phi + \frac{\lambda}{3!}\phi^3 = 0.$$

Unlike the free Klein–Gordon equation, this equation is nonlinear. The nonlinearity is the classical shadow of particle interactions in the quantum theory.

14.3 Dimensional Analysis

In four spacetime dimensions, the action is dimensionless in units $\hbar = c = 1$. Since

$$S = \int \mathrm{d}^4x \, \mathcal{L},$$

the Lagrangian density has mass dimension four. From the kinetic term one finds

$$[\phi] = 1.$$

Therefore

$$[\lambda] = 0.$$

The coupling λ is dimensionless in four dimensions. This is one reason ϕ^4 theory is important: it is power-counting renormalizable in four dimensions.

In dimensions other than four, the mass dimension of λ changes. This will become important in the discussion of renormalization and effective field theory.

14.4 Path-Integral Generating Functional

The generating functional is

$$Z[J] = \int \mathcal{D}\phi \, \exp\left(i \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 + J\phi\right]\right).$$

It can be written in terms of the free generating functional as

$$Z[J] = \exp\left[-i\frac{\lambda}{4!}\int \mathrm{d}^4x \left(\frac{1}{i}\frac{\delta}{\delta J(x)}\right)^4\right] Z_0[J].$$

Expanding the exponential in powers of λ generates perturbation theory. The free functional $Z_0[J]$ supplies propagators; the derivative operator supplies interaction vertices.

This is the functional-integral version of the Dyson expansion and Wick's theorem.

14.5 Feynman Rules in Position Space

The basic ingredients are simple. The free propagator is

$$D_F(x-y) = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}.$$

Each interaction vertex contributes

$$-i\lambda \int \mathrm{d}^4x.$$

Four scalar lines meet at each vertex. Correlation functions are obtained by summing over all contractions, or equivalently all diagrams, with the appropriate symmetry factors.

Position-space diagrams are useful conceptually because they show that each vertex is integrated over spacetime. Momentum-space diagrams are usually more convenient for scattering amplitudes because translation invariance turns those integrals into momentum-conserving delta functions.

14.6 Feynman Rules in Momentum Space

In momentum space, the propagator is

$$\frac{i}{k^2 - m^2 + i\epsilon}.$$

The four-point vertex contributes

$$-i\lambda.$$

At each vertex, four-momentum is conserved. For each undetermined internal momentum, one integrates

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}.$$

Finally, one includes the symmetry factor of the diagram.

These rules compute contributions to correlation functions and, after LSZ reduction, to scattering amplitudes.

14.7 Tree-Level $2 \to 2$ Scattering

The simplest nontrivial process is

$$\phi\phi \longrightarrow \phi\phi.$$

At lowest order there is one four-point vertex. The amplitude is

$$i\mathcal{M}_{\text{tree}} = -i\lambda, \qquad \mathcal{M}_{\text{tree}} = -\lambda.$$

This is a contact interaction. There is no internal propagator at this order.

The full matrix element contains the momentum-conserving delta function:

$$\langle p_3, p_4 | i\hat{T} | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4)\mathcal{M}_{\text{tree}}.$$

This is the same structure introduced in scattering theory, now obtained from a specific interacting Lagrangian.

14.8 First Loop Correction: The Tadpole

The two-point function receives a one-loop correction from the tadpole diagram. In momentum space, the loop integral has the form

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\epsilon}.$$

This integral diverges at large $|\ell|$. The divergence is ultraviolet: it comes from arbitrarily short-distance fluctuations.

The tadpole correction shifts the mass parameter. This is the first appearance of an important lesson: the parameters in the Lagrangian are not automatically the physical measured parameters. Quantum fluctuations correct them.

14.9 One-Loop Four-Point Correction

The four-point function receives one-loop corrections from diagrams with two vertices and two internal lines. A typical loop integral has the schematic form

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m^2 + i\epsilon} \frac{i}{(\ell + p)^2 - m^2 + i\epsilon}.$$

There are contributions in the s-, t-, and u-channels, corresponding to different ways of routing the external momenta through the loop.

These loop corrections modify the effective strength of the four-point interaction. In other words, they correct the coupling λ.

14.10 Ultraviolet Divergences

The loop integrals of ϕ^4 theory are typically divergent. This does not mean the theory is useless. It means that the parameters appearing in the Lagrangian must be related carefully to measured quantities.

A regulator is introduced to make integrals finite. Examples include a momentum cutoff Λ or dimensional regularization. A regulated divergent integral might depend on Λ, or contain poles in $4-d$ when computed in d dimensions.

The goal is not to keep the regulator as a physical object. The goal is to absorb regulator dependence into redefinitions of parameters and then express predictions in terms of finite measured quantities.

14.11 Counterterms

To renormalize the theory, write the Lagrangian in terms of renormalized parameters plus counterterms:

$$\mathcal{L} = \frac{1}{2}Z_\phi \partial_\mu\phi\partial^\mu\phi - \frac{1}{2}Z_m m^2\phi^2 - \frac{Z_\lambda \lambda}{4!}\phi^4.$$

Equivalently, one may write

$$\mathcal{L} = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}},$$

where

$$\mathcal{L}_{\text{ct}} = \frac{1}{2}\delta Z\, \partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta\lambda}{4!}\phi^4.$$

The counterterms are chosen to cancel the divergences from loop diagrams.

In ϕ^4 theory in four dimensions, the required counterterms have the same form as terms already present in the Lagrangian. This is the practical meaning of perturbative renormalizability.

14.12 Renormalized Parameters

The mass and coupling in the Lagrangian are not directly observable until a renormalization prescription is chosen. A renormalization condition defines the parameters in terms of physical or reference quantities.

For example, one may define the physical mass by the pole of the full propagator. The renormalized coupling may be defined by the value of the four-point amplitude at a chosen reference momentum scale.

Different schemes define parameters differently, but physical predictions agree when expressed in terms of measured quantities. The scheme dependence of intermediate parameters is the beginning of the renormalization group story.

14.13 Connected and One-Particle-Irreducible Functions

The generating functional $Z[J]$ produces all diagrams. The functional $W[J] = -i\ln Z[J]$ produces connected diagrams. The effective action $\Gamma[\phi_c]$ produces one-particle-irreducible diagrams.

In ϕ^4 theory:

- the connected two-point function gives the full propagator;
- the 1PI two-point function gives the self-energy;
- the 1PI four-point function gives the quantum-corrected vertex;
- counterterms are fixed by imposing conditions on these functions.

This is the language used systematically in renormalization.

14.14 Perturbative Expansion and Its Meaning

A typical observable is computed as a formal power series:

$$\mathcal{M} = \mathcal{M}^{(0)} + \lambda \mathcal{M}^{(1)} + \lambda^2 \mathcal{M}^{(2)} + \cdots .$$

The superscript does not always count loops directly, but higher powers of λ generally correspond to more vertices and more complicated diagrams.

Perturbation theory is useful when λ is small enough that the first few terms give a good approximation. It is not a complete nonperturbative definition of the theory. Nevertheless, it is the main calculational tool for weakly coupled QFT.

14.15 Physical Lessons from ϕ^4 Theory

The theory teaches several lessons that recur throughout QFT:

- interactions are represented by vertices;
- amplitudes are built from propagators and vertices;
- loop diagrams express virtual quantum fluctuations;
- loop integrals can diverge at high momentum;
- counterterms absorb divergences into parameter definitions;
- measured masses and couplings are renormalized quantities.

These ideas reappear in QED, non-Abelian gauge theory, the Standard Model, and effective field theory.

Summary

The ϕ^4 model is the first complete interacting QFT laboratory. Its single four-scalar vertex gives tree-level scattering. Its loop diagrams produce ultraviolet divergences that require regularization and counterterms. The theory introduces the distinction between bare and renormalized parameters and shows why physical predictions must be expressed in terms of measured quantities rather than unrenormalized Lagrangian parameters.

14.16 Chapter Checklist

After reading this chapter, one should be able to:

- write the ϕ^4 Lagrangian and identify free and interacting parts;
- explain why the factor 4! is included;

- derive the basic propagator and vertex Feynman rules;
- compute the tree-level $2 \to 2$ amplitude;
- identify the tadpole as a one-loop mass correction;
- describe the one-loop correction to the four-point function;
- explain what an ultraviolet divergence is;
- state why counterterms are introduced;
- distinguish bare and renormalized parameters;
- explain why ϕ^4 theory is the first useful laboratory for renormalization.

Chapter 15

Yukawa Theory

Remark

Yukawa theory is the simplest model in which a scalar field interacts with a fermion field. It introduces the basic scalar–fermion vertex, shows how scalar exchange generates an effective force, and provides a bridge between ϕ^4 theory and QED.

15.1 Why Yukawa Theory Comes Next

The ϕ^4 model introduced self-interactions of a scalar field. The next natural step is to let different kinds of fields interact. Yukawa theory couples a real scalar field ϕ to a Dirac field ψ. The basic interaction has the form

$$\bar{\psi}\psi\phi.$$

This is the simplest Lorentz-invariant coupling between a scalar and a fermion bilinear.

Historically, Yukawa introduced scalar exchange as a model for nuclear forces. In modern QFT, Yukawa interactions also appear in the Standard Model, where the Higgs field couples to fermions and gives them masses after electroweak symmetry breaking.

For us, Yukawa theory is useful because it combines several structures already introduced:

- scalar propagators;
- Dirac propagators;
- fermionic signs;
- vertices involving different fields;
- scattering and decay amplitudes.

15.2 The Lagrangian

A simple Yukawa theory has Lagrangian density

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m_\phi^2\phi^2 + \bar{\psi}(i\gamma^\mu\partial_\mu - m_\psi)\psi - g\phi\bar{\psi}\psi.$$

The parameter g is the Yukawa coupling. In four spacetime dimensions,

$$[\phi] = 1, \qquad [\psi] = \frac{3}{2}, \qquad [g] = 0.$$

Thus the Yukawa coupling is dimensionless in four dimensions, just like the ϕ^4 coupling.

The interaction term is

$$\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi.$$

Equivalently, in the interaction Hamiltonian one uses

$$\mathcal{H}_{\text{int}} = g\phi\bar{\psi}\psi,$$

up to the usual relation between Lagrangian and Hamiltonian conventions for nonderivative interactions.

15.3 Lorentz Structure of the Interaction

The scalar field ϕ is a Lorentz scalar. The bilinear

$$\bar{\psi}\psi$$

is also a Lorentz scalar. Therefore

$$\phi\bar{\psi}\psi$$

is Lorentz invariant.

This matters because not every product of spinor fields is a scalar. Other bilinears have different transformation properties:

$$\bar{\psi}\gamma^{\mu}\psi \quad \text{is a vector,} \qquad \bar{\psi}i\gamma^{5}\psi \quad \text{is a pseudoscalar.}$$

Yukawa theory in this chapter uses the scalar bilinear $\bar{\psi}\psi$. Later, pseudoscalar and chiral Yukawa couplings will appear in more advanced theories.

15.4 Feynman Rules

The free propagators are the scalar propagator

$$\frac{i}{k^2 - m_\phi^2 + i\epsilon}$$

and the Dirac propagator

$$\frac{i(\gamma^{\mu}p_{\mu} + m_\psi)}{p^2 - m_\psi^2 + i\epsilon}.$$

The Yukawa vertex connects one scalar line and two fermion lines. With the interaction convention above, the vertex factor is

$$-ig.$$

Fermion lines carry spinor indices and arrows indicating fermion-number flow. External fermions are represented by spinors $u_s(p)$, $\bar{u}_s(p)$, $v_s(p)$, and $\bar{v}_s(p)$, depending on whether the external particle is a fermion or antifermion and whether it is incoming or outgoing.

The basic diagrammatic fact is:

$$\text{one Yukawa vertex} = \text{one scalar leg plus two fermion legs.}$$

15.5 Scalar Decay into a Fermion Pair

If the scalar is heavy enough,

$$m_\phi > 2m_\psi,$$

it can decay into a fermion–antifermion pair:

$$\phi \longrightarrow \psi + \bar{\psi}.$$

At tree level, the amplitude is

$$i\mathcal{M} = -ig\,\bar{u}_s(p_1)v_r(p_2),$$

so

$$\mathcal{M} = -g\,\bar{u}_s(p_1)v_r(p_2).$$

The decay rate is obtained from

$$\mathrm{d}\Gamma = \frac{1}{2m_\phi}|\mathcal{M}|^2\,\mathrm{d}\Pi_2.$$

Spin sums convert products of spinors into traces of gamma matrices. This is the first example where spinor algebra directly enters a measurable rate.

15.6 Fermion Scattering by Scalar Exchange

Consider fermion–fermion scattering mediated by scalar exchange:

$$\psi\psi \longrightarrow \psi\psi.$$

At tree level, one contribution contains an internal scalar propagator. If the momentum transfer is q, the propagator contributes

$$\frac{i}{q^2 - m_\phi^2 + i\epsilon}.$$

The amplitude contains two Yukawa vertices and two fermion bilinears. A typical factor has the form

$$(-ig)^2 \frac{i}{q^2 - m_\phi^2 + i\epsilon} [\bar{u}(p_3)u(p_1)][\bar{u}(p_4)u(p_2)].$$

For identical fermions, one must also include the exchange diagram and the relative minus sign required by fermionic statistics.

This example illustrates how an exchanged field mediates an effective force between matter particles.

15.7 The Yukawa Potential

The low-energy, nonrelativistic limit of scalar exchange gives the Yukawa potential. In this limit, the energy transfer is small and the scalar propagator is approximately

$$\frac{-i}{\mathbf{q}^2 + m_\phi^2}.$$

The Fourier transform gives

$$V(r) = -\frac{g^2}{4\pi} \frac{e^{-m_\phi r}}{r}.$$

The range of the force is set by the inverse scalar mass:

$$R \sim \frac{1}{m_\phi}.$$

A massless exchanged scalar would give a long-range $1/r$ potential. A massive exchanged scalar gives a short-range force.

This result is a useful physical interpretation of a propagator: in an appropriate limit, exchange of a virtual quantum produces an effective potential.

15.8 Fermion Self-Energy

Yukawa theory also produces loop corrections. The fermion propagator receives a one-loop correction from emitting and reabsorbing a scalar. The corresponding self-energy has schematic form

$$\Sigma(p) \sim g^2 \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{\ell^2 - m_\phi^2 + i\epsilon} \frac{i(\gamma^\mu(p_\mu - \ell_\mu) + m_\psi)}{(p-\ell)^2 - m_\psi^2 + i\epsilon}.$$

This correction modifies the fermion propagator. The full propagator has the schematic form

$$S(p) = \frac{i}{\gamma^\mu p_\mu - m_\psi - \Sigma(p)}.$$

The pole of the full propagator determines the physical fermion mass. Thus loop corrections shift the mass parameter, just as in ϕ^4 theory.

15.9 Scalar Self-Energy

The scalar propagator receives a one-loop correction from a closed fermion loop. Schematically,

$$\Pi(p^2) \sim -g^2 \int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \operatorname{tr}\left[\frac{i(\gamma^\mu \ell_\mu + m_\psi)}{\ell^2 - m_\psi^2 + i\epsilon} \frac{i(\gamma^\mu(\ell_\mu + p_\mu) + m_\psi)}{(\ell + p)^2 - m_\psi^2 + i\epsilon}\right].$$

The minus sign is the closed-fermion-loop sign. The trace is over spinor indices.

This diagram illustrates a new feature compared with pure scalar theory: fermions can appear virtually in loops and modify bosonic propagation. In the path-integral language, the same effect is encoded in the fermion determinant.

15.10 Vertex Correction

The Yukawa vertex itself receives loop corrections. The full interaction is not just the tree-level coupling g; quantum fluctuations modify the effective vertex. The corrected vertex depends on external momenta and on the renormalization scale.

This leads to the need for a coupling counterterm. One writes

$$g_0 = Z_g g,$$

or equivalently introduces a counterterm interaction

$$-\delta g\, \phi\bar{\psi}\psi.$$

The renormalized coupling is fixed by a chosen renormalization condition, such as the value of a three-point function at a reference momentum configuration.

15.11 Counterterms and Renormalization

A renormalized Yukawa theory contains counterterms for both fields, both masses, and the coupling:

$$\mathcal{L}_{\text{ct}} = \frac{1}{2}\delta Z_\phi \partial_\mu\phi\partial^\mu\phi - \frac{1}{2}\delta m_\phi^2\phi^2 + \delta Z_\psi \bar{\psi} i\gamma^\mu\partial_\mu\psi - \delta m_\psi \bar{\psi}\psi - \delta g\, \phi\bar{\psi}\psi.$$

These counterterms cancel divergences in the scalar propagator, fermion propagator, and Yukawa vertex.

The structure is more complex than in ϕ^4 theory because there are now several kinds of fields and several parameters. Nevertheless, the principle is the same: divergences are absorbed into redefinitions of fields, masses, and couplings.

15.12 Symmetries

The simplest Yukawa theory with a real scalar and one Dirac fermion has limited symmetry. It preserves Lorentz invariance and, if ψ carries a conserved fermion number, the global transformation

$$\psi \longrightarrow e^{i\alpha}\psi, \qquad \bar{\psi} \longrightarrow \bar{\psi}e^{-i\alpha}$$

leaves $\bar{\psi}\psi$ invariant.

If the scalar is assigned transformation properties under a discrete symmetry, additional constraints are possible. For example, a transformation

$$\phi \longrightarrow -\phi$$

would forbid the term $\phi\bar{\psi}\psi$ unless the fermions also transform in a compensating way.

In chiral theories, Yukawa couplings can connect left- and right-handed fermions. This is the structural origin of fermion mass generation in the Standard Model.

15.13 Relation to the Standard Model

In the Standard Model, Yukawa couplings connect fermions to the Higgs field. After electroweak symmetry breaking, the Higgs field develops a vacuum expectation value. A Yukawa term then produces a fermion mass.

Schematically,

$$-y\bar{\psi}_L H \psi_R + \text{h.c.} \quad \longrightarrow \quad -m_f \bar{\psi}\psi - \frac{m_f}{v} h \bar{\psi}\psi.$$

Here h is the physical Higgs excitation and v is the Higgs vacuum expectation value. The same coupling that gives the fermion a mass also controls its interaction with the Higgs boson.

Thus the simple Yukawa model studied here is not only a toy model. It is the prototype for one of the central mechanisms of the Standard Model.

Summary

Yukawa theory couples a scalar field to a Dirac fermion through $g\phi\bar{\psi}\psi$. It introduces the scalar–fermion–fermion vertex, fermion spinor chains, scalar-mediated forces, and loop corrections involving both bosons and fermions. It is the simplest setting in which fermion self-energy, scalar self-energy, vertex renormalization, and closed-fermion-loop signs all appear together.

15.14 Chapter Checklist

After reading this chapter, one should be able to:

- write the Yukawa Lagrangian;
- explain why $\phi\bar{\psi}\psi$ is Lorentz invariant;
- state the scalar, fermion, and Yukawa-vertex Feynman rules;
- compute the structure of $\phi \to \psi\bar{\psi}$ at tree level;
- describe fermion scattering by scalar exchange;
- derive the qualitative Yukawa potential;
- identify fermion and scalar self-energy diagrams;
- explain the closed-fermion-loop minus sign;
- list the counterterms needed in Yukawa theory;
- explain how Yukawa couplings are related to fermion masses in the Standard Model.

Chapter 16

Quantum Electrodynamics

Remark

Quantum electrodynamics is the quantum field theory of charged Dirac matter and the electromagnetic field. It combines the Dirac field, gauge invariance, photons, fermion spinor chains, and perturbation theory into the first realistic relativistic quantum field theory of the course.

16.1 QED as a Gauge Theory

The free Dirac Lagrangian is

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

It is invariant under the global phase transformation

$$\psi \longrightarrow e^{i\alpha}\psi.$$

QED is obtained by promoting this symmetry to a local one,

$$\psi(x) \longrightarrow e^{i\alpha(x)}\psi(x).$$

Ordinary derivatives do not transform covariantly under a local phase rotation, so one introduces the covariant derivative

$$D_\mu = \partial_\mu + ieA_\mu.$$

The gauge field transforms as

$$A_\mu \longrightarrow A_\mu - \frac{1}{e}\partial_\mu \alpha.$$

The QED Lagrangian is

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi.$$

Expanding the covariant derivative gives

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu\psi A_\mu.$$

The last term is the interaction between charged matter and photons.

16.2 The Electromagnetic Current

The interaction can be written as

$$\mathcal{L}_{\text{int}} = -j^\mu A_\mu,$$

where

$$j^\mu = e\bar{\psi}\gamma^\mu\psi.$$

The current is conserved:

$$\partial_\mu j^\mu = 0,$$

using the Dirac equation. Current conservation is deeply connected to gauge invariance. In perturbation theory it leads to Ward identities, which enforce cancellations of unphysical photon polarizations.

The electric charge operator counts electrons and positrons with opposite signs. With the convention that the electron has charge $-e$, one may write schematically

$$\hat{Q}_{\mathrm{em}} = -e \sum_s \int \frac{\mathrm{d}^3 p}{(2\pi)^3} \left(\hat{b}_s^\dagger(\mathbf{p}) \hat{b}_s(\mathbf{p}) - \hat{d}_s^\dagger(\mathbf{p}) \hat{d}_s(\mathbf{p}) \right).$$

16.3 Gauge Fixing and the Photon Propagator

To define a photon propagator, one must fix a gauge. In covariant gauges, add

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi} (\partial_\mu A^\mu)^2.$$

The photon propagator is then

$$D_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon} \left(\eta_{\mu\nu} - (1-\xi) \frac{k_\mu k_\nu}{k^2 + i\epsilon} \right).$$

In Feynman gauge, $\xi = 1$, this becomes

$$D_{\mu\nu}(k) = \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The gauge parameter ξ may appear in intermediate expressions, but physical observables must be independent of it.

In Abelian QED, the Faddeev–Popov ghosts decouple in linear covariant gauges, so they do not appear in ordinary QED Feynman rules. This is special to Abelian gauge theory.

16.4 QED Feynman Rules

The basic free propagators are the electron propagator

$$\frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon}$$

and the photon propagator, for example in Feynman gauge,

$$\frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}.$$

The electron–photon vertex is

$$-ie\gamma^\mu.$$

Momentum is conserved at each vertex. External electron and positron lines are represented by spinors $u, \bar{u}, v, \bar{v}$. External photons are represented by polarization vectors $\epsilon_\mu(k)$.

The fine-structure constant is

$$\alpha = \frac{e^2}{4\pi}.$$

Perturbation theory in QED is an expansion in powers of e, or equivalently in powers of α for probabilities.

16.5 Spinor Chains

QED amplitudes contain spinor chains. A typical factor has the form

$$\bar{u}(p')\gamma^\mu u(p).$$

This is the matrix element of the electromagnetic current between electron states. For processes with internal fermion propagators, longer chains appear, for example

$$\bar{u}(p')\gamma^\mu \frac{i(\gamma^\rho q_\rho + m)}{q^2 - m^2 + i\epsilon}\gamma^\nu u(p).$$

Spinor chains are evaluated using gamma-matrix algebra and spin sums. For unpolarized scattering, one averages over initial spins and sums over final spins. The completeness relations are

$$\sum_s u_s(p)\bar{u}_s(p) = \gamma^\mu p_\mu + m,$$

and

$$\sum_s v_s(p)\bar{v}_s(p) = \gamma^\mu p_\mu - m.$$

These identities turn squared amplitudes into traces of gamma matrices.

16.6 Ward Identities

Gauge invariance implies Ward identities. The simplest tree-level statement is that replacing an external photon polarization vector by its momentum gives zero for a physical amplitude:

$$\epsilon_\mu(k) \longrightarrow k_\mu \qquad \Longrightarrow \qquad \mathcal{M} \longrightarrow 0.$$

This expresses the fact that unphysical longitudinal photon polarizations do not contribute to physical processes.

At the level of correlation functions, Ward identities relate current conservation to identities among Green functions. At loop level, they constrain renormalization constants and protect gauge invariance.

A practical lesson is that gauge invariance is not optional bookkeeping. It is a powerful consistency condition on amplitudes.

16.7 Electron–Muon Scattering

A clean tree-level QED process is

$$e^-\mu^- \longrightarrow e^-\mu^-.$$

At leading order, the particles exchange one virtual photon. If the momentum transfer is q, the amplitude in Feynman gauge is

$$i\mathcal{M} = [\bar{u}_e(p_3)(-ie\gamma^\mu)u_e(p_1)]\frac{-i\eta_{\mu\nu}}{q^2 + i\epsilon}[\bar{u}_\mu(p_4)(-ie\gamma^\nu)u_\mu(p_2)].$$

This example shows the basic QED structure:

$$\text{current} \times \text{photon propagator} \times \text{current}.$$

In the nonrelativistic limit, this exchange reproduces the Coulomb interaction. The photon propagator is therefore the quantum-field-theoretic origin of the electromagnetic force between charged particles.

16.8 Electron–Positron Annihilation

Another fundamental process is

$$e^- e^+ \longrightarrow \mu^- \mu^+.$$

At tree level, the electron and positron annihilate into a virtual photon, which then creates a muon pair. The amplitude has the form

$$i\mathcal{M} = [\bar{v}_e(p_2)(-ie\gamma^\mu)u_e(p_1)]\frac{-i\eta_{\mu\nu}}{q^2 + i\epsilon}[\bar{u}_\mu(p_3)(-ie\gamma^\nu)v_\mu(p_4)].$$

Here

$$q = p_1 + p_2 = p_3 + p_4.$$

This process is a prototype for collider physics: an initial particle–antiparticle pair produces an intermediate virtual gauge boson, which then decays into a new pair of particles.

16.9 Compton Scattering

Compton scattering is

$$e^- \gamma \longrightarrow e^- \gamma.$$

At tree level, two diagrams contribute. They correspond to the two possible orderings of photon absorption and emission along the electron line. The amplitude contains two electron propagators in different channels.

This example is important because gauge invariance requires the sum of both diagrams. Replacing an external photon polarization by its momentum gives a cancellation only after both diagrams are included. A single diagram by itself is not gauge invariant.

Compton scattering therefore illustrates a recurring lesson: physical amplitudes are often gauge invariant only after summing all diagrams required at a given order.

16.10 Pair Creation and Crossing

QED processes are related by crossing symmetry. A particle moved from the initial state to the final state is reinterpreted as an antiparticle with reversed momentum, and vice versa. For example, electron–positron annihilation and pair creation by a virtual photon are crossing-related.

Crossing symmetry is not a new Feynman rule. It follows from the analytic structure of relativistic amplitudes and from the fact that fields contain both particle annihilation and antiparticle creation operators.

This idea was already visible in the mode expansion of the Dirac field:

$$\hat{\psi} \sim \hat{b} + \hat{d}^\dagger.$$

QED makes this structure operational in scattering calculations.

16.11 Loop Corrections in QED

Beyond tree level, QED has several important loop corrections:

- electron self-energy corrections to the fermion propagator;
- photon self-energy corrections from fermion loops;
- vertex corrections to the electron–photon interaction.

These corrections are ultraviolet divergent and require renormalization. They also produce finite physical effects, such as anomalous magnetic moments and the running of the electromagnetic coupling.

The next chapter studies loop effects in QED in more detail. For now, the key point is that QED is not just tree-level photon exchange. Quantum fluctuations of electrons, positrons, and photons modify the parameters and predictions of the theory.

16.12 Renormalization in QED

The renormalized QED Lagrangian is written with field and parameter renormalization constants:

$$\mathcal{L} = -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} + Z_2 \bar{\psi} i \gamma^\mu \partial_\mu \psi - Z_m m \bar{\psi} \psi - Z_1 e \bar{\psi} \gamma^\mu \psi A_\mu.$$

Gauge invariance imposes relations among renormalization constants. In particular, Ward identities imply a relation between the vertex and fermion wave-function renormalization.

The physical electric charge is defined by a renormalization condition. At higher orders, the effective strength of the electromagnetic interaction depends on the energy scale. This scale dependence is the running of the coupling.

16.13 Infrared Issues

Because the photon is massless, QED also has infrared subtleties. Soft photons with very small energy can be emitted at little energy cost. As a result, exclusive amplitudes with a fixed number of photons can contain infrared divergences.

Physical detectors have finite energy resolution. Observable cross sections must include processes with unresolved soft photons. When this is done correctly, infrared divergences cancel between virtual and real soft-photon contributions.

This is one reason the particle interpretation in QED is more subtle than in a massive scalar theory. Charged particles are accompanied by long-range electromagnetic fields and soft photon clouds.

16.14 What QED Teaches

QED is the first fully realistic QFT in the course. It teaches several general lessons:

- local gauge symmetry determines the interaction;
- photons couple to conserved currents;
- amplitudes are built from spinor chains and photon propagators;
- gauge invariance appears as Ward identities;
- loop corrections require renormalization;
- massless gauge bosons create infrared subtleties.

These lessons reappear in non-Abelian gauge theory and the Standard Model.

Summary

QED describes charged Dirac matter interacting with the electromagnetic gauge field. Local $U(1)$ gauge invariance requires the covariant derivative and produces the vertex

$-ie\gamma^\mu$. Tree-level amplitudes are built from spinor currents and photon propagators. Ward identities enforce gauge invariance and remove unphysical photon polarizations. Loop corrections lead to renormalization and infrared effects, making QED the first realistic example of perturbative quantum field theory.

16.15 Chapter Checklist

After reading this chapter, one should be able to:

- derive QED from local $U(1)$ gauge invariance;
- write the QED Lagrangian and identify the interaction term;
- state the photon and electron propagators;
- state the QED vertex rule;
- construct spinor chains for simple amplitudes;
- explain the meaning of Ward identities;
- write the tree-level structure of electron–muon scattering;
- write the tree-level structure of electron–positron annihilation;
- explain why Compton scattering requires a sum of diagrams;
- describe the main loop corrections and renormalization issues in QED;
- explain why massless photons produce infrared subtleties.

Chapter 17

Loop Effects in QED

Remark

Tree-level QED describes photon exchange, annihilation, and emission. Loop corrections reveal the genuinely quantum structure of the theory: particles interact with their own fields, the vacuum is polarizable, charges run with scale, and infrared physics forces us to define inclusive observables.

17.1 Why Loop Effects Matter

The previous chapter introduced QED at tree level. Tree diagrams already explain many qualitative features of electromagnetic scattering, but they are not the whole theory. The perturbative expansion of QED contains loop diagrams, and these diagrams have two important roles.

First, loops produce finite measurable corrections. Examples include the anomalous magnetic moment of the electron and radiative corrections to scattering cross sections. Second, loops produce ultraviolet divergences. These are not ignored; they are absorbed into the definitions of physical masses, fields, and charges through renormalization.

The main one-loop structures in QED are

- the electron self-energy;
- the photon self-energy, also called vacuum polarization;
- the vertex correction;
- real and virtual soft-photon effects.

Together they show why QED is both highly predictive and conceptually subtle.

17.2 Electron Self-Energy

The electron self-energy is the correction to the electron propagator from the emission and reabsorption of a virtual photon. The one-loop diagram has an internal photon line and an internal electron line. Its contribution is written schematically as

$$-i\Sigma(p) = (-ie)^2 \int \frac{\mathrm{d}^4 k}{(2\pi)^4}\, \gamma^\mu \frac{i(\gamma^\rho(p_\rho - k_\rho) + m)}{(p-k)^2 - m^2 + i\epsilon} \gamma^\nu \frac{-i\eta_{\mu\nu}}{k^2 + i\epsilon}$$

in Feynman gauge.

The full electron propagator is then formally

$$S(p) = \frac{i}{\gamma^\mu p_\mu - m - \Sigma(p)}.$$

The self-energy shifts the location and residue of the propagator pole. The pole position is used to define the physical electron mass, while the residue is related to field-strength renormalization.

17.3 Mass and Wave-Function Renormalization

The bare mass appearing in the Lagrangian is not directly the measured mass. The measured mass is associated with the pole of the full propagator. One imposes a renormalization condition such as

$$\gamma^\mu p_\mu - m - \Sigma(p) = 0 \qquad \text{at} \qquad p^2 = m_{\text{phys}}^2.$$

This condition fixes the relation between the bare mass and the physical mass.

The residue of the pole defines the wave-function renormalization. Near the physical pole, the full propagator has the form

$$S(p) \sim \frac{iZ_2}{\gamma^\mu p_\mu - m_{\text{phys}}} + \text{regular terms}.$$

The constant Z_2 is the electron field-strength renormalization. It tells us how the field in the Lagrangian is normalized relative to the one-particle pole of the full propagator.

17.4 Vacuum Polarization

The photon self-energy arises from a fermion loop with two external photon legs. It is called vacuum polarization because it describes how virtual charged pairs modify the propagation of the electromagnetic field.

The one-loop tensor has the structure

$$i\Pi^{\mu\nu}(q) = -(-ie)^2 \int \frac{\mathrm{d}^4 k}{(2\pi)^4}\, \mathrm{tr}\left[\gamma^\mu \frac{i(\gamma^\rho k_\rho + m)}{k^2 - m^2 + i\epsilon} \gamma^\nu \frac{i(\gamma^\sigma (k_\sigma + q_\sigma) + m)}{(k+q)^2 - m^2 + i\epsilon}\right].$$

The initial minus sign is the closed-fermion-loop sign.

Gauge invariance constrains this tensor. It must be transverse:

$$q_\mu \Pi^{\mu\nu}(q) = 0.$$

Therefore it has the form

$$\Pi^{\mu\nu}(q) = (q^\mu q^\nu - q^2 \eta^{\mu\nu})\Pi(q^2).$$

This transversality is a direct consequence of current conservation and the Ward identity.

17.5 Charge Renormalization and the Running Coupling

Vacuum polarization modifies the photon propagator. In effect, the electric charge measured at one momentum scale differs from the charge measured at another scale. The physical picture is that the vacuum contains virtual charged pairs that screen or antiscreen charges depending on the theory.

In QED, the effective electromagnetic coupling increases slowly with energy. One writes a scale-dependent coupling

$$\alpha(\mu) = \frac{e(\mu)^2}{4\pi}.$$

At leading order with one charged fermion, the beta function has the form

$$\beta(e) = \mu \frac{\mathrm{d}e}{\mathrm{d}\mu} = \frac{e^3}{12\pi^2} + \cdots.$$

The positive sign means that the coupling grows at shorter distances.

This running is usually tiny at ordinary energies, but conceptually it is one of the central discoveries of renormalized QFT: coupling constants depend on the scale at which they are defined.

17.6 Vertex Correction

The QED vertex also receives a one-loop correction. Instead of the tree-level factor

$$-ie\gamma^\mu,$$

the full vertex has the form

$$-ie\Gamma^\mu(p', p),$$

where

$$\Gamma^\mu(p', p) = \gamma^\mu + \Lambda^\mu(p', p).$$

Here Λ^μ denotes loop corrections.

The most general on-shell vertex for a spin-1/2 particle can be decomposed into form factors:

$$\bar{u}(p')\Gamma^\mu u(p) = \bar{u}(p') \left[\gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} F_2(q^2) \right] u(p),$$

where $q = p' - p$. The function F_1 is related to electric charge, while $F_2(0)$ gives the anomalous magnetic moment.

17.7 The Anomalous Magnetic Moment

For a Dirac particle at tree level, the gyromagnetic factor is

$$g = 2.$$

Loop corrections modify this. The anomalous magnetic moment is defined by

$$a_e = \frac{g-2}{2}.$$

At one loop in QED,

$$a_e = \frac{\alpha}{2\pi} + \cdots .$$

This finite prediction is one of the most famous successes of perturbative QFT. It comes from the vertex correction after renormalization.

The important conceptual point is that loops are not merely divergent nuisances. They also contain finite physical effects that can be measured with extraordinary precision.

17.8 Ward Identity

Gauge invariance relates the electron self-energy and the vertex correction. The Ward–Takahashi identity is

$$q_\mu \Gamma^\mu(p+q, p) = S^{-1}(p+q) - S^{-1}(p).$$

In the limit $q \to 0$, this implies a relation between the vertex renormalization and the electron wave-function renormalization. In common notation,

$$Z_1 = Z_2.$$

Here Z_1 renormalizes the electron–photon vertex, while Z_2 renormalizes the electron field.

This relation is a powerful consequence of gauge invariance. It ensures that charge renormalization is controlled by the photon field renormalization rather than by independent vertex and electron-field effects.

17.9 Ultraviolet and Infrared Divergences

QED loop integrals can diverge in two different regimes. Ultraviolet divergences come from large loop momentum:

$$|k| \to \infty.$$

They reflect sensitivity to short-distance physics and are handled by renormalization.

Infrared divergences come from small photon momentum:

$$|k| \to 0.$$

They occur because the photon is massless. Soft virtual photons can produce divergent loop corrections, while real photons of arbitrarily low energy can be emitted without being experimentally resolved.

The physical treatment of these two kinds of divergences is different. Ultraviolet divergences are absorbed into parameters. Infrared divergences cancel only after defining observables inclusively enough.

17.10 Soft Photon Emission

Consider a process involving charged particles. In addition to the process with no extra photons, there is a process with an additional soft photon:

$$\text{charged process} \longrightarrow \text{same process} + \gamma_{\text{soft}}.$$

If the photon energy is below detector resolution, the two outcomes cannot be experimentally distinguished.

The amplitude for soft photon emission factorizes in a universal way. For external charged particles with momenta p_i, the soft factor has the form

$$e \sum_i \eta_i \frac{p_i \cdot \epsilon(k)}{p_i \cdot k},$$

where η_i depends on whether the charged particle is incoming or outgoing and on the sign of its charge.

The denominator $p_i \cdot k$ becomes singular as the photon momentum k goes to zero. This is the source of soft divergences in real emission.

17.11 Inclusive Observables

Virtual soft-photon loops and real soft-photon emission produce divergences with opposite signs. Physical cross sections must include all experimentally indistinguishable final states. Therefore one computes inclusive quantities such as

$$\sigma_{\text{inclusive}} = \sigma_0 + \sigma_{\text{virtual}} + \sigma_{\text{soft real}} + \cdots .$$

The infrared divergences cancel in this inclusive sum.

This cancellation is an example of a general principle: the mathematical object that is finite is not always the exclusive amplitude for a fixed number of massless particles. It is the observable that matches what a detector can actually distinguish.

17.12 Physical Meaning of Radiative Corrections

Radiative corrections change how charged particles interact. They shift masses, renormalize fields, modify the effective charge, correct magnetic moments, and alter cross sections. They also teach us that the vacuum is not empty in a classical sense: it supports virtual fluctuations that affect observable processes.

In QED these effects are small because α is small at low energies. This makes QED one of the most precise theories in physics. The same logic applies in other quantum field theories, although the coupling may be larger and the calculations more difficult.

Summary

Loop effects in QED include electron self-energy, vacuum polarization, and vertex corrections. They produce ultraviolet divergences that require renormalization and finite effects such as the anomalous magnetic moment. Gauge invariance gives Ward identities, including the relation $Z_1 = Z_2$. Because the photon is massless, QED also has infrared divergences, which cancel only in inclusive observables that include unresolved soft radiation.

17.13 Chapter Checklist

After reading this chapter, one should be able to:

- identify the main one-loop diagrams in QED;
- explain electron self-energy and its relation to mass renormalization;
- explain vacuum polarization and transversality;
- describe charge renormalization and the running coupling;
- write the form-factor decomposition of the QED vertex;
- state the one-loop result $a_e = \alpha/(2\pi) + \cdots$;
- state the Ward–Takahashi identity and the implication $Z_1 = Z_2$;
- distinguish ultraviolet and infrared divergences;
- explain why real soft-photon emission must be included;
- explain why inclusive observables are infrared finite.

Chapter 18

Regularization

Remark

Loop integrals in QFT are often divergent before they are given a precise meaning. Regularization is the temporary procedure that makes such expressions well defined. It does not by itself solve the physical problem; it prepares the calculation for renormalization.

18.1 Why Divergent Integrals Appear

Loop diagrams require integration over internal momenta. A typical one-loop integral has the form

$$\int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{k^2 - m^2 + i\epsilon}.$$

At large $|k|$, this behaves too slowly to give a finite answer. The integral is ultraviolet divergent. Other integrals may diverge at small momentum, giving infrared divergences.

Divergences arise because field theory has degrees of freedom at arbitrarily short distances and, for massless particles, arbitrarily long distances. A loop integral sums over all virtual momenta. Without further definition, this sum may not exist as an ordinary integral.

Regularization modifies the calculation so that every intermediate expression is finite or at least well defined. Only then can one isolate divergent pieces and renormalize them.

18.2 Regularization Versus Renormalization

Regularization and renormalization are related but distinct.

Regularization answers the question:

How do we make divergent expressions mathematically well defined?

Renormalization answers the question:

How do we express predictions in terms of finite measured parameters?

A regulator is temporary. Physical predictions should not depend on the details of the regulator after renormalization. If they did, the regulator would be a new physical assumption rather than a calculational device.

A good regulator should preserve as many symmetries as possible. In gauge theories, this is especially important because a regulator that violates gauge invariance can obscure Ward identities and require additional care.

18.3 Momentum Cutoff

The most intuitive regulator is a momentum cutoff. One restricts loop momenta to satisfy

$$|k| < \Lambda.$$

For example,

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4} f(k) \quad \longrightarrow \quad \int_{|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4} f(k).$$

The cutoff Λ represents a maximum momentum or minimum distance scale. Divergences then appear as powers or logarithms of Λ, such as

$$\Lambda^2, \qquad \ln\Lambda.$$

The advantage of a cutoff is physical transparency. The disadvantage is that a simple cutoff can break Lorentz invariance or gauge invariance unless it is implemented carefully.

18.4 Power Divergences and Logarithmic Divergences

A cutoff makes the degree of divergence visible. An integral may behave as

$$I(\Lambda) \sim \Lambda^2$$

for a quadratic divergence, or

$$I(\Lambda) \sim \ln\Lambda$$

for a logarithmic divergence.

Power divergences are sensitive to the highest momentum scales. Logarithmic divergences are milder but especially important because they lead directly to scale dependence and renormalization group running.

For example, a logarithm often appears in the form

$$\ln\frac{\Lambda}{\mu},$$

where μ is a reference scale. After renormalization, dependence on μ becomes the scale dependence of renormalized parameters.

18.5 Pauli–Villars Regularization

Pauli–Villars regularization modifies propagators by subtracting fictitious heavy fields. A scalar propagator may be replaced schematically by

$$\frac{i}{k^2 - m^2 + i\epsilon} \quad \longrightarrow \quad \frac{i}{k^2 - m^2 + i\epsilon} - \frac{i}{k^2 - M^2 + i\epsilon},$$

where M is a large regulator mass. At large k, the two terms cancel the leading ultraviolet behavior.

The regulator field is not physical. One takes

$$M \to \infty$$

after renormalization. Pauli–Villars is useful conceptually because it shows how high-momentum behavior can be softened while preserving Lorentz covariance.

However, Pauli–Villars can be awkward in non-Abelian gauge theories and chiral theories. It is no longer the most common regulator for modern perturbative calculations.

18.6 Dimensional Regularization

Dimensional regularization changes the number of spacetime dimensions from four to

$$d = 4 - \epsilon.$$

Loop integrals are evaluated as analytic functions of d. Divergences appear as poles in ϵ:

$$\frac{1}{\epsilon}.$$

A typical integral becomes

$$\int \frac{\mathrm{d}^4 k}{(2\pi)^4} \quad \longrightarrow \quad \mu^{4-d} \int \frac{\mathrm{d}^d k}{(2\pi)^d}.$$

The scale μ is introduced so that couplings keep their usual mass dimensions.

Dimensional regularization is widely used because it preserves Lorentz invariance and is especially compatible with gauge invariance. It also sets many power divergences to zero, leaving logarithmic divergences as poles in ϵ.

18.7 A Standard Dimensionally Regularized Integral

A common Euclidean-type integral has the form

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{(k^2 + \Delta)^\alpha} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(\alpha - d/2)}{\Gamma(\alpha)} \Delta^{d/2-\alpha}.$$

Divergences appear when the gamma function has a pole. Near $d = 4$, one often finds terms such as

$$\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) - \ln \frac{\Delta}{\mu^2}.$$

Different subtraction schemes decide which combination of constants is removed.

The most common schemes are minimal subtraction, abbreviated MS, and modified minimal subtraction, abbreviated $\overline{\mathrm{MS}}$. In $\overline{\mathrm{MS}}$, one subtracts not only $1/\epsilon$ but also the standard constants $-\gamma_E + \ln 4\pi$.

18.8 Feynman Parameters

Loop integrals often contain products of denominators. Feynman parameters combine them. The basic identity is

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x \, \frac{1}{[xA + (1-x)B]^2}.$$

More generally,

$$\frac{1}{A_1 \cdots A_n} = (n-1)! \int_0^1 \mathrm{d}x_1 \cdots \mathrm{d}x_n \, \delta\left(1 - \sum_i x_i\right) \frac{1}{(x_1 A_1 + \cdots + x_n A_n)^n}.$$

After combining denominators, one shifts the loop momentum and evaluates a standard Gaussian-type integral.

Feynman parameters are not a regulator by themselves. They are an algebraic tool that makes regulated loop integrals easier to compute.

18.9 Infrared Regularization

Not all divergences are ultraviolet. In theories with massless particles, integrals may diverge when loop momentum becomes small. To regulate infrared divergences, one may introduce a small photon mass m_γ, work in $d \neq 4$ dimensions, or impose an energy resolution for soft radiation.

For example, a soft divergence may appear as

$$\ln m_\gamma$$

if a small photon mass is used. This mass is not physical; it is only an infrared regulator. Physical inclusive observables must be finite when

$$m_\gamma \to 0.$$

Ultraviolet and infrared divergences can appear in the same calculation, but they have different physical meanings and are removed in different ways.

18.10 Regulators and Symmetries

A regulator can preserve or violate symmetries. This matters. If a regulator breaks a symmetry that the quantum theory should preserve, additional work is needed to restore the symmetry in renormalized quantities.

Dimensional regularization is popular because it usually preserves gauge invariance. A hard momentum cutoff is intuitive but may break gauge invariance if applied naively. Pauli–Villars can preserve some symmetries but may struggle with chiral gauge theories.

Sometimes a symmetry cannot be preserved by any regulator. Then the symmetry is anomalous. Anomalies will be discussed later. For now, the lesson is that the choice of regulator is not only technical; it can reveal or obscure important symmetry structure.

18.11 Subtraction Schemes

After regularizing, one subtracts divergent pieces. A subtraction scheme defines what is removed and what finite parts are included in the definition of renormalized parameters.

In minimal subtraction, one removes only pole terms such as

$$\frac{1}{\epsilon}.$$

In $\overline{\mathrm{MS}}$, one removes

$$\frac{1}{\epsilon} - \gamma_E + \ln 4\pi.$$

In an on-shell scheme, parameters are fixed by physical pole masses and physical charges. Different schemes give different numerical values for renormalized parameters at intermediate stages, but physical predictions agree when computed consistently.

Scheme dependence is not a contradiction. It reflects the freedom to choose how finite pieces are assigned between loop corrections and parameter definitions.

18.12 What Regularization Does Not Do

Regularization does not make a theory predictive by itself. A cutoff can make an integral finite, but the result may depend on the cutoff. Predictivity comes only after one shows that cutoff dependence can be absorbed into a finite number of measured parameters, or else interprets the theory as an effective field theory with cutoff-suppressed corrections.

Regularization also does not mean that infinities are physical. Divergent terms usually indicate that the parameters in the Lagrangian are not yet related to physical observables. Renormalization provides that relation.

Thus the correct sequence is

$$\text{divergent expression} \longrightarrow \text{regularized expression} \longrightarrow \text{renormalized prediction.}$$

Summary

Regularization is the temporary step that makes divergent loop integrals well defined. Momentum cutoffs are intuitive but can obscure symmetries. Pauli–Villars softens high-momentum behavior with heavy regulator fields. Dimensional regularization evaluates integrals in $d = 4 - \epsilon$ dimensions and represents divergences as poles in ϵ. Subtraction schemes define which pieces are absorbed into renormalized parameters. Physical predictions should not depend on the regulator.

18.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain why loop integrals diverge;
- distinguish regularization from renormalization;
- describe a momentum cutoff;
- distinguish power and logarithmic divergences;
- explain Pauli–Villars regularization;
- explain dimensional regularization and the role of $d = 4 - \epsilon$;
- use the basic idea of Feynman parameters;
- distinguish ultraviolet and infrared regularization;
- explain why preserving symmetries matters;
- distinguish MS, $\overline{\text{MS}}$, and on-shell subtraction ideas;
- state why a regulator is not itself a physical prediction.

Chapter 19

Renormalization

Remark

Regularization makes divergent expressions well defined. Renormalization turns regularized expressions into finite predictions by relating the parameters in the Lagrangian to measured quantities. It is not just a trick for removing infinities; it is the statement that physics depends on scale.

19.1 The Basic Problem

Perturbative loop calculations produce divergent expressions. After introducing a regulator, these expressions depend on an auxiliary scale or cutoff. For example, a one-loop correction may contain

$$\ln \frac{\Lambda}{m}$$

with a cutoff regulator, or

$$\frac{1}{\epsilon}$$

in dimensional regularization.

A physical prediction cannot depend on an arbitrary regulator. The way out is to recognize that the parameters in the original Lagrangian are not directly the measured parameters. Bare masses, fields, and couplings must be related to renormalized quantities.

Renormalization is the systematic procedure for making this relation precise.

19.2 Bare and Renormalized Quantities

Start with a scalar theory. The bare Lagrangian may be written as

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu \phi_0 \partial^\mu \phi_0 - \frac{1}{2}m_0^2\phi_0^2 - \frac{\lambda_0}{4!}\phi_0^4.$$

The subscript 0 denotes bare quantities. Introduce renormalized quantities by

$$\phi_0 = Z_\phi^{1/2}\phi, \qquad m_0^2 = m^2 + \delta m^2, \qquad \lambda_0 = \lambda + \delta\lambda$$

with the precise form depending on convention.

The Lagrangian is rewritten as

$$\mathcal{L}_0 = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}}.$$

The renormalized part contains finite parameters m, λ. The counterterm part contains $\delta Z_\phi, \delta m^2, \delta\lambda$, chosen to cancel loop divergences.

19.3 Counterterm Lagrangian

For ϕ^4 theory, a common counterterm Lagrangian is

$$\mathcal{L}_{\text{ct}} = \frac{1}{2}\delta Z_\phi \partial_\mu \phi \partial^\mu \phi - \frac{1}{2}\delta m^2 \phi^2 - \frac{\delta\lambda}{4!}\phi^4.$$

Each counterterm gives an additional Feynman rule. For example:

- δm^2 gives a two-point insertion;
- δZ_ϕ gives a momentum-dependent two-point insertion;
- $\delta\lambda$ gives a four-point counterterm vertex.

The counterterm diagrams are included together with loop diagrams. Their role is to cancel regulator-dependent divergent pieces.

This may look artificial at first, but it expresses a physical fact: the parameters we write in a Lagrangian must be calibrated by measurements.

19.4 Renormalization Conditions

Counterterms are fixed by renormalization conditions. In an on-shell scheme for a scalar field, one may require that the full propagator has a pole at the physical mass:

$$p^2 = m_{\text{phys}}^2,$$

and that the residue of the pole is one. These conditions define the physical mass and the normalization of the field.

For the coupling, one may impose a condition on the four-point amplitude at a chosen kinematic point. Schematically,

$$\mathcal{M}(s_0, t_0, u_0) = -\lambda_{\text{ren}}.$$

This equation defines the renormalized coupling at that reference point.

Different renormalization schemes impose different conditions. The numerical values of renormalized parameters may differ from scheme to scheme, but physical predictions agree when expressed consistently in terms of measured inputs.

19.5 Two-Point Function and Self-Energy

The full scalar propagator can be written as

$$G(p) = \frac{i}{p^2 - m^2 - \Pi(p^2) + i\epsilon},$$

where $\Pi(p^2)$ is the self-energy. The pole of the propagator is determined by

$$p^2 - m^2 - \Pi(p^2) = 0.$$

The self-energy contains loop contributions and counterterm contributions. Renormalization conditions fix the counterterms so that the pole occurs at the chosen physical mass and with the chosen residue.

In perturbation theory, one expands Π order by order. At each order, divergences are canceled by the counterterms allowed by the Lagrangian.

19.6 Four-Point Function and Coupling Renormalization

The four-point function determines the interaction strength. In ϕ^4 theory, the tree-level amplitude is

$$\mathcal{M}_{\text{tree}} = -\lambda.$$

Loop corrections modify this amplitude. The renormalized four-point function has schematic form

$$\mathcal{M} = -\lambda + \mathcal{M}_{\text{loop}} + \mathcal{M}_{\text{ct}}.$$

The loop contribution is divergent before renormalization. The counterterm contribution $\mathcal{M}_{\text{ct}}$ cancels the divergent part and fixes how λ is related to a measured scattering amplitude.

This is coupling renormalization. It is the reason the coupling appearing in the Lagrangian must be associated with a scale or a physical renormalization condition.

19.7 Field-Strength Renormalization

The field-strength renormalization Z_ϕ is fixed by the residue of the full propagator. Near the physical pole,

$$G(p) \sim \frac{iZ}{p^2 - m_{\text{phys}}^2 + i\epsilon}.$$

In an on-shell scheme, one often chooses the renormalized field so that

$$Z = 1.$$

This condition determines δZ_ϕ.

Field-strength renormalization affects external legs in scattering amplitudes. In LSZ reduction, the residue of the pole controls the normalization of one-particle states. Thus Z_ϕ is not merely a formal coefficient; it is connected to how fields create physical particles from the vacuum.

19.8 Minimal Subtraction

In minimal subtraction, counterterms are chosen to remove only divergent pole terms in dimensional regularization. For example, a counterterm may contain

$$\delta\lambda = \frac{a_1(\lambda)}{\epsilon} + \frac{a_2(\lambda)}{\epsilon^2} + \cdots .$$

In the $\overline{\text{MS}}$ scheme, one also subtracts the standard combination

$$-\gamma_E + \ln 4\pi.$$

Minimal subtraction schemes are not defined directly by physical pole masses or scattering amplitudes. Instead, they define parameters at a renormalization scale μ.

This makes them especially convenient for high-energy calculations and for the renormalization group.

19.9 Renormalization Scale

Dimensional regularization introduces a scale μ. Renormalized parameters therefore depend on μ:

$$m = m(\mu), \qquad \lambda = \lambda(\mu), \qquad e = e(\mu).$$

Physical quantities cannot depend on the arbitrary scale μ. Therefore the explicit μ-dependence of loop corrections must cancel the implicit μ-dependence of renormalized parameters.

This requirement leads to renormalization group equations. For a coupling $g(\mu)$, one defines

$$\beta(g) = \mu \frac{\mathrm{d}g}{\mathrm{d}\mu}.$$

The beta function tells us how the coupling changes with scale. The next chapter will develop this idea systematically.

19.10 Renormalizable and Nonrenormalizable Interactions

A theory is perturbatively renormalizable if all divergences can be absorbed into a finite number of counterterms already present in the Lagrangian. In four dimensions, ϕ^4, Yukawa theory, and QED are renormalizable in this sense.

Interactions with couplings of negative mass dimension generate infinitely many new counterterms at higher orders. Such theories are called nonrenormalizable in the traditional sense. This does not mean they are useless. It means they should be interpreted as effective field theories valid below some scale.

The modern viewpoint is that renormalizability is not simply a yes-or-no mark of validity. It tells us how predictive a theory is at a given energy scale and how many parameters are needed to reach a chosen accuracy.

19.11 Physical Predictions

A renormalized calculation follows a clear pattern:

1. choose a regulator;
2. compute loop diagrams and counterterm diagrams;
3. impose renormalization conditions or choose a subtraction scheme;
4. express bare parameters in terms of renormalized parameters;
5. compute observables in terms of measured inputs.

The final observable must be finite and independent of the regulator.

For example, a scattering cross section should not depend on $1/\epsilon$ or on a cutoff Λ. It may depend on physical masses, physical charges, external momenta, and the chosen renormalization scale in a way compensated by running parameters.

19.12 What Renormalization Means Physically

Renormalization teaches that parameters are scale-dependent summaries of physics. A mass or coupling in a Lagrangian is not an isolated number with meaning at all scales. It is defined by a measurement procedure or a subtraction convention.

Short-distance fluctuations affect long-distance parameters. Instead of trying to remove this fact, renormalization organizes it. The same theory can be described at different scales with different parameter values while predicting the same physical observables.

This is why renormalization is central to modern QFT. It is not merely a repair of infinities; it is a theory of how physics changes with scale.

Summary

Renormalization relates bare Lagrangian parameters to finite measured quantities. Loop divergences are canceled by counterterms, whose finite parts are fixed by a renormalization scheme or physical conditions. Two-point functions determine mass and field-strength renormalization, while four-point functions determine coupling renormalization. The dependence of renormalized parameters on the scale μ leads to the renormalization group.

19.13 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish bare and renormalized quantities;
- write a counterterm Lagrangian for ϕ^4 theory;
- explain how renormalization conditions fix counterterms;
- relate the two-point function to mass renormalization;
- relate the pole residue to field-strength renormalization;
- explain coupling renormalization through the four-point function;
- distinguish on-shell and minimal subtraction schemes;
- explain the role of the renormalization scale μ;
- define the beta function qualitatively;
- distinguish renormalizable interactions from effective interactions;
- explain why renormalization is about scale dependence, not only about removing infinities.

Chapter 20

Renormalization Group

Remark

Renormalization introduced a scale μ. The renormalization group explains how physics remains independent of this arbitrary scale while renormalized parameters run with it. It is the language of scale dependence in quantum field theory.

20.1 Why a Group of Scale Transformations?

In the previous chapter, renormalized parameters were defined at a scale μ. A coupling written as $g(\mu)$ is not a fixed number independent of context. It depends on the scale at which it is measured or defined.

This does not mean that physical predictions depend on an arbitrary convention. A physical observable $\mathcal{O}$ may depend explicitly on μ through loop logarithms, and implicitly on μ through the running parameters:

$$\mathcal{O} = \mathcal{O}(p_i, m(\mu), g(\mu), \mu).$$

The total dependence on μ must cancel:

$$\mu \frac{\mathrm{d}}{\mathrm{d}\mu} \mathcal{O} = 0.$$

The renormalization group is the formalism that organizes this cancellation.

The word "group" reflects the fact that changing from one scale to another can be composed: running from μ_1 to μ_2, and then from μ_2 to μ_3, is equivalent to running directly from μ_1 to μ_3.

20.2 Running Couplings

A running coupling satisfies an equation of the form

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = \beta(g).$$

The function $\beta(g)$ is the beta function. It is determined by quantum loop corrections and depends on the theory and renormalization scheme.

If $\beta(g) > 0$, the coupling grows as the scale increases. If $\beta(g) < 0$, the coupling decreases as the scale increases. The sign and zeros of the beta function therefore encode important physical information.

For QED with one charged fermion, the leading beta function is

$$\beta(e) = \frac{e^3}{12\pi^2} + \cdots .$$

The positive sign means that the effective electric charge grows slowly at short distances.

20.3 Solving a Simple RG Equation

Suppose the beta function has the form

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = bg^3.$$

Then

$$\frac{\mathrm{d}g}{g^3} = b\frac{\mathrm{d}\mu}{\mu}.$$

Integrating between μ_0 and μ gives

$$\frac{1}{g(\mu)^2} = \frac{1}{g(\mu_0)^2} - 2b\ln\frac{\mu}{\mu_0}.$$

This equation shows how logarithms are reorganized into scale-dependent couplings.

If $b > 0$, the coupling grows with μ. If $b < 0$, the coupling becomes weaker at high energies. The latter behavior is called asymptotic freedom.

20.4 The Callan–Symanzik Equation

Correlation functions also satisfy renormalization group equations. For an n-point renormalized correlation function $G^{(n)}$, the Callan–Symanzik equation has the schematic form

$$\left[\mu\frac{\partial}{\partial\mu} + \beta(g)\frac{\partial}{\partial g} + m\gamma_m(g)\frac{\partial}{\partial m} + n\gamma_\phi(g)\right] G^{(n)} = 0.$$

Here γ_ϕ is the anomalous dimension of the field, and γ_m is the anomalous dimension of the mass.

The equation says that explicit scale dependence is compensated by the running of couplings, masses, and field normalizations. It is a precise version of the statement that physics does not depend on the arbitrary renormalization scale.

20.5 Anomalous Dimensions

Classically, a field has an engineering dimension determined by dimensional analysis. For a scalar field in four dimensions,

$$[\phi]_{\text{classical}} = 1.$$

Quantum corrections can modify scaling behavior. The correction is called an anomalous dimension.

If Z_ϕ is the field-strength renormalization, the anomalous dimension is schematically

$$\gamma_\phi(g) = \frac{1}{2}\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\ln Z_\phi.$$

The full scaling dimension is then

$$\Delta_\phi = \Delta_{\text{classical}} + \gamma_\phi.$$

Anomalous dimensions are especially important near fixed points, where they give universal critical exponents.

20.6 Fixed Points

A fixed point is a value g_* for which

$$\beta(g_*) = 0.$$

At a fixed point, the coupling no longer runs. The theory may become scale invariant, and often conformally invariant under additional assumptions.

The Gaussian fixed point is

$$g_* = 0.$$

It describes a free theory. Interacting fixed points, if they exist, describe scale-invariant interacting quantum field theories.

The behavior near a fixed point is determined by linearizing the beta function:

$$\beta(g) \approx \beta'(g_*)(g - g_*).$$

If perturbations grow as one moves to low energies, they are relevant. If they shrink, they are irrelevant. If they remain of comparable size, they are marginal.

20.7 Relevant, Irrelevant, and Marginal Operators

Operators are classified by how their couplings scale. In four dimensions, an operator $\mathcal{O}$ of dimension Δ appears as

$$\mathcal{L} \supset c\,\mathcal{O}.$$

The coupling has dimension

$$[c] = 4 - \Delta.$$

If $\Delta < 4$, the operator is relevant. If $\Delta = 4$, it is marginal. If $\Delta > 4$, it is irrelevant.

Relevant operators become more important in the infrared. Irrelevant operators are suppressed at low energies by powers of a high scale. Marginal operators require quantum analysis: their beta functions decide whether they are marginally relevant, marginally irrelevant, or exactly marginal.

This classification is the bridge between renormalization group and effective field theory.

20.8 Dimensional Transmutation

A theory may begin with a dimensionless coupling but generate a physical mass scale through running. This phenomenon is called dimensional transmutation.

Suppose

$$\mu \frac{\mathrm{d}g}{\mathrm{d}\mu} = -bg^3, \qquad b > 0.$$

Then the running coupling can be written in terms of a scale Λ:

$$g(\mu)^2 \sim \frac{1}{2b \ln(\mu/\Lambda)}.$$

The dimensionless coupling has been exchanged for a dimensionful scale Λ. This idea is central in non-Abelian gauge theories and QCD, where a strong-coupling scale emerges from a classically scale-invariant gauge coupling.

20.9 Asymptotic Freedom

A theory is asymptotically free if its coupling becomes small at high energies:

$$g(\mu) \longrightarrow 0 \qquad \text{as} \qquad \mu \to \infty.$$

This occurs when the beta function is negative at small coupling:

$$\beta(g) = -bg^3 + \cdots, \qquad b > 0.$$

Non-Abelian gauge theories can have this property. It is the reason quarks and gluons behave approximately as weakly interacting particles at very short distances, even though they are confined at long distances.

QED is not asymptotically free. Its coupling grows slowly at short distances. This contrast between Abelian and non-Abelian gauge theories is one of the most important lessons of the renormalization group.

20.10 Landau Poles

If a coupling grows without bound at a finite scale in perturbation theory, the scale is called a Landau pole. In a simple running equation with positive b,

$$\frac{1}{g(\mu)^2} = \frac{1}{g(\mu_0)^2} - 2b \ln \frac{\mu}{\mu_0},$$

the denominator can vanish at finite μ. Perturbation theory then breaks down.

A Landau pole may signal that the theory cannot be extended to arbitrarily high energies without new physics, or that perturbation theory is no longer reliable. In effective field theory, this is not necessarily a problem; it simply indicates the scale at which the description must be replaced or completed.

20.11 RG Improvement of Perturbation Theory

Perturbative calculations often contain large logarithms such as

$$\lambda \ln \frac{E}{\mu}.$$

If E is very different from μ, these logarithms can spoil the convergence of perturbation theory even when λ is small.

The renormalization group improves perturbation theory by choosing μ near the physical scale and using running couplings. This resums towers of logarithms into scale-dependent parameters.

The practical rule is:

$$\text{choose } \mu \text{ of order the characteristic energy scale.}$$

Then the explicit logarithms are small, and the running coupling carries the large-scale dependence.

20.12 Physical Meaning of Running

Running couplings do not mean that charge or interaction strength changes with time. They mean that different experiments probe the theory at different length or momentum scales. A coupling is an effective parameter summarizing the response of the quantum vacuum at a chosen scale.

At long distances, high-energy fluctuations have already been integrated into the parameters. At short distances, more fluctuations are resolved explicitly. The renormalization group describes how the description changes as the resolution changes.

This is the conceptual heart of modern QFT: the same physics can be described at many scales, but the parameters and relevant degrees of freedom may change.

Summary

The renormalization group organizes the dependence of renormalized quantities on the scale μ. Couplings run according to beta functions, fields and masses acquire anomalous dimensions, and physical observables remain independent of the arbitrary renormalization scale. Fixed points, relevant and irrelevant operators, dimensional transmutation, asymptotic freedom, and RG improvement are all consequences of scale-dependent quantum fluctuations.

20.13 Chapter Checklist

After reading this chapter, one should be able to:

- explain why renormalized parameters depend on μ;
- define a beta function;
- solve a simple one-coupling RG equation;
- state the meaning of the Callan–Symanzik equation;
- explain anomalous dimensions;
- define fixed points;
- classify relevant, irrelevant, and marginal operators;
- explain dimensional transmutation;
- distinguish asymptotic freedom from Landau-pole behavior;
- explain how RG improvement resums large logarithms;
- describe the physical meaning of running couplings.

Chapter 21

Effective Field Theory

Remark

Effective field theory is the modern language of QFT at finite resolution. It accepts that a theory may be valid only below some scale and organizes all allowed interactions as an expansion in powers of energy over that scale.

21.1 The Effective Field Theory Viewpoint

The renormalization group teaches that physics changes with scale. Effective field theory, or EFT, turns this lesson into a practical method. Instead of requiring a theory to be fundamental at all distances, one asks what degrees of freedom and symmetries are relevant at the energies being studied.

An EFT is valid below a cutoff scale Λ. It contains the fields that are light compared with Λ, and it includes every interaction compatible with the symmetries:

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{ren}} + \sum_i \frac{c_i}{\Lambda^{\Delta_i - 4}} \mathcal{O}_i.$$

Here $\mathcal{O}_i$ are local operators of dimension $\Delta_i > 4$, and c_i are dimensionless coefficients.

The expansion parameter is usually

$$\frac{E}{\Lambda},$$

where E is the characteristic energy of the process.

21.2 Separation of Scales

EFT works when there is a hierarchy of scales. Suppose a theory contains a heavy particle of mass M and we study processes with

$$E \ll M.$$

At such energies, the heavy particle cannot be produced on shell. Its effects are still present, but they appear indirectly through local interactions among the light fields.

The heavy propagator illustrates the idea. For momentum transfer $p^2 \ll M^2$,

$$\frac{i}{p^2 - M^2} = -\frac{i}{M^2}\left(1 + \frac{p^2}{M^2} + \frac{p^4}{M^4} + \cdots\right).$$

The nonlocal exchange of a heavy particle becomes a series of local operators suppressed by powers of M.

This is the basic mechanism behind effective interactions.

21.3 Integrating Out Heavy Fields

To integrate out a heavy field means to remove it as an explicit degree of freedom while keeping its low-energy effects. In the path integral, if H is a heavy field and ℓ denotes light fields, one writes schematically

$$e^{iS_{\text{eff}}[\ell]} = \int \mathcal{D}H \, e^{iS[\ell,H]}.$$

The result is an effective action for the light fields only.

At tree level, integrating out a heavy field often amounts to solving its classical equation of motion and substituting back. At loop level, it also produces determinants and quantum corrections.

The effective action is generally nonlocal if written exactly. At low energies, it is expanded in local operators with increasing numbers of derivatives or fields.

21.4 Matching

Matching determines the coefficients of the EFT operators. One computes the same low-energy quantity in the full theory and in the EFT, then chooses the EFT coefficients so that the two agree at the matching scale.

Schematically,

$$\mathcal{A}_{\text{full}}(E \ll M) = \mathcal{A}_{\text{EFT}}(E \ll M).$$

This equality fixes the Wilson coefficients c_i.

Matching can be done at tree level or loop level. Tree-level matching captures classical heavy-particle exchange. Loop-level matching captures virtual heavy particle effects.

The matching scale is usually chosen near the heavy mass:

$$\mu \sim M.$$

Then large logarithms are avoided at the matching step. Renormalization group evolution can then run the EFT coefficients down to lower scales.

21.5 Power Counting

Power counting tells us which operators matter at a given accuracy. If an operator has dimension Δ, its contribution scales roughly as

$$\left(\frac{E}{\Lambda}\right)^{\Delta-4}.$$

Operators of higher dimension are more suppressed at low energies.

This means that EFT can be predictive even though it contains infinitely many operators. To compute to a fixed order in E/Λ, only finitely many operators are needed.

For example, if one works up to order $1/\Lambda^2$, only dimension-six operators are needed beyond the renormalizable terms. Dimension-eight operators enter at order $1/\Lambda^4$.

21.6 Symmetries and Operator Bases

The EFT Lagrangian must include all operators allowed by the symmetries. It must exclude operators forbidden by symmetries. Symmetries therefore organize the operator basis.

Operators related by integration by parts, equations of motion, or algebraic identities are not independent. An operator basis is a minimal independent set of operators after these redundancies are removed.

The construction follows a clear logic:

1. choose the light fields;

2. choose the symmetries;

3. list local operators ordered by dimension;
4. remove redundant operators;
5. assign Wilson coefficients;
6. match coefficients to data or to a UV theory.

The result is the most general low-energy theory compatible with the chosen assumptions.

21.7 Wilson Coefficients

The coefficients c_i multiplying EFT operators are called Wilson coefficients. They summarize short-distance physics not resolved explicitly in the EFT.

A Wilson coefficient may be computed from a known high-energy theory by matching. If the high-energy theory is unknown, the coefficients are treated as parameters to be measured or bounded experimentally.

The separation is

$$\text{long-distance physics} \longrightarrow \text{matrix elements of EFT operators,}$$

$$\text{short-distance physics} \longrightarrow \text{Wilson coefficients.}$$

Renormalization group evolution moves information between scales by changing the coefficients as μ changes.

21.8 Decoupling

The decoupling principle says that heavy particles have suppressed effects at low energies. If a heavy particle has mass M, its virtual effects often appear as powers of $1/M$.

However, decoupling must be interpreted carefully. Heavy particles can also modify renormalized parameters of lower-dimensional operators. Such effects are not necessarily suppressed by powers of $1/M$; they may appear as threshold corrections to masses, couplings, or kinetic terms.

After these parameters are matched, genuinely new low-energy effects from heavy physics are organized by higher-dimensional operators.

21.9 Naturalness

Naturalness asks whether small parameters remain small under quantum corrections. Suppose a scalar mass receives a correction of order

$$\delta m^2 \sim \frac{\lambda}{16\pi^2}\Lambda^2.$$

If the physical mass is much smaller than Λ, a cancellation between the bare mass and the correction may be required. This is called fine-tuning.

Fermion masses and gauge boson masses can be protected by symmetries. Scalar masses are less protected in generic theories. This is the origin of naturalness questions surrounding scalar fields, including the Higgs boson.

Naturalness is not a theorem. It is a diagnostic about sensitivity to short- distance physics.

21.10 The Fermi Theory of Weak Interactions

A classic EFT example is Fermi's four-fermion theory. At energies much below the W-boson mass, weak interactions can be described by a local operator:

$$\mathcal{L}_{\text{Fermi}} \sim -\frac{G_F}{\sqrt{2}}(\bar{\psi}\gamma^\mu(1-\gamma^5)\psi)(\bar{\psi}\gamma_\mu(1-\gamma^5)\psi).$$

The coefficient scales as

$$G_F \sim \frac{1}{M_W^2}.$$

The full electroweak theory contains W-boson exchange. At low energies, that exchange becomes a local four-fermion interaction.

This example shows how an EFT can be extremely successful while not being fundamental at arbitrarily high energies.

21.11 Euler–Heisenberg as a Photon EFT

Another example is the low-energy effective theory of photons after integrating out the electron. At energies much smaller than the electron mass, virtual electron loops generate photon self-interactions:

$$\mathcal{L}_{\text{EH}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + c_1\frac{(F_{\mu\nu}F^{\mu\nu})^2}{m_e^4} + c_2\frac{(F_{\mu\nu}\widetilde{F}^{\mu\nu})^2}{m_e^4} + \cdots .$$

These terms describe light-by-light scattering at energies below the electron mass. They are strongly suppressed by powers of $1/m_e^4$.

The EFT makes clear why photon self-interactions are tiny at low energies even though they are allowed quantum mechanically.

21.12 EFT and Renormalization

EFTs are renormalized like any other quantum field theory. Loop diagrams involving higher-dimensional operators generate divergences, which are absorbed into coefficients of operators allowed by the symmetries.

Because infinitely many operators are present, the theory is not renormalizable in the old finite-counterterm sense. But it is renormalizable order by order in E/Λ. At any fixed order, only finitely many counterterms are needed.

This is the modern meaning of predictivity in EFT: choose an accuracy, include all operators up to the required dimension, renormalize them, and estimate the size of neglected terms.

21.13 EFT as the Default View of QFT

Modern QFT often treats every theory as effective unless proven otherwise. The Standard Model itself may be an EFT below a much higher scale. Gravity is also successfully described as an EFT at energies far below the Planck scale.

This viewpoint changes the meaning of nonrenormalizable interactions. They are not automatically forbidden. They are suppressed by a high scale and become important when experiments reach enough precision or energy.

The EFT attitude is therefore practical and humble:

write what is allowed, organize by scale, measure what is unknown.

Summary

Effective field theory describes physics below a cutoff scale Λ using light fields, symmetries, and a systematic expansion in E/Λ. Heavy fields are integrated out, leaving local operators with Wilson coefficients. Matching fixes those coefficients, power counting determines which operators are needed, and renormalization group evolution moves the theory between scales. EFT is predictive order by order even when infinitely many operators are allowed.

21.14 Chapter Checklist

After reading this chapter, one should be able to:

- explain the EFT viewpoint;
- describe separation of scales;
- explain what it means to integrate out a heavy field;
- define matching;
- use power counting to rank operators;
- explain the role of symmetries in building an operator basis;
- define Wilson coefficients;
- explain decoupling and threshold effects;
- describe naturalness as sensitivity to high scales;
- explain the Fermi theory as an EFT;
- state why EFTs are predictive order by order.

Chapter 22

Non-Abelian Gauge Theory as a Quantum Theory

Remark

QED is based on an Abelian gauge group. Non-Abelian gauge theory is richer: the gauge fields themselves carry charge and interact with one another. This chapter introduces the quantum structure behind Yang–Mills theory, preparing the way for BRST symmetry and QCD.

22.1 From Abelian to Non-Abelian Gauge Symmetry

In QED, the gauge group is $U(1)$. The gauge field is neutral, and the field strength is linear in A_μ:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

In a non-Abelian gauge theory, the gauge group has noncommuting generators T^a:

$$[T^a, T^b] = if^{abc}T^c.$$

The constants f^{abc} are the structure constants of the Lie algebra.

The gauge field carries an adjoint index:

$$A_\mu = A_\mu^a T^a.$$

The covariant derivative is

$$D_\mu = \partial_\mu + igA_\mu^a T^a.$$

Because the generators do not commute, the gauge field becomes self-interacting.

22.2 Field Strength and Yang–Mills Action

The non-Abelian field strength is defined by the commutator of covariant derivatives:

$$[D_\mu, D_\nu] = igF_{\mu\nu}^a T^a.$$

In components,

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf^{abc}A_\mu^b A_\nu^c,$$

up to sign conventions for the covariant derivative.

The Yang–Mills Lagrangian is

$$\mathcal{L}_{\text{YM}} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}.$$

Unlike the Maxwell Lagrangian, this contains interaction terms even without matter. Expanding $F_{\mu\nu}^a F^{a\mu\nu}$ gives cubic and quartic powers of A_μ^a. Therefore gauge bosons interact with each other.

22.3 Gauge Transformations

A finite gauge transformation is

$$U(x) = e^{i\alpha^a(x)T^a}.$$

Matter fields transform as

$$\psi(x) \longrightarrow U(x)\psi(x).$$

The gauge field transforms as

$$A_\mu \longrightarrow UA_\mu U^{-1} + \frac{i}{g}U\partial_\mu U^{-1},$$

with the precise sign depending on the convention for D_μ.

Infinitesimally, the gauge field changes by

$$\delta A^a_\mu = -D^{ab}_\mu \alpha^b,$$

where the adjoint covariant derivative is

$$D^{ab}_\mu = \delta^{ab}\partial_\mu + gf^{acb}A^c_\mu.$$

The nonlinearity of this transformation is another sign that non-Abelian gauge theory is structurally different from QED.

22.4 Gauge Fixing

The path integral over A^a_μ overcounts gauge-equivalent configurations. As in the Abelian case, one must fix a gauge. A common covariant gauge-fixing term is

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial_\mu A^{a\mu})^2.$$

The gauge-fixed quadratic part of the action gives the gauge-boson propagator

$$D^{ab}_{\mu\nu}(k) = \frac{-i\delta^{ab}}{k^2+i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2+i\epsilon}\right).$$

In Feynman gauge,

$$D^{ab}_{\mu\nu}(k) = \frac{-i\delta^{ab}\eta_{\mu\nu}}{k^2+i\epsilon}.$$

The color index a is conserved along a free gauge-boson propagator.

22.5 Faddeev–Popov Ghosts

For non-Abelian gauge theory, the Faddeev–Popov determinant depends on the gauge field. It is represented using ghost fields $c^a, \bar{c}^a$. The ghost Lagrangian in covariant gauge is

$$\mathcal{L}_{\text{gh}} = \bar{c}^a\partial^\mu D^{ab}_\mu c^b.$$

Expanding the covariant derivative gives a free ghost kinetic term and a ghost–gauge-boson interaction.

Ghosts are anticommuting scalar fields. They are not physical particles and do not appear as external states. They are required in covariant gauges to preserve unitarity and gauge independence of physical amplitudes.

In Abelian QED, ghosts decouple in linear gauges. In non-Abelian theories, they do not decouple because the determinant depends on A^a_μ.

22.6 Gauge-Boson Self-Interactions

The Yang–Mills field strength contains the nonlinear term

$$-gf^{abc}A_\mu^b A_\nu^c.$$

Therefore the action contains cubic and quartic gauge-field interactions. Diagrammatically, there are two new types of vertices:

- a three-gauge-boson vertex proportional to gf^{abc};
- a four-gauge-boson vertex proportional to g^2 and products of structure constants.

These vertices have no analogue in QED. They are responsible for many of the special features of non-Abelian gauge theories, including asymptotic freedom.

The physical meaning is simple but profound: gauge bosons themselves carry the charge associated with the gauge group.

22.7 Matter Couplings

Matter fields couple through the covariant derivative. For a Dirac fermion in a representation R, the Lagrangian is

$$\mathcal{L}_{\text{matter}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi.$$

Expanding the derivative gives the interaction

$$\mathcal{L}_{\text{int}} = -g\bar{\psi}\gamma^\mu T_R^a \psi A_\mu^a,$$

up to sign conventions. The corresponding vertex factor is

$$-ig\gamma^\mu T_R^a.$$

This is the non-Abelian analogue of the QED vertex $-ie\gamma^\mu$, with the charge e replaced by a matrix gT^a.

The group generators track how matter transforms under the gauge group. In QCD, quarks transform in the fundamental representation of $SU(3)$, and gluons transform in the adjoint representation.

22.8 Feynman Rules: The Basic Set

In a covariant gauge, the basic perturbative ingredients are:

- gauge-boson propagator $D_{\mu\nu}^{ab}(k)$;
- ghost propagator;
- fermion propagator, if matter is present;
- fermion–gauge-boson vertex;
- ghost–gauge-boson vertex;
- three-gauge-boson vertex;
- four-gauge-boson vertex.

Every diagram carries both Lorentz structure and group-theory structure. Momentum conservation holds at each vertex, while group indices are contracted according to the generators and structure constants.

The algebra is more complex than QED, but the logic is the same: propagators come from quadratic terms, and vertices come from interaction terms in the gauge-fixed action.

22.9 Ward Identities and Slavnov–Taylor Identities

In QED, gauge invariance leads to Ward identities. In non-Abelian gauge theory, the corresponding identities are more complicated because gauge transformations are nonlinear and ghosts are present. They are called Slavnov–Taylor identities.

These identities relate Green functions and ensure that unphysical polarizations and ghosts cancel from physical observables. They also constrain renormalization, ensuring that the gauge-fixed quantum theory preserves the content of the original gauge symmetry.

BRST symmetry provides the most efficient modern derivation of these identities. The next chapter develops BRST symmetry more systematically.

22.10 Unitarity and Physical States

Covariant quantization introduces unphysical gauge-field polarizations and ghost fields. The full gauge-fixed state space is larger than the physical Hilbert space. Physical states must be selected so that negative-norm and ghost degrees of freedom do not appear as observable particles.

The cancellation of unphysical contributions is not diagram-by-diagram obvious. It follows from the gauge symmetry encoded after gauge fixing, especially through BRST symmetry. This is why ghosts, although unphysical, are essential in loop calculations.

The physical S-matrix is unitary when restricted to the physical subspace. This statement is one of the central consistency requirements of quantum gauge theory.

22.11 One-Loop Beta Function

Non-Abelian gauge theories have a remarkable one-loop beta function. For an $SU(N)$ gauge theory with n_f Dirac fermions in the fundamental representation, the leading beta function has the form

$$\beta(g) = -\frac{g^3}{16\pi^2}\left(\frac{11}{3}N - \frac{2}{3}n_f\right) + \cdots .$$

The gauge-boson self-interactions give the positive $11N/3$ contribution inside the parentheses, leading to a negative beta function when n_f is not too large.

This is the origin of asymptotic freedom. At high energies, the coupling becomes small. At low energies, it grows, leading toward strong-coupling physics.

22.12 Comparison with QED

The contrast with QED is sharp. In QED, photons do not carry electric charge and there are no photon self-interactions at the fundamental level. The leading beta function is positive, and the coupling grows at high energies.

In non-Abelian gauge theory, gauge bosons carry the gauge charge and interact with one another. Their contribution to the beta function can dominate over matter screening and make the theory asymptotically free.

This difference is not a technical detail. It explains why QCD behaves very differently from QED: quarks and gluons are weakly coupled at short distances but strongly coupled at long distances.

22.13 Toward QCD

Quantum chromodynamics is the non-Abelian gauge theory with gauge group $SU(3)$. Its gauge bosons are gluons, and its matter fields are quarks. The gluons carry color charge because they transform in the adjoint representation.

The Yang–Mills structure introduced in this chapter provides the foundation of QCD:

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + \sum_f \bar{q}_f (i\gamma^\mu D_\mu - m_f) q_f.$$

The later QCD chapter will build on this structure to discuss color, asymptotic freedom, confinement, jets, and hadrons.

Summary

Non-Abelian gauge theories are quantum theories of gauge fields whose symmetry generators do not commute. Their gauge bosons self-interact, ghosts do not decouple in covariant gauges, and physical consistency is encoded in Slavnov–Taylor identities and BRST symmetry. The one-loop beta function can be negative, producing asymptotic freedom. This structure is the foundation of QCD and of the gauge sector of the Standard Model.

22.14 Chapter Checklist

After reading this chapter, one should be able to:

- distinguish Abelian and non-Abelian gauge symmetry;
- write the non-Abelian field strength;
- explain why Yang–Mills theory contains gauge-boson self-interactions;
- write the covariant gauge-fixing term;
- explain why ghosts do not decouple in non-Abelian theories;
- identify the basic non-Abelian Feynman rules;
- write the matter–gauge-boson vertex;
- explain the role of Slavnov–Taylor identities;
- explain why BRST symmetry is needed for physical-state selection;
- state the qualitative origin of asymptotic freedom;
- connect non-Abelian gauge theory to QCD.

Chapter 23

BRST Symmetry

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

23.1 Planned Role

This chapter will introduce BRST transformations, ghost number, nilpotency, physical states as cohomology, gauge independence, and Slavnov–Taylor identities.

Chapter 24

Quantum Chromodynamics

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

24.1 Planned Role

This chapter will introduce QCD field content, color charge, QCD Feynman rules, gluon self-interactions, asymptotic freedom, confinement, jets, and perturbative versus nonperturbative regimes.

Chapter 25

Symmetries in Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

25.1 Planned Role

This chapter will translate classical symmetries into quantum operators, conserved charges, Ward identities, current correlation functions, selection rules, and constraints on observables.

Chapter 26

Anomalies

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

26.1 Planned Role

This chapter will study symmetries that fail after quantization, chiral currents, triangle diagrams, path-integral measure effects, gauge anomalies, anomaly cancellation, and physical consequences.

Chapter 27

Vacuum Structure and Nonperturbative Effects

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

27.1 Planned Role

This chapter will discuss the quantum vacuum, degenerate vacua, tunneling, instantons, theta vacua, false vacuum decay, semiclassical approximation, and nonperturbative contributions.

Chapter 28

The Quantum Electroweak Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

28.1 Planned Role

This chapter will introduce the quantum electroweak sector, gauge fixing after symmetry breaking, massive vector boson propagators, currents, weak processes, and the low-energy Fermi theory.

Chapter 29

The Standard Model Lagrangian as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

29.1 Planned Role

This chapter will organize the Standard Model field content, representations, gauge interactions, Yukawa sector, Higgs sector, flavor mixing, neutrino masses, anomaly cancellation, and EFT viewpoint.

Chapter 30

Selected Standard Model Processes

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

30.1 Planned Role

This chapter will connect QFT calculations with representative Standard Model predictions and collider observables.

Chapter 31

Euclidean Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

31.1 Planned Role

This chapter will introduce Wick rotation, Euclidean correlation functions, reflection positivity, statistical field theory connections, finite temperature, boundary conditions, and Matsubara frequencies.

Chapter 32

Lattice Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

32.1 Planned Role

This chapter will introduce lattice regularization, scalar fields on a lattice, gauge fields on links, Wilson action, Wilson loops, fermion difficulties, Monte Carlo sampling, and the continuum limit.

Chapter 33

Large-N and Semiclassical Methods

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

33.1 Planned Role

This chapter will introduce large-N counting, planar diagrams, saddle-point methods, semiclassical expansion, collective fields, solitons, instanton calculus, and reliability of approximations.

Chapter 34

Conceptual Synthesis I

34.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 35

Conceptual Synthesis II

35.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 36

Summary: The Architecture of Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

36.1 Planned Role

This chapter will summarize quantization, Fock space, propagators, interactions, renormalization, gauge theory, anomalies, nonperturbative effects, the Standard Model, and open directions.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \quad \longrightarrow \quad \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \qquad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\mathrm{PB}} \quad \longrightarrow \quad \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word “distribution” is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green’s Function to Propagator

A classical Green’s function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle.$$

It is related to Green’s functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

Appendix B

Reference B

Placeholder appendix.

Appendix C

Reference C

Placeholder appendix.

Appendix D

Reference D

Placeholder appendix.

Appendix E

Reference E

Placeholder appendix.

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